



US012412164B2

(12) **United States Patent**
Nelms et al.

(10) **Patent No.:** **US 12,412,164 B2**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **METHOD FOR SELF-CHECKOUT WITH A MOBILE DEVICE**

(71) Applicant: **Walmart Apollo, LLC**, Bentonville, AR (US)

(72) Inventors: **David Martin Nelms**, Rogers, AR (US); **Jason Todd**, Lowell, AR (US); **Tim Webb**, Rogers, AR (US); **Philip Wayne Marbut**, Bentonville, AR (US); **Douglas Jahe Ryner**, Cave Springs, AR (US)

(73) Assignee: **Walmart Apollo, LLC**, Bentonville, AR (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/427,556**

(22) Filed: **Jan. 30, 2024**

(65) **Prior Publication Data**

US 2024/0394682 A1 Nov. 28, 2024

Related U.S. Application Data

(60) Continuation of application No. 18/217,945, filed on Jul. 3, 2023, now Pat. No. 11,961,061, which is a continuation of application No. 17/004,819, filed on Aug. 27, 2020, now Pat. No. 11,853,988, which is a continuation of application No. 16/730,567, filed on Dec. 30, 2019, now Pat. No. 10,803,435, which is a continuation of application No. 15/956,400, filed on (Continued)

(51) **Int. Cl.**
G06Q 20/00 (2012.01)
G06Q 20/20 (2012.01)
G06Q 30/0601 (2023.01)

(52) **U.S. Cl.**
CPC **G06Q 20/20** (2013.01); **G06Q 30/0633** (2013.01); **G06Q 30/0641** (2013.01)

(58) **Field of Classification Search**
CPC . G06Q 20/20; G06Q 30/0641; G06Q 30/0633
USPC 705/16
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,055,660 A 10/1991 Bertagna
5,457,307 A 10/1995 Dumont
(Continued)

FOREIGN PATENT DOCUMENTS

CA 2809523 4/2012
CN 108805546 A 11/2018
(Continued)

OTHER PUBLICATIONS

U.S. Appl. No. 18/203,433, filed May 30, 2023, David Sica.
(Continued)

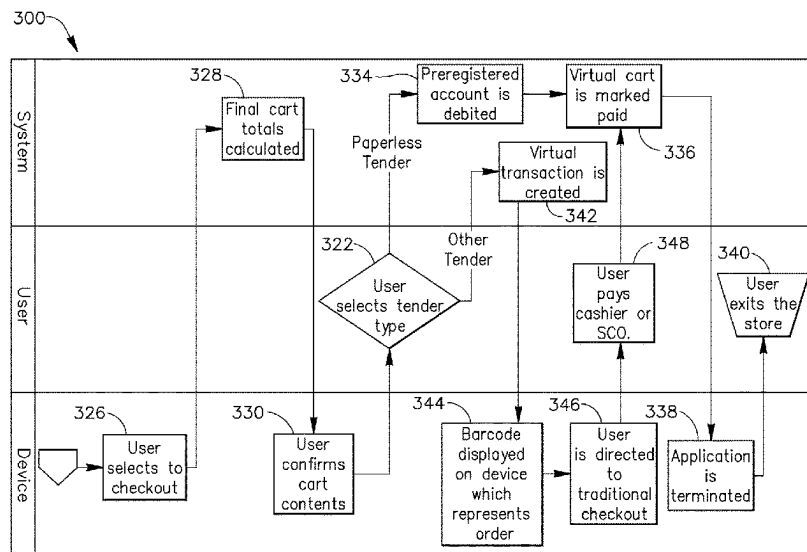
Primary Examiner — Rokib Masud

(74) *Attorney, Agent, or Firm* — Fitch, Even, Tabin & Flannery LLP

(57) **ABSTRACT**

A method for purchasing an item using a mobile device is provided. The method includes but is not limited to inputting identification information into the mobile device which identifies the item for purchase and generating an order for the item identified by the identification information. The method also includes but is not limited to displaying on a display of the mobile device an optical machine-readable representation of the order.

20 Claims, 24 Drawing Sheets



Related U.S. Application Data

Apr. 18, 2018, now abandoned, which is a division of application No. 12/947,545, filed on Nov. 16, 2010, now Pat. No. 10,121,133.

- (60) Provisional application No. 61/392,908, filed on Oct. 13, 2010.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,489,773	A	2/1996	Kumar	8,820,633	B2	9/2014	Bishop
5,689,101	A	11/1997	Kikuchi	8,851,369	B2	10/2014	Bishop
5,736,721	A	4/1998	Swartz	9,033,217	B2	5/2015	Williams
5,804,807	A	9/1998	Murrah	9,195,981	B2	11/2015	Annan
5,825,002	A	10/1998	Roslak	10,068,276	B2	9/2018	Karmakar
5,918,211	A	6/1999	Sloane	10,121,133	B2	11/2018	Nelms
5,923,735	A	7/1999	Swartz	10,496,951	B1	12/2019	Kandukuri
5,978,772	A	11/1999	Mold	10,803,435	B2	10/2020	Nelms
5,979,753	A	11/1999	Roslak	10,872,326	B2	12/2020	Garner
5,979,757	A	11/1999	Tracy	10,915,906	B2	2/2021	Keith
6,018,397	A	1/2000	Cloutier	10,990,960	B2	4/2021	Koeppel
6,131,814	A	10/2000	Swartz	11,263,682	B2	3/2022	Karmakar
6,179,206	B1	1/2001	Matsumori	11,532,149	B1	12/2022	Floerkemeier
6,354,496	B1	3/2002	Murphy	11,853,988	B2	12/2023	Nelms
6,367,694	B1	4/2002	Roslak	11,907,998	B2	2/2024	Karmakar
6,382,357	B1	5/2002	Morrison	11,961,061	B2	4/2024	Nelms
6,507,279	B2	1/2003	Loof	2002/0060246	A1	5/2002	Gobburu
6,550,672	B1	4/2003	Tracy	2002/0096564	A1	7/2002	Bellis
6,571,218	B1	5/2003	Sadler	2002/0143593	A1	10/2002	Takata
6,598,791	B2	7/2003	Bellis, Jr.	2002/0145038	A1	10/2002	O'Hagan
6,601,759	B2	8/2003	Fife	2002/0178088	A1	11/2002	Lurie
6,606,411	B1	8/2003	Loui	2003/0015585	A1	1/2003	Wike
6,685,093	B2	2/2004	Challa	2003/0057284	A1	3/2003	Challa
6,745,186	B1	6/2004	Testa	2003/0132298	A1	7/2003	Swartz
6,804,528	B1	10/2004	Laroia	2003/0149630	A1	8/2003	Jacobs
6,810,149	B1	10/2004	Squilla	2003/0149663	A1	8/2003	Vonholm
6,820,062	B1	11/2004	Gupta	2003/0158796	A1	8/2003	Balent
6,837,436	B2	1/2005	Swartz	2003/0195818	A1	10/2003	Howell
6,854,651	B2	2/2005	Smith	2004/0056101	A1	3/2004	Barkan
6,876,977	B1	4/2005	Marks	2004/0103034	A1	5/2004	Reade
6,886,101	B2	4/2005	Glazer	2004/0171378	A1	9/2004	Rautila
6,926,202	B2	8/2005	Noonan	2004/0193499	A1	9/2004	Ortiz
6,937,989	B2	8/2005	Mcintyre	2005/0017071	A1	1/2005	Noonan
7,047,041	B2	5/2006	Vaenskae	2005/0075940	A1	4/2005	Deangelis
7,114,656	B1	10/2006	Garver	2005/0246714	A1	11/2005	Moore
7,124,940	B2	10/2006	Som	2006/0043175	A1	3/2006	Fu
7,177,820	B2	2/2007	Mcintyre	2006/0108419	A1	5/2006	Som
7,194,430	B2	3/2007	Vonholm	2006/0124733	A1	6/2006	Matthews
7,207,477	B1	4/2007	Ramachandran	2006/0149626	A1	7/2006	Bennett
7,209,903	B1	4/2007	Mamdani	2006/0283941	A1	12/2006	Singer-Harter
7,222,363	B2	5/2007	Rice	2006/0293968	A1	12/2006	Brice
7,376,583	B1	5/2008	Rolf	2007/0051801	A1	3/2007	Garver
7,431,208	B2	10/2008	Feldman	2007/0069016	A1	3/2007	Garver
7,493,284	B2	2/2009	Celi, Jr.	2007/0080219	A1	4/2007	Garver
7,587,756	B2	9/2009	Peart	2007/0080220	A1	4/2007	Garver
7,617,128	B2	11/2009	Greak	2007/0239552	A1	10/2007	Sundaresan
7,635,084	B2	12/2009	Wang	2007/0241189	A1	10/2007	Slavin
7,716,064	B2	5/2010	Mcintyre	2007/0255620	A1	11/2007	Tumminaro
7,725,326	B1	5/2010	Tracy	2008/0011825	A1	1/2008	Giordano
7,725,362	B2	5/2010	Weathers, Jr.	2008/0059341	A1	3/2008	Narayanaswami
7,730,216	B1	6/2010	Issa	2008/0133366	A1	6/2008	Evans
7,970,661	B1	6/2011	Abraham	2008/0177641	A1	7/2008	Herniak
7,997,486	B2	8/2011	Ulrich	2008/0228591	A1	9/2008	Watanabe
8,069,092	B2	11/2011	Bryant	2008/0237340	A1	10/2008	Emmons
8,091,780	B2	1/2012	Todd	2008/0296392	A1	12/2008	Connell, II
8,355,711	B2	1/2013	Heins	2009/0043658	A1	2/2009	Webb
8,417,572	B1	4/2013	Chenault	2009/0055278	A1	2/2009	Nemani
8,572,712	B2	10/2013	Rice	2009/0088203	A1	4/2009	Havens
8,583,512	B1	11/2013	Gupta	2009/0094100	A1 *	4/2009	Xavier
8,645,221	B1	2/2014	Sarma				G06Q 20/204 705/16
8,682,973	B2	3/2014	Kikin-Gil	2009/0099961	A1	4/2009	Ogilvy
8,720,771	B2	5/2014	Mackinnon Keith	2009/0101713	A1	4/2009	Ulrich
8,751,316	B1 *	6/2014	Fletchall	2009/0192892	A1	7/2009	Cason
				2009/0250515	A1	10/2009	Todd
				2009/0261164	A1	10/2009	Sato
				2010/0030578	A1	2/2010	Siddique
				2010/0030624	A1	2/2010	Marko
				2010/0063891	A1 *	3/2010	Townsend
							G06Q 30/0222 705/26.1
				2010/0082419	A1	4/2010	Au-Yeung
				2010/0082444	A1	4/2010	Lin
				2010/0082445	A1	4/2010	Hodge
				2010/0082447	A1	4/2010	Lin
				2010/0125510	A1 *	5/2010	Smith
							G06Q 20/425 705/17
				2010/0138344	A1	6/2010	Wong
				2010/0185504	A1	7/2010	Rajan
				2010/0259549	A1	10/2010	Brown
				2010/0287057	A1	11/2010	Aihara
8,814,039	B2	8/2014	Bishop				

(56) References Cited

U.S. PATENT DOCUMENTS

2010/0293474	A1	11/2010	York
2010/0299616	A1	11/2010	Chen
2011/0075213	A1	3/2011	Murayama
2011/0106655	A1	5/2011	Kreiner
2011/0225057	A1	9/2011	Webb
2011/0251892	A1	10/2011	Laracey
2011/0289113	A1	11/2011	Arling
2012/0054046	A1	3/2012	Albisu
2012/0072304	A1	3/2012	Homan
2012/0095853	A1	4/2012	Von Bose
2012/0173347	A1	7/2012	De Almeida Neves
2012/0173351	A1	7/2012	Hanson
2012/0185355	A1	7/2012	Kilroy
2012/0203605	A1	8/2012	Morgan
2012/0209749	A1	8/2012	Hammad
2012/0253972	A1	10/2012	Oskolkov
2012/0267432	A1	10/2012	Kuttuva
2013/0013438	A1	1/2013	Chen
2013/0054328	A1	2/2013	Chavie
2013/0117378	A1	5/2013	Kotorov
2013/0132220	A1	5/2013	Baum
2013/0211953	A1	8/2013	Abraham
2013/0268331	A1	10/2013	Bitz
2013/0282581	A1	10/2013	Singh
2013/0290234	A1	10/2013	Harris
2013/0335273	A1	12/2013	Pakzad
2014/0108178	A1	4/2014	Cao
2014/0214564	A1	7/2014	Argue
2015/0106213	A1	4/2015	Guen
2015/0161706	A1	6/2015	Karmakar
2016/0005019	A1	1/2016	Paulo Rodrigues
2016/0342993	A1	11/2016	Morgan
2017/0161720	A1	6/2017	Xing
2017/0161735	A1	6/2017	Ortiz
2018/0240088	A1	8/2018	Nelms
2020/0019743	A1	1/2020	Lei
2020/0134588	A1	4/2020	Nelms
2021/0081914	A1	3/2021	Nelms
2022/0148068	A1	5/2022	Karmakar
2023/0342744	A1	10/2023	Nelms
2024/0152991	A1	5/2024	Karmakar

FOREIGN PATENT DOCUMENTS

EP	1120071	8/2001
EP	1170690 A1	1/2002
GB	2497842	6/2013
JP	2001325468	11/2001
JP	2002189785	7/2002
JP	2002197384	7/2002
JP	2003140865	5/2003
JP	2003323570	11/2003
JP	2004021607	1/2004
JP	2006146774	6/2006
JP	2010092146	4/2010
JP	2010182105	8/2010
JP	2013541107	11/2013
JP	2016219034	12/2016
JP	6419926	11/2018
JP	6615972	12/2019
KR	1020070087811	8/2007
WO	2003060646	7/2003
WO	2003090023	10/2003
WO	2009116954	9/2009
WO	2012051316	4/2012
WO	2015085101	6/2015
WO	2018035677 A1	3/2018
WO	2012172568 A1	12/2021

OTHER PUBLICATIONS

U.S. Appl. No. 18/217,945, filed Jul. 3, 2023, Nelms David Martin.
 U.S. Appl. No. 29/993,452, filed May 30, 2023, David Sica.
 ABC7; “Is it legal to search shopping bags when leaving a store?”;

<<https://abc7news.com/archive/6498833>>; Nov. 11, 2008; pp. 1-5.
 Adobe; “JPEG vs. RAW”; <<https://www.adobe.com/creativecloud/file-types/image/comparison/jpeg-vs-raw.html>>; available at least as early as Jul. 10, 2023; pp. 1-3.

Barbaro, Michael, et al.; “Wal-Mart at Forefront of Hurricane Relief”; Sep. 6, 2005; The Washington Post; pp. 1-4.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01496; Exhibit/Paper No. 2006; Walmart Apollo, LLC; “Plaintiff’s Claim Construction Brief”; United States District Court for the Middle District of Florida, Orlando Division; Filing Date Oct. 21, 2022; 27 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01496; Exhibit/Paper No. 2007; Walmart Apollo, LLC; “Defendant BJ’s Wholesale Club Holdings, Inc.’s Responsive Claim Construction Brief”; United States District Court for the Middle District of Florida, Orlando Division; Filing Date Nov. 21, 2022; 24 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review IPR2022-01528, IPR2022-01563, IPR2022-01564; Exhibit/Paper No. 0027; USPTO Patent Trial and Appeal Board; “Leap Practitioner Request and Verification Form (Patent Owner)”; USPTO Patent Trial and Appeal Board; Filing Date Jan. 8, 2024; 4 pages.
BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review IPR2022-01528, IPR2022-01563, IPR2022-01564; Exhibit/Paper No. 0028; Walmart Apollo, LLC; “Patent Owner’s Exhibit List”; USPTO Patent Trial and Appeal Board; Filing Date Jan. 8, 2024; 6 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review IPR2022-01528, IPR2022-01563, IPR2022-01564; Exhibit/Paper No. 0030; USPTO Patent Trial and Appeal Board; “Hearing Transcript”; USPTO Patent Trial and Appeal Board; Filing Date Apr. 5, 2024; pp. 1-75.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01471; Exhibit/Paper No. 0002; BJ’s Wholesale Club Holdings, Inc.; “Petition for Inter Partes Review of U.S. Pat. No. 10,121,133”; USPTO Patent Trial and Appeal Board; Filing Date Sep. 29, 2022; pp. iii-vii and 1-83.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01471; Exhibit/Paper No. 0006; Walmart Apollo, LLC; “Patent Owner’s Preliminary Response”; USPTO Patent Trial and Appeal Board; Filing Date Jan. 13, 2023; pp. i-v and 1-74.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01471; Exhibit/Paper No. 0007; USPTO Patent Trial and Appeal Board; “Denying Institution of Inter Partes Review 35 U.S.C. § 314”; USPTO Patent Trial and Appeal Board; Filing Date Apr. 12, 2023; 19 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01471; Exhibit/Paper No. 0010; USPTO Patent Trial and Appeal Board; “Notice of Refund”; USPTO Patent Trial and Appeal Board; Filing Date Apr. 5, 2024; pp. 1-3.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01471; Exhibit/Paper No. 1002; BJ’s Wholesale Club Holdings, Inc.; “Declaration of Micheal I. Shamos, Ph.D.”; USPTO Patent Trial and Appeal Board; Filing Date Sep. 29, 2022; pp. i-iii and 1-130.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01471; Exhibit/Paper No. 2004; USPTO Patent Trial and Appeal Board; “Exhibit, Decision on Appeal in Ex Parte Burns”; USPTO Patent Trial and Appeal Board; Filing Date Jan. 13, 2023; 8 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01496, IPR2022-01509; Exhibit/Paper No. 2008; Walmart Apollo, LLC; “Deposition of David Hilliard Williams”; USPTO Patent Trial and Appeal Board; Filing Date Jun. 14, 2023; pp. 1-62.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01496; Exhibit/Paper No. 2004; BJ’s Wholesale Club Holdings, Inc.; “Declaration of Dr. Darrell Long”; USPTO Patent Trial and Appeal Board; Filing Date Jul. 14, 2023; 126 pages.
BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01496; Exhibit/Paper No. 2021; Walmart

(56)

References Cited**OTHER PUBLICATIONS**

Apollo, LLC; "Declaration of Matthew W. Johnson"; USPTO Patent Trial and Appeal Board; Filing Date Jul. 14, 2023; 5 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01561, IPR2022-01563, IPR2022-01564; Exhibit/Paper No. 0007; BJ's Wholesale Club Holdings, Inc.; "Petitioner's Notice of Ranking Petitions for Inter Partes Review of U.S. Pat. No. 10,803,435"; USPTO Patent Trial and Appeal Board; Filing Date Oct. 17, 2022; pp. 1-6.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01561, IPR2022-01563, IPR2022-01564; Exhibit/Paper No. 0002; Walmart Apollo, LLC; "Exhibit, Declaration of Matthew W. Johnson"; USPTO Patent Trial and Appeal Board; Filing Date Jan. 13, 2023; pp. 1-5.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01561, IPR2022-01563, IPR2022-01564; Exhibit/Paper No. 0003; Walmart Apollo, LLC; "Exhibit, Claim 1 to Claim 15 Comparison"; USPTO Patent Trial and Appeal Board; Filing Date Jan. 13, 2023; pp. 1.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01561, IPR2022-01563; Exhibit/Paper No. 0010; Walmart Apollo, LLC; "Patent Owner's Preliminary Sur-Reply"; USPTO Patent Trial and Appeal Board; Filing Date Feb. 22, 2023; pp. 1-8.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01561, IPR2022-01563; Exhibit/Paper No. 1027; BJ's Wholesale Club Holdings, Inc.; "Combined Declaration from Translator Hyunjo Kim and Patent Attorney Seongsu Hong For Original Certified Translation in Exhibit 1010"; USPTO Patent Trial and Appeal Board; Filing Date Jun. 8, 2023; pp. 1-4.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01561, IPR2022-01563; Exhibit/Paper No. 1028; BJ's Wholesale Club Holdings, Inc.; "Certified English Translation of Korean Patent Pub. No. KR 10-2007-0087811 Presented in Exhibit 1009"; USPTO Patent Trial and Appeal Board; Filing Date Jun. 16, 2023; pp. 1-52.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01561, IPR2022-01563; Exhibit/Paper No. 1029; BJ's Wholesale Club Holdings, Inc.; "Certified English Translation of Korean Patent Pub. No. KR 10-2007-0087811 Presented in Exhibit 2001"; USPTO Patent Trial and Appeal Board; Filing Date Jun. 16, 2023; pp. 1-37.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01563, IPR2022-01564; Exhibit/Paper No. 0013; Walmart Apollo, LLC; "Notice of Cross-Examination of Dr. Shamos"; USPTO Patent Trial and Appeal Board; Filing Date May 24, 2023; pp. 3 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01563, IPR2022-01564; Exhibit/Paper No. 0014; BJ's Wholesale Club Holdings, Inc.; "Petitioner's Updated Mandatory Notices Under 37 C.F.R. § 42.8"; USPTO Patent Trial and Appeal Board; Filing Date Jun. 15, 2023; 3 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01563, IPR2022-01564; Exhibit/Paper No. 0015; Walmart Apollo, LLC; "Notice of Stipulation of Due Dates"; USPTO Patent Trial and Appeal Board; Filing Date Jun. 21, 2023; 11 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01563, IPR2022-01564; Exhibit/Paper No. 0017; Walmart Apollo, LLC; "Patent Owner's Updated Mandatory Notices"; USPTO Patent Trial and Appeal Board; Filing Date Jul. 20, 2023; pp. 1-4.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01563, IPR2022-01564; Exhibit/Paper No. 0019; BJ's Wholesale Club Holdings, Inc.; "Notice of Stipulation of Due Dates"; USPTO Patent Trial and Appeal Board; Filing Date Aug. 17, 2023; 9 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01563, IPR2022-01564; Exhibit/

Paper No. 0022; Walmart Apollo, LLC; "Notice of Stipulation of Due Dates"; USPTO Patent Trial and Appeal Board; Filing Date Oct. 26, 2023; 7 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01563, IPR2022-01564; Exhibit/Paper No. 0024; Walmart Apollo, LLC; "Patent Owner's Request for Oral Argument"; USPTO Patent Trial and Appeal Board; Filing Date Nov. 28, 2023; 5 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01563, IPR2022-01564; Exhibit/Paper No. 0026; BJ's Wholesale Club Holdings, Inc.; "Petitioner Amended Notice of Cross-Examination of Dr. Williams"; USPTO Patent Trial and Appeal Board; Filing Date Sep. 18, 2023; pp. 1-3.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01563, IPR2022-01564; Exhibit/Paper No. 0026; USPTO Patent Trial and Appeal Board; "Setting Oral Argument 37 C.F.R. § 42.70"; USPTO Patent Trial and Appeal Board; Filing Date Dec. 2023; 10 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01563, IPR2022-01564; Exhibit/Paper No. 1022; BJ's Wholesale Club Holdings, Inc.; "Remote Deposition of Tim Williams, Ph.D."; USPTO Patent Trial and Appeal Board; Filing Date Oct. 11, 2023; 103 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01563, IPR2022-01564; Exhibit/Paper No. 1035; BJ's Wholesale Club Holdings, Inc.; "Petitioner's Demonstratives"; USPTO Patent Trial and Appeal Board; Filing Date Jan. 8, 2024; pp. 1-89.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01563, IPR2022-01564; Exhibit/Paper No. 2009; Walmart Apollo, LLC; "Declaration of Tim A. Williams, Ph.D. Under 37 C.F.R. § 42.100 Regarding U.S. Pat. No. 10,803,435"; USPTO Patent Trial and Appeal Board; Filing Date Jul. 19, 2023; pp. i-ii and 1-105.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01563, IPR2022-01564; Exhibit/Paper No. 2009; Walmart Apollo, LLC; "Declaration of Tim A. Williams"; USPTO Patent Trial and Appeal Board; Filing Date Jul. 19, 2023; 108 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01563, IPR2022-01564; Exhibit/Paper No. 2011; Walmart Apollo, LLC; "Michael Shamos Ph.D Deposition Transcript"; USPTO Patent Trial and Appeal Board; Filing Date Jul. 19, 2023; pp. 1-188.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01563, IPR2022-01564; Exhibit/Paper No. 2012; Walmart Apollo, LLC; "Barcode Merriam Webster 11th"; USPTO Patent Trial and Appeal Board; Filing Date Jul. 19, 2023; pp. 1-4.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01563, IPR2022-01564; Exhibit/Paper No. 2013; Walmart Apollo, LLC; "Barcode American Heritage 4th"; USPTO Patent Trial and Appeal Board; Filing Date Jul. 19, 2023; pp. 1-4.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01563, IPR2022-01564; Exhibit/Paper No. 2014; Walmart Apollo, LLC; "Receipt American Heritage 4th"; USPTO Patent Trial and Appeal Board; Filing Date Jul. 19, 2023; pp. 1-4.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01563, IPR2022-01564; Exhibit/Paper No. 2017; Walmart Apollo, LLC; "Declaration of Matthew W. Johnson"; USPTO Patent Trial and Appeal Board; Filing Date Jul. 14, 2023; 4 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01563, IPR2022-01564; Exhibit/Paper No. 2017; Walmart Apollo, LLC; "Declaration of Matthew W. Johnson Under 37 C.F.R. § 1.68"; USPTO Patent Trial and Appeal Board; Filing Date Jul. 19, 2023; 4 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01563, IPR2022-01564; Exhibit/

(56)

References Cited**OTHER PUBLICATIONS**

Paper No. 2018; Walmart Apollo, LLC; "Patent Owner's Demonstratives"; USPTO Patent Trial and Appeal Board; Filing Date Jan. 11, 2024; pp. 1-81.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01563; Exhibit/Paper No. 0017; Walmart Apollo, LLC; "Patent Owner's Notice Regarding Petitioner's Motions to Submit Supplemental Information"; USPTO Patent Trial and Appeal Board; Filing Date Jun. 1, 2023; 3 pages. *BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC*; AIA Review No. IPR2022-01528; Exhibit/Paper No. 0002; BJ's Wholesale Club Holdings, Inc.; "Petition for Inter Partes Review of U.S. Pat. No. 10,803,435"; USPTO Patent Trial and Appeal Board; Filing Date Sep. 22, 2022; pp. ii-viii and 1-72.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528; Exhibit/Paper No. 0006; BJ's Wholesale Club Holdings, Inc.; "Petitioner's Updated Exhibit List"; USPTO Patent Trial and Appeal Board; Filing Date Oct. 17, 2022; pp. i-ii and 1-1.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528; Exhibit/Paper No. 0008; Walmart Apollo, LLC; "Patent Owner's Preliminary Response"; USPTO Patent Trial and Appeal Board; Filing Date Jan. 13, 2023; pp. i-iv and 1-44.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528; Exhibit/Paper No. 0009; BJ's Wholesale Club Holdings, Inc.; "Petitioner's Reply to Patent Owner Preliminary Response"; USPTO Patent Trial and Appeal Board; Filing Date Feb. 15, 2023; pp. 1-6.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528; Exhibit/Paper No. 0011; USPTO Patent Trial and Appeal Board; "Institution Decision: Decision Granting Institution of Inter Partes Review 35 U.S.C. § 314"; USPTO Patent Trial and Appeal Board; Filing Date Apr. 12, 2023; pp. 2-35.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528; Exhibit/Paper No. 0013; Walmart Apollo, LLC; "Patent Owner's Objections to Evidence"; USPTO Patent Trial and Appeal Board; Filing Date Apr. 24, 2023; pp. 1-5.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528; Exhibit/Paper No. 0014; BJ's Wholesale Club Holdings, Inc.; "Objections to Patent Owner's Under 37 C.F.R. 42.64(b)(1) To Evidence Served"; USPTO Patent Trial and Appeal Board; Filing Date Apr. 24, 2023; 3 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528; Exhibit/Paper No. 0016; BJ's Wholesale Club Holdings, Inc.; "Petitioner's Motion to Submit Supplemental Information Pursuant to 37 C.F.R. sec. 42.123"; USPTO Patent Trial and Appeal Board; Filing Date May 26, 2023; 11 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528; Exhibit/Paper No. 0018; USPTO Patent Trial and Appeal Board; "Order Granting Petitioner's Motion to Submit Supplemental Information"; USPTO Patent Trial and Appeal Board; Filing Date Jun. 6, 2023; 4 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528; Exhibit/Paper No. 0019; BJ's Wholesale Club Holdings, Inc.; "Updated List of Exhibits"; USPTO Patent Trial and Appeal Board; Filing Date Jun. 8, 2023; 5 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528; Exhibit/Paper No. 0022; Walmart Apollo, LLC; "Patent Owner Response"; USPTO Patent Trial and Appeal Board; Filing Date Jul. 19, 2023; 84 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528; Exhibit/Paper No. 0027; BJ's Wholesale Club Holdings, Inc.; "Petitioner's Reply"; USPTO Patent Trial and Appeal Board; Filing Date Oct. 11, 2023; 34 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528; Exhibit/Paper No. 0029; Walmart

Apollo, LLC; "Patent Owner's Sur-Reply in IPR2022-01528"; USPTO Patent Trial and Appeal Board; Filing Date Nov. 28, 2023; pp. i-v and 1-35.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528; Exhibit/Paper No. 0031; BJ's Wholesale Club Holdings, Inc.; "Petitioner's Request for Oral Argument"; USPTO Patent Trial and Appeal Board; Filing Date Nov. 29, 2023; pp. 1-5.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528; Exhibit/Paper No. 0035; BJ's Wholesale Club Holdings, Inc.; "Updated List of Exhibits"; USPTO Patent Trial and Appeal Board; Filing Date Jan. 8, 2024; 5 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528; Exhibit/Paper No. 0037; USPTO Patent Trial and Appeal Board; "Final Written Decision Determining All Challenged Claims Unpatentable 35 U.S.C. § 318(a)"; USPTO Patent Trial and Appeal Board; Filing Date Apr. 11, 2024; pp. 1-79.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528; Exhibit/Paper No. 1002; BJ's Wholesale Club Holdings, Inc.; "Declaration of Michael Shamos, Ph.D."; USPTO Patent Trial and Appeal Board; Filing Date Sep. 22, 2022; pp. i-v and 1-118.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528; Exhibit/Paper No. 1033; BJ's Wholesale Club Holdings, Inc.; "Integrated Solution with DB2"; USPTO Patent Trial and Appeal Board; Filing Date Oct. 11, 2023; 4 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528; Exhibit/Paper No. 1034; BJ's Wholesale Club Holdings, Inc.; "Declaration of Michael V. Messinger"; USPTO Patent Trial and Appeal Board; Filing Date Oct. 11, 2023; 2 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01561; Exhibit/Paper No. 0002; BJ's Wholesale Club Holdings, Inc.; "Petition for Inter Partes Review of U.S. Pat. No. 10,803,435"; USPTO Patent Trial and Appeal Board; Filing Date Sep. 29, 2022; pp. iii-ix and 1-66.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01561; Exhibit/Paper No. 0006; BJ's Wholesale Club Holdings, Inc.; "Petitioner's Updated Exhibit List"; USPTO Patent Trial and Appeal Board; Filing Date Oct. 17, 2023; pp. i-ii and 1-1.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01561; Exhibit/Paper No. 0008; Walmart Apollo, LLC; "Patent Owner's Preliminary Response"; USPTO Patent Trial and Appeal Board; Filing Date Jan. 13, 2023; pp. i-iii and 1-44.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01561; Exhibit/Paper No. 0009; BJ's Wholesale Club Holdings, Inc.; "Petitioner's Reply to Patent Owner Preliminary Response"; USPTO Patent Trial and Appeal Board; Filing Date Feb. 15, 2023; pp. 1-6.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01561; Exhibit/Paper No. 0011; USPTO Patent Trial and Appeal Board; "Institution Decision: Deny"; USPTO Patent Trial and Appeal Board; Filing Date Apr. 12, 2023; pp. 2-30.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01561; Exhibit/Paper No. 0012; BJ's Wholesale Club Holdings, Inc.; "Petitioner's Request for Refund of Post-Institution Fees"; USPTO Patent Trial and Appeal Board; Filing Date Apr. 2, 2024; 3 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01561; Exhibit/Paper No. 1002; BJ's Wholesale Club Holdings, Inc.; "Declaration of Michael Shamos, Ph.D."; USPTO Patent Trial and Appeal Board; Filing Date Sep. 22, 2022; pp. i-vi and 1-148.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01561; Exhibit/Paper No. 1026; BJ's Wholesale Club Holdings, Inc.; "Plaintiff's Preliminary Disclosure of Asserted Claims and Infringement Contentions to BJ's Wholesale Club Holdings, Inc."; USPTO Patent Trial and Appeal Board; Filing Date Oct. 17, 2022; pp. 1-7.

(56)

References Cited**OTHER PUBLICATIONS**

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01561; Exhibit/Paper No. 3001; USPTO Patent Trial and Appeal Board; "Denying Institution of Inter Partes Review 35 U.S.C. § 314"; USPTO Patent Trial and Appeal Board; Filing Date Apr. 11, 2024; pp. 1-30.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01561; Exhibit/Paper No. 3002; USPTO Patent Trial and Appeal Board; "Petition for Inter Partes Review of U.S. Pat. No. 10,803,435"; USPTO Patent Trial and Appeal Board; Filing Date Apr. 11, 2024; pp. iii-ix and 1-66.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01563; Exhibit/Paper No. 0002; BJ's Wholesale Club Holdings, Inc.; "Petition for Inter Partes Review of U.S. Pat. No. 10,803,435"; USPTO Patent Trial and Appeal Board; Filing Date Sep. 22, 2022; pp. iii-viii and 1-50.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01563; Exhibit/Paper No. 0006; BJ's Wholesale Club Holdings, Inc.; "Petitioner's Updated Exhibit List"; USPTO Patent Trial and Appeal Board; Filing Date Oct. 17, 2022; pp. 1-3.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01563; Exhibit/Paper No. 0007; BJ's Wholesale Club Holdings, Inc.; "Petitioner's Notice of Ranking Petitions for Inter Partes Review of U.S. Pat. No. 10,803,435"; USPTO Patent Trial and Appeal Board; Filing Date Oct. 17, 2022; pp. 1-6.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01563; Exhibit/Paper No. 0008; Walmart Apollo, LLC; "Patent Owner's Preliminary Response"; USPTO Patent Trial and Appeal Board; Filing Date Jan. 13, 2023; pp. i-iii and 1-35.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01563; Exhibit/Paper No. 0009; BJ's Wholesale Club Holdings, Inc.; "Petitioner's Reply to Patent Owner Preliminary Response"; USPTO Patent Trial and Appeal Board; Filing Date Feb. 15, 2023; pp. 1-6.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01563; Exhibit/Paper No. 0011; USPTO Patent Trial and Appeal Board; "Institution Decision: Granting Institution of Inter Partes Review 35 U.S.C. § 314"; USPTO Patent Trial and Appeal Board; Filing Date Apr. 12, 2023; pp. 1-28.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01563; Exhibit/Paper No. 0013; Walmart Apollo, LLC; "Patent Owner's Objections Under 37 C.F.R. § 42.64(b)(1) to Evidence Served"; USPTO Patent Trial and Appeal Board; Filing Date Apr. 24, 2023; pp. 1-5.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01563; Exhibit/Paper No. 0014; BJ's Wholesale Club Holdings, Inc.; "Petitioner's Objections to Patent Owner's Evidence"; USPTO Patent Trial and Appeal Board; Filing Date Apr. 24, 2023; 3 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01563; Exhibit/Paper No. 0016; BJ's Wholesale Club Holdings, Inc.; "Petitioner's Motion to Submit Supplemental Information Pursuant to 37 C.F.R. sec. 42.123"; USPTO Patent Trial and Appeal Board; Filing Date May 26, 2023; 11 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01563; Exhibit/Paper No. 0018; USPTO Patent Trial and Appeal Board; "Order Granting Petitioner's Motion to Submit Supplemental Information"; USPTO Patent Trial and Appeal Board; Filing Date Jun. 13, 2023; 4 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01563; Exhibit/Paper No. 0020; BJ's Wholesale Club Holdings, Inc.; "Updated Exhibit List"; USPTO Patent Trial and Appeal Board; Filing Date Jun. 16, 2023; 5 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01563; Exhibit/Paper No. 0027; BJ's Wholesale Club Holdings, Inc.; "Petitioner's Reply"; USPTO Patent Trial and Appeal Board; Filing Date Oct. 11, 2023; pp. i-v and 1-24.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01563; Exhibit/Paper No. 0029; Walmart Apollo, LLC; "Patent Owner's Sur-Reply in IPR2022-01563"; USPTO Patent Trial and Appeal Board; Filing Date Nov. 28, 2023; pp. i-iv and 1-27.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01563; Exhibit/Paper No. 0031; BJ's Wholesale Club Holdings, Inc.; "Petitioner's Request for Oral Argument"; USPTO Patent Trial and Appeal Board; Filing Date Nov. 29, 2023; pp. 1-4.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01563; Exhibit/Paper No. 0035; BJ's Wholesale Club Holdings, Inc.; "Updated List of Exhibits"; USPTO Patent Trial and Appeal Board; Filing Date Jan. 8, 2024; 5 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01563; Exhibit/Paper No. 0037; USPTO Patent Trial and Appeal Board; "Final Written Decision Determining All Challenged Claims Unpatentable 35 U.S.C. § 318(a)"; USPTO Patent Trial and Appeal Board; Filing Date Apr. 11, 2024; pp. 1-45.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01563; Exhibit/Paper No. 1002; BJ's Wholesale Club Holdings, Inc.; "Exhibit, Declaration of Michael I. Shamos, Ph.D."; USPTO Patent Trial and Appeal Board; Filing Date Sep. 22, 2022; pp. i-vi and 1-148.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01563; Exhibit/Paper No. 1030; BJ's Wholesale Club Holdings, Inc.; "Letter dated Jan. 31, 2023 issued by Korean Intellectual Property Office (KIPO) with a certified English translation (identifying the publication of Korean Patent Pub. No. KR 10-2007-0087811 (Ex. 1009) by KIPO)"; USPTO Patent Trial and Appeal Board; Filing Date Jun. 16, 2023; pp. 1-52.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01564; Exhibit/Paper No. 0002; BJ's Wholesale Club Holdings, Inc.; "Petition for Inter Partes Review of U.S. Pat. No. 10,803,435"; USPTO Patent Trial and Appeal Board; Filing Date Sep. 22, 2022; pp. i-vii and 1-75.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01564; Exhibit/Paper No. 0006; BJ's Wholesale Club Holdings, Inc.; "Petitioner's Updated Exhibit List"; USPTO Patent Trial and Appeal Board; Filing Date Oct. 17, 2022; pp. i-iii and 1.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01564; Exhibit/Paper No. 0008; Walmart Apollo, LLC; "Patent Owner's Preliminary Response"; USPTO Patent Trial and Appeal Board; Filing Date Jan. 13, 2023; pp. i-vi and 1-34.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01564; Exhibit/Paper No. 0009; USPTO Patent Trial and Appeal Board; "Institution Decision: Granting Institution of Inter Partes Review 35 U.S.C. § 314"; USPTO Patent Trial and Appeal Board; Filing Date Apr. 11, 2023; pp. 1-48.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01564; Exhibit/Paper No. 0012; BJ's Wholesale Club Holdings, Inc.; "Petitioner's Objections to Patent Owner's Evidence"; USPTO Patent Trial and Appeal Board; Filing Date Apr. 24, 2023; pp. 3 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01564; Exhibit/Paper No. 0016; Walmart Apollo, LLC; "Patent Owner Response to Petition for Inter Partes Review"; USPTO Patent Trial and Appeal Board; Filing Date Jul. 19, 2023; pp. i-v and 1-49.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01564; Exhibit/Paper No. 0021; BJ's Wholesale Club Holdings, Inc.; "Petitioner's Reply"; USPTO Patent Trial and Appeal Board; Filing Date Oct. 11, 2023; 31 pages *BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC*; AIA Review No.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01564; Exhibit/Paper No. 0023; Walmart Apollo, LLC; "Patent Owner's Sur-Reply in IPR2022-01564"; USPTO Patent Trial and Appeal Board; Filing Date Nov. 28, 2023; 34 pages.

(56)

References Cited

OTHER PUBLICATIONS

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01564; Exhibit/Paper No. 0025; BJ's Wholesale Club Holdings, Inc.; "Petitioner's Request for Oral Argument"; USPTO Patent Trial and Appeal Board; Filing Date Nov. 29, 2023; 4 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01564; Exhibit/Paper No. 0029; BJ's Wholesale Club Holdings, Inc.; "Updated List of Exhibits"; USPTO Patent Trial and Appeal Board; Filing Date Jan. 8, 2024; 5 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01564; Exhibit/Paper No. 0031; USPTO Patent Trial and Appeal Board; "Judgement Final Written Decision"; USPTO Patent Trial and Appeal Board; Filing Date Apr. 10, 2024; 58 pages.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01564; Exhibit/Paper No. 1021; BJ's Wholesale Club Holdings, Inc.; "Yong English Annotated Drawings Figures 1-11"; USPTO Patent Trial and Appeal Board; Filing Date Sep. 22, 2022; pp. 1-7.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01564; Exhibit/Paper No. 1026; Walmart Apollo, LLC; "Plaintiff's Preliminary Disclosure of Asserted Claims and Infringement Contentions to BJ's Wholesale Club Holdings, Inc."; USPTO Patent Trial and Appeal Board; Filing Date Oct. 17, 2022; pp. 1-7.

BJ's Wholesale Club Holdings, Inc. v. Walmart Apollo, LLC; AIA Review No. IPR2022-01528, IPR2022-01563, IPR2022-01564; Exhibit/Paper No. 2011; Walmart Apollo, LLC; "Deposition of: Michael Shamos Ph.D."; USPTO Patent Trial and Appeal Board; Filing Date Jul. 14, 2023; pp. 1-108.

Chinese Office Action from related Chinese patent application No. 201180045758.1 issued Dec. 29, 2017; 18 pages.

Colapinto, John; "Stop, Thief!"; *Annals of Retail*; Aug. 25, 2008; The New Yorker; New York City; pp. 1-24.

Digipay.guru; "Scan & Thru Solution—Fast, Secure & Convenient | DigiPay.Guru"; <<https://www.digipay.guru/scan-and-go-self-checkout-payment-solution/>>; available at least as early as Dec. 12, 2022; pp. 1-26.

Emigh, Jacqueline et al; "Windows May be coming to Your Shopping Cart This Year" *Betanews*; Published Jan. 14, 2008; Accessed Aug. 12, 2010; <http://www.betanews.com/windows-may-be-coming-to-your-shopping-cart-this-year/1200339171>.

Examination Report from related Great Britain patent application No. GB1221399.7 issued Jan. 10, 2018; 8 pages.

Examination Search Report from related Canadian patent application No. 2,809,523 issued Feb. 19, 2018; 5 pages.

Gao, Jerry, et al.; "A 2D Barcode-based Mobile Payment System"; 2009 Third International Conference on Multimedia and Ubiquitous Engineering; 2009; San Jose State University; San Jose; pp. 1-13.

Gast, Matthew S.; "802.11 Wireless Networks: The Definitive Guide"; 2002; O'Reilly Media; pp. 1-335.

HHAPPE, Inc. Shop 'n Stock; "Grocery Shopping—A Whole New Way!" 2009-2010, Accessed Aug. 12, 2010; <http://www.shop-n-stock.com>; 3 pages.

Houghton Mifflin Company, "The American Heritage College Dictionary"; 2007; Fourth Edition; Houghton Mifflin Company; pp. 1-4.

IBM; "Public key cryptography"; <<https://www.ibm.com/docs/en/ztzf1.1.0.15?topic=concepts-public-key-cryptography>>; available at least as early as Jul. 10, 2023; pp. 1.

IBM; IP.com No. IPCOM000139258D "System and Method for Retail Store Self Checkout via Cellular Phone Technology" Published Aug. 17, 2006; www.ip.com, 5 pages.

Illum Magazine; "The Magic of Scanning: Science Behind the QR Code"; USC Viterbi; Feb. 14, 2022; University of Southern California; <<https://illum.usc.edu/the-magic-of-scanning-science-behind-the-qr-code/>>; pp. 1-4.

International Search Report and Written Opinion from related international patent application No. PCT/US2011/055984 (filed Oct. 21, 2011); 14 pages.

Library of Congress; "MP3 (MPEG Layer III Audio Encoding)"; <<https://www.loc.gov/preservation/digital/formats/ffd/fdd000012.shtml>>; available at least as early as Jul. 10, 2023; 1 page.

Libretexts; "1.5: Coding Theory"; <[https://math.libretexts.org/Bookshelves/Combinatorics_and_Discrete_Mathematics/Combinatorics_\(Morris\)/01%3A_Introduction/01%3A_What_is_Combinatorics/1.05%3A_Coding_Theory](https://math.libretexts.org/Bookshelves/Combinatorics_and_Discrete_Mathematics/Combinatorics_(Morris)/01%3A_Introduction/01%3A_What_is_Combinatorics/1.05%3A_Coding_Theory)>; available at least as early as Jul. 10, 2023; pp. 1.

Lichtenstein, Nelson, "The Retail Revolution: How Wal-Mart Created a Brave New World of Business"; 2009; Henry Holt and Company; United States of America; 19 pages.

Merriam-Webster, Inc.; "Webster's New College Dictionary"; 2007; Wiley Publishing, Inc.; Cleveland; 3 pages.

Merriam-Webster, Incorporated, "Merriam-Webster's Collegiate Dictionary"; 2000; Tenth Edition; Merriam-Webster, Incorporated; 4 pages.

Merriam-Webster, Incorporated, "Merriam-Webster's Collegiate Dictionary"; 2006; Eleventh Edition; Merriam-Webster, Incorporated; pp. 1-4.

Microsoft, "Microsoft Computer Dictionary"; 2002; Fifth Edition; Microsoft Press; Redmond; pp. 1-3.

National Ocean Service; "The Global Positioning System"; <https://oceanservice.noaa.gov/education/tutorial_geodesy/geo09_gps.html>; Mar. 1, 2021; pp. 1-2.

Office Action from related Brazilian Patent Application No. BR112013008325-5 issued Sep. 18, 2019; 6 pages.

Office Action from related Chinese patent application No. 201180045758.1 dated Jun. 13, 2016; 8 pages.

Office Action from related Great Britain patent application No. 1221399.7 issued Mar. 20, 2018; 9 pages.

Office Action from related Japanese patent application No. 2013-533976 dated Mar. 29, 2016; 5 pages.

Office Action issued Aug. 4, 2015 in related Japanese patent application No. 2013-533976; 25 pages.

Oxford English Dictionary; "algorithm"; Oxford University Press; Mar. 2023; <<https://www.oed.com/view/Entry/4959?redirectedFrom=algorithm#eid#2>>; pp. 1.

Oxford English Dictionary; "code"; Oxford University Press; Sep. 2022; <<https://www.oed.com/view/Entry/35578?rskey=4dwr9F&result=1#eid>>; pp. 1-6.

Sales Shark; "Walmart Scan & Go | How it Works & Important Things to Know"; <<https://www.youtube.com/watch?v=W3fZ6hRFAe0>>; Feb. 11, 2022; 13 pages.

Sam's Club; "Scan & Go"; <<https://www.samsclub.com/content/scan-and-go>>; available at least as early as May 2, 2023; 9 pages.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 001; Sam's West, Inc.; "Complaint and Demand for Jury Trial and Injunctive Relief with Exhibits"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; Mar. 22, 2022; 247 pages.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 035; BJ's Wholesale Club Holdings, Inc.; "Defendant BJ's Wholesale Club Holdings, Inc.'s Motion to Dismiss, with Appendix"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; Apr. 29, 2022; 30 pages.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 036; Sam's West, Inc.; "Plaintiffs' Response to Defendant's Motion to Dismiss Under Rule 12(b)(6)"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; May 20, 2022; pp. 1-20.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 037-01; Sam's West, Inc.; "Proposed First Amended Complaint and Demand for Jury Trial and Injunctive Relief"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; May 23, 2022; pp. 1-71.

(56)

References Cited**OTHER PUBLICATIONS**

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 037-06; Sam's West, Inc.; "Exhibit 5"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; May 23, 2022; pp. 1-33.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 037-07; Sam's West, Inc.; "Exhibit 6"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; May 23, 2022; pp. 1-22.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 037-08; Sam's West, Inc.; "Exhibit 7"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; May 23, 2022; pp. 1-13.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 037-09; Sam's West, Inc.; "Exhibit 8"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; May 23, 2022; pp. 1-14.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 049; Baker, Magistrate Judge David A.; "Order Denying as Moot 35 Motion to Dismiss for Failure to State a Claim; Granting 37 Motion to Amend Complaint; Denying as Moot 39 Motion for Leave to File a Reply Memorandum in Support of Motion to Dismiss"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; Jun. 8, 2022; pp. 1-3.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 054; BJ's Wholesale Club Holdings, Inc.; "Defendant BJ's Wholesale Club Holdings, Inc.'s Motion to Dismiss"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; Jun. 27, 2022; pp. 1-27.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 062; Sam's West, Inc.; "Plaintiffs' Response to Defendant's Motion to Dismiss Under Rule 12(B)(6)"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; Jul. 18, 2022; pp. 1-21.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 067; BJ's Wholesale Club Holdings, Inc.; "Defendant BJ's Wholesale Club Holdings, Inc.'s Reply in Support of its Motion to Dismiss"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; Aug. 5, 2022; pp. 1-7.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 078; Sam's West, Inc. "Plaintiffs' Statement Regarding Defendant's IPR Petitions Pursuant to Court's Notice of Hearing"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; Sep. 7, 2022; pp. 1-10.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 079; BJ's Wholesale Club Holdings, Inc. "Defendant BJ's Wholesale Club Holdings, Inc.'s Statement Pursuant to Sep. 1, 2022 Order, with Exhibit"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; Sep. 7, 2022; 21 pages.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 083; Sam's West, Inc.; "Corrected First Amended Complaint and Demand for Jury Trial and Injunctive Relief"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; Sep. 19, 2022; pp. 1-72.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 085; Sam's West, Inc.; "Plaintiffs' Short Form Motion to Compel Responses to Interrogatory Nos. 2, 5, 6, and 11, with Exhibits"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; Sep. 22, 2022; 48 pages.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 088; Sam's West, Inc. and BJ's Wholesale Club Holdings, Inc.; "Joint Claim Construction Statement, with Exhibit"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; Sep. 23, 2022; 6 pages.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 091; BJ's Wholesale Club Holdings, Inc.; "Defendant BJ's Wholesale Club Holdings, Inc.'s Motion to Stay Proceedings Pending Inter Partes Review"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; Sep. 26, 2022; pp. 1-19.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 094; BJ's Wholesale Club Holdings, Inc.; "Defendant BJ's Wholesale Club Holdings, Inc.'s Opposition to Plaintiffs' Short Form Motion to Compel, with Exhibits"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; Sep. 29, 2022; 85 pages.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 096; Sam's West, Inc.; "Plaintiffs' Response to Defendant BJ's Wholesale Club Holdings, Inc.'s Motion to Stay Proceedings Pending Inter Partes Review, with Exhibits"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; Oct. 11, 2022; 60 pages.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 101; Chao-Lee Lin, Joel; "Declaration of Joel Chao-Lee Lin in Support of Plaintiffs' Claim Construction Brief"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; Oct. 21, 2022; pp. 1-4.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 102; Sam's West, Inc.; "Plaintiffs' Claim Construction Brief"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; Oct. 21, 2022; pp. 1-27.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 108; Baker, Magistrate Judge David A.; "Order Granting in Part and Denying in Part 91 Motion to Stay"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; Nov. 21, 2022; pp. 1-17.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 112; BJ's Wholesale Club Holdings, Inc.; "Defendant BJ's Wholesale Club Holdings, Inc.'s Responsive Claim Construction Brief"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; Nov. 21, 2022; pp. 1-24.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 114; Baker, Magistrate Judge David A.; "Order Denying Without Prejudice 54 Motion to Dismiss; adopting 109 Report and Recommendations"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; Feb. 2, 2023; pp. 1-2.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 117; United States District Court for the Middle District of Florida; "Transcript of hearing RE: Defendant BJ's Wholesale Club Holdings, Inc.'s Motion to Stay Proceedings Pending Inter Partes Review"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; Mar. 18, 2023; pp. 1-54.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 124; Sam's West, Inc.; "Plaintiff's Motion to Life the Stay"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; Jun. 14, 2023; 202 pages.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 126; BJ's Wholesale Club Holdings, Inc.; "Defendant BJ's Wholesale Club Holdings, Inc.'s Response to Plaintiff's Motion to Life the Stay"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; Jun. 28, 2023; 221 pages.

(56)

References Cited**OTHER PUBLICATIONS**

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 131; Norway, Magistrate Judge Robert M.; "Order Denying Plaintiffs' Motion to Lift the Stay"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; Jul. 31, 2023; 11 pages. Schematic "Stop & Shop: Scan It!—Elevating Interactivity" Published Jan. 27, 2010; Accessed Aug. 12, 2010 <http://touch.schematic.com/2010/01/stop-shop-scan-it/>; 10 pages.

SN Staff; Supermarket News Kroger Store Deploys Tunnel Scanners' at Checkout; Published Jul. 21, 2010; Accessed Nov. 17, 2010 http://supermarketnews.com/news/tunnel_scanner_0721/; 2 pages.

Tanenbaum, Andrew S., et al.; "Modern Operating Systems"; 2008; Fourth Edition; Pearson Education, Inc.; 1137 pages.

Truta, Filip; "AisleBuyer Announces Self-Checkout iPhone Application"; <<https://news.softpedia.com/news/AisleBuyer-Announces-Self-Checkout-iPhone-Application-152569.shtml>>; Aug. 17, 2010; pp. 1-2.

U.S. Appl. No. 12/947,545; Notice of Allowance mailed Jan. 18, 2018; 11 pages.

U.S. Appl. No. 12/947,545; Notice of Allowance mailed Feb. 8, 2018; 4 pages.

U.S. Appl. No. 12/947,545; Notice of Allowance mailed Jun. 25, 2018; 10 pages.

U.S. Appl. No. 12/947,545; Office Action mailed Jan. 13, 2017; 23 pages.

U.S. Appl. No. 12/947,545; Office Action mailed Jan. 15, 2016; 19 pages.

U.S. Appl. No. 12/947,545; Office Action mailed Mar. 9, 2012; 9 pages.

U.S. Appl. No. 12/947,545; Office Action mailed Mar. 26, 2015; 14 pages.

U.S. Appl. No. 12/947,545; Office Action mailed Jun. 30, 2014; 14 pages.

U.S. Appl. No. 12/947,545; Office Action mailed Jul. 10, 2015; 16 pages.

U.S. Appl. No. 12/947,545; Office Action mailed Aug. 25, 2016; 17 pages.

U.S. Appl. No. 12/947,545; Office Action mailed Sep. 25, 2012; 11 pages.

U.S. Appl. No. 12/947,545; Office Action mailed Nov. 21, 2013; 11 pages.

U.S. Appl. No. 16/730,567; Notice of Allowance mailed Aug. 6, 2020; 8 pages.

U.S. Appl. No. 16/730,567; Office Action mailed Apr. 17, 2020; 10 pages.

U.S. Appl. No. 17/004,819; Non-Final Rejection mailed Nov. 22, 2022; (pp. 1-9).

U.S. Appl. No. 17/004,819; Notice of Allowance and Fees Due (PTOL-85) mailed Aug. 4, 2023; (pp. 1-10).

U.S. Appl. No. 17/004,819; Notice of Allowance and Fees Due (PTOL-85) mailed Sep. 25, 2023; (pp. 1-10).

Walsh, Mark; "Target Teams With Shop kick" MediaPost Blogs; Published Nov. 16, 2010; Accessed Nov. 17, 2010; http://mediapost.com/publications?fa=Articles.showArticle&art_aid=13961 0. (pp. 1-2).

Wikipedia; "Pull Technology"; <https://en.wikipedia.org/wiki/Pull_technology>; available at least as early as Jul. 10, 2023; pp. 1.

Wikipedia; "Push Technology"; <https://en.wikipedia.org/wiki/Push_technology>; available at least as early as Jul. 10, 2023; pp. 1-2.

Dragt, B. "Universal Commerce: A seamless, Personalized Purchase Experience for Today's Connected Consumers," 2012.

Examination Report in GB Patent Application No. GB1609525.9, mailed on Aug. 10, 2020; 7 pages.

Final Office Action in U.S. Appl. No. 14/098,318, mailed on Jun. 29, 2015; 20 pages.

Final Office Action in U.S. Appl. No. 14/098,318, mailed on Nov. 2, 2016; 38 pages.

J.-S. Yih, F. Pinel, Y.-H. Liu and T. Chieu, "Pervasive computing technologies for retail in-store shopping," ICPS '05. Proceedings. International Conference on Pervasive Services, 2005., 2005, pp. 111-116, doi: 0.1109/PERSER.2005.1506399 (Year: 2005).

J.-S. Yih, F. Pinel, Y.-H. Liu and T. Chieu, "Pervasive computing technologies for retail in-store shopping," ICPS '05. Proceedings. International Conference on Pervasive Services, 2005., Santorini, Greece, 2005, pp. 111-116, doi: 10.1109/PERSER.2005.1506399 (Year: 2005).

Non-Final Office Action in U.S. Appl. No. 14/098,318, mailed on Apr. 22, 2016; 29 pages.

Non-Final Office Action in U.S. Appl. No. 14/098,318, mailed on Dec. 4, 2015; 33 pages.

Non-Final Office Action in U.S. Appl. No. 14/098,318, mailed on Jan. 16, 2015; 15 pages.

Non-Final Office Action in U.S. Appl. No. 14/098,318, mailed on Sep. 11, 2017; 53 pages.

Notice of Allowance in U.S. Appl. No. 14/098,318, mailed on May 1, 2018; 23 pages.

Office Action in Canadian Patent Application No. 2,932,195 mailed on Dec. 14, 2018; 3 pages.

Office Action in Canadian Patent Application No. 2,932,195, dated Feb. 1, 2018; 3 pages.

Office Action in counterpart Canadian Patent Application No. 2,932,195 dated Mar. 10, 2017; 6 pages.

P. Urien and S. Piramuthu, "Framework and authentication protocols for smartphone, NFC, and RFID in retail transactions," 2013 IEEE Eighth International Conference on Intelligent Sensors, Sensor Networks and Information Processing, Melbourne, VIC, Australia, 2013, pp. 77-82, doi: 10.1109/ISSNIP.2013. (Year: 2013).

PCT; App. No. PCT/US2014/068638; International Search Report and Written Opinion mailed Mar. 6, 2015; p. 5.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 139; BJ's Wholesale Club Holdings, Inc.; "Defendant BJ's Wholesale Club Holdings, Inc.'s Motion to Lift the Stay"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; Jun. 12, 2024; 211 pages.

Sam's West, Inc. and Walmart Apollo, LLC v. BJ's Wholesale Club Holdings, Inc.; Document No. 141; Sam's West, Inc.; "Plaintiff Sam's West, Inc. And Walmart Apollo, LLC's Opposition to Motion to Lift the Stay"; United States District Court for the Middle District of Florida, Orlando Division; Case No. 6:22-cv-00587; Jun. 26, 2024; 12 pages.

USPTO; U.S. Appl. No. 16/050,026; Final Office Action mailed Jul. 23, 2020; (pp. 1-29).

USPTO; U.S. Appl. No. 16/050,026; Non-Final Office Action mailed Jan. 10, 2020; (pp. 1-29).

USPTO; U.S. Appl. No. 16/050,026; Non-Final Office Action mailed Mar. 31, 2021; (pp. 1-12).

USPTO; U.S. Appl. No. 16/050,026; Notice of Allowance mailed Oct. 25, 2021; (pp. 1-9).

USPTO; U.S. Appl. No. 17/585,052; Non-Final Rejection mailed May 24, 2023; (8 pages).

USPTO; U.S. Appl. No. 17/585,052; Notice of Allowance and Fees Due (PTOL-85) mailed Oct. 10, 2023; (11 pages).

USPTO; U.S. Appl. No. 18/412,273; Non-Final Rejection mailed Oct. 1, 2024; (pp. 1-12).

USPTO; U.S. Appl. No. 18/412,273; Notice of Allowance and Fees Due (PTOL-85) mailed Mar. 4, 2025; (pp. 1-12).

* cited by examiner

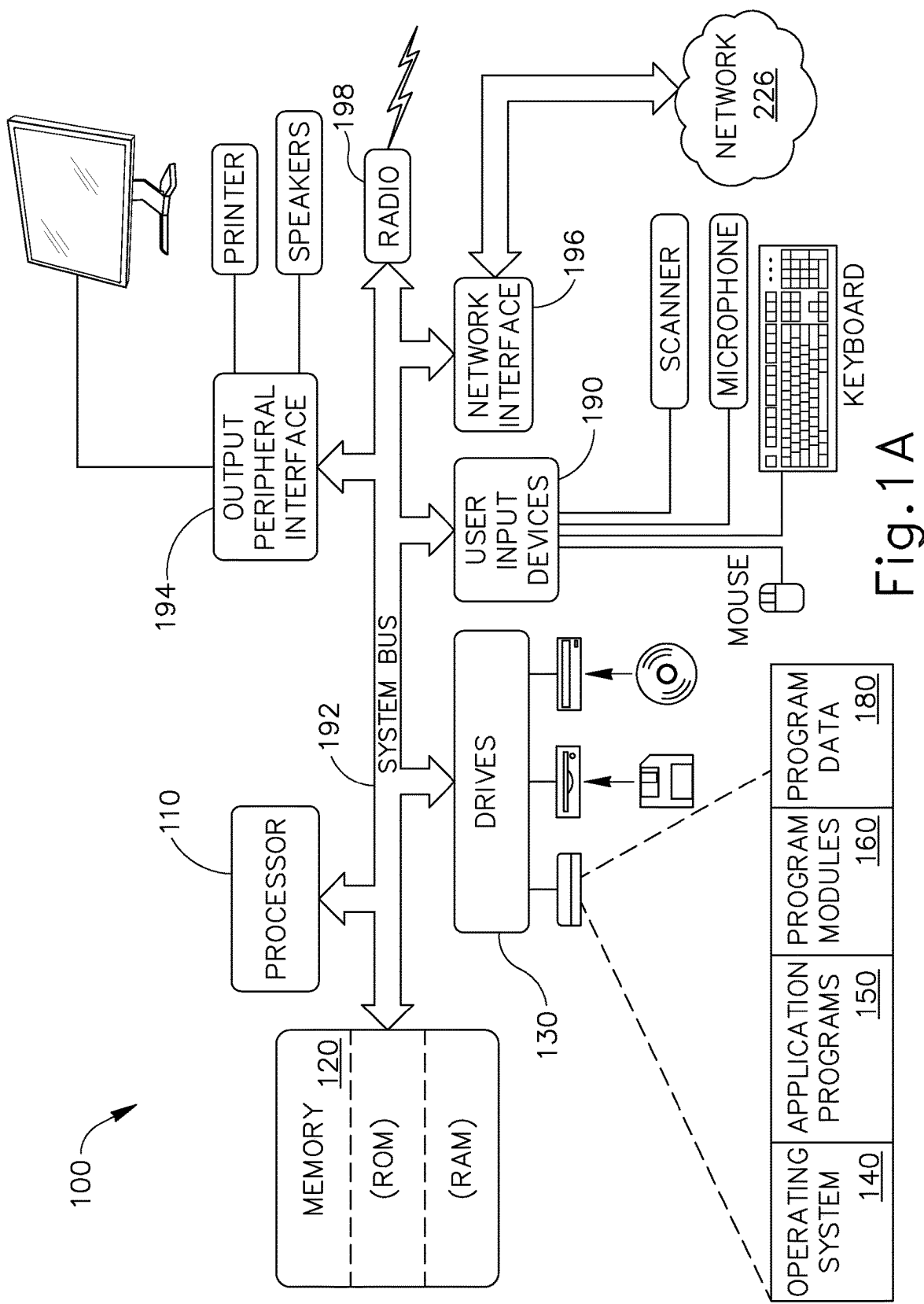


Fig. 1A

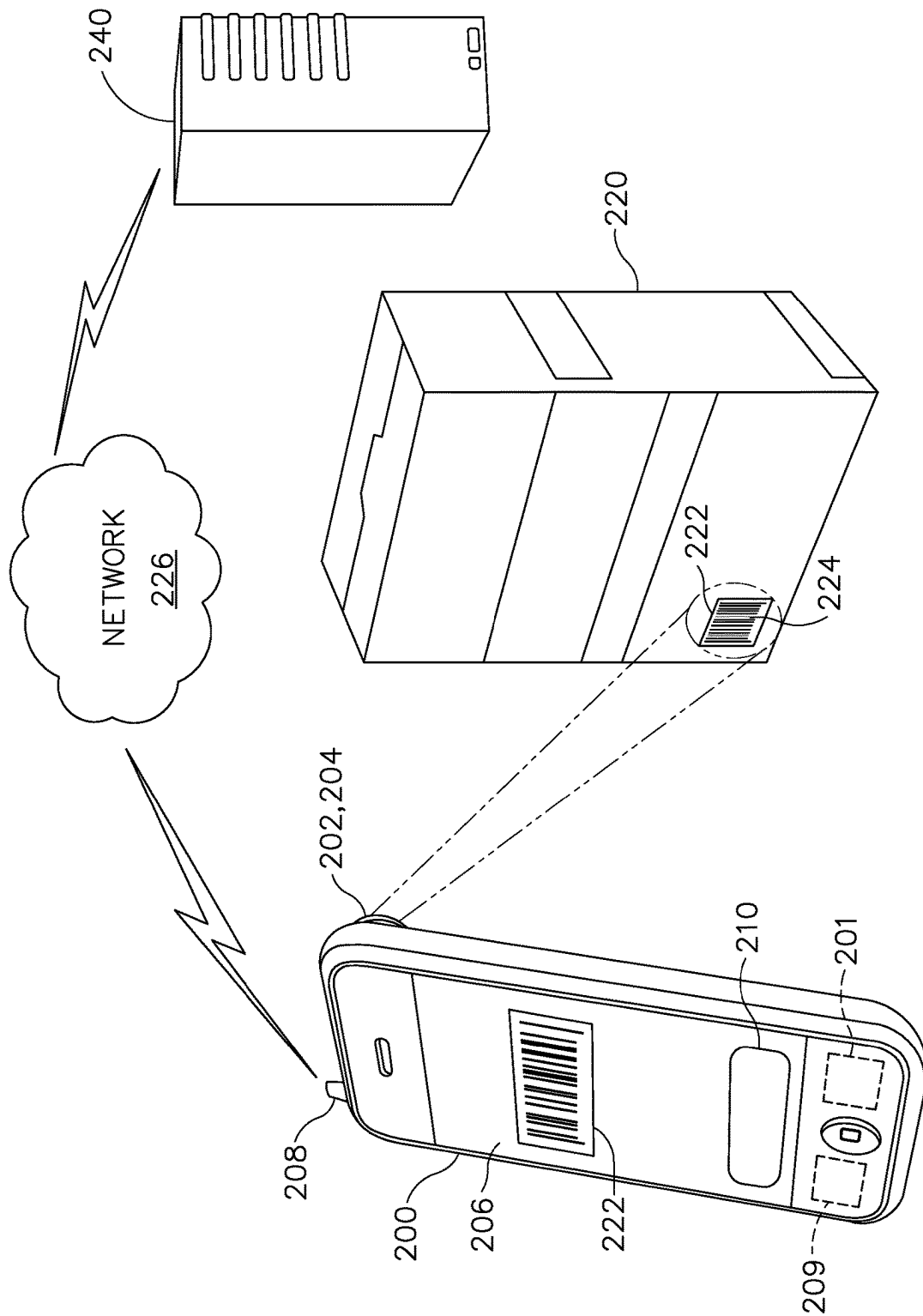


Fig. 1 B

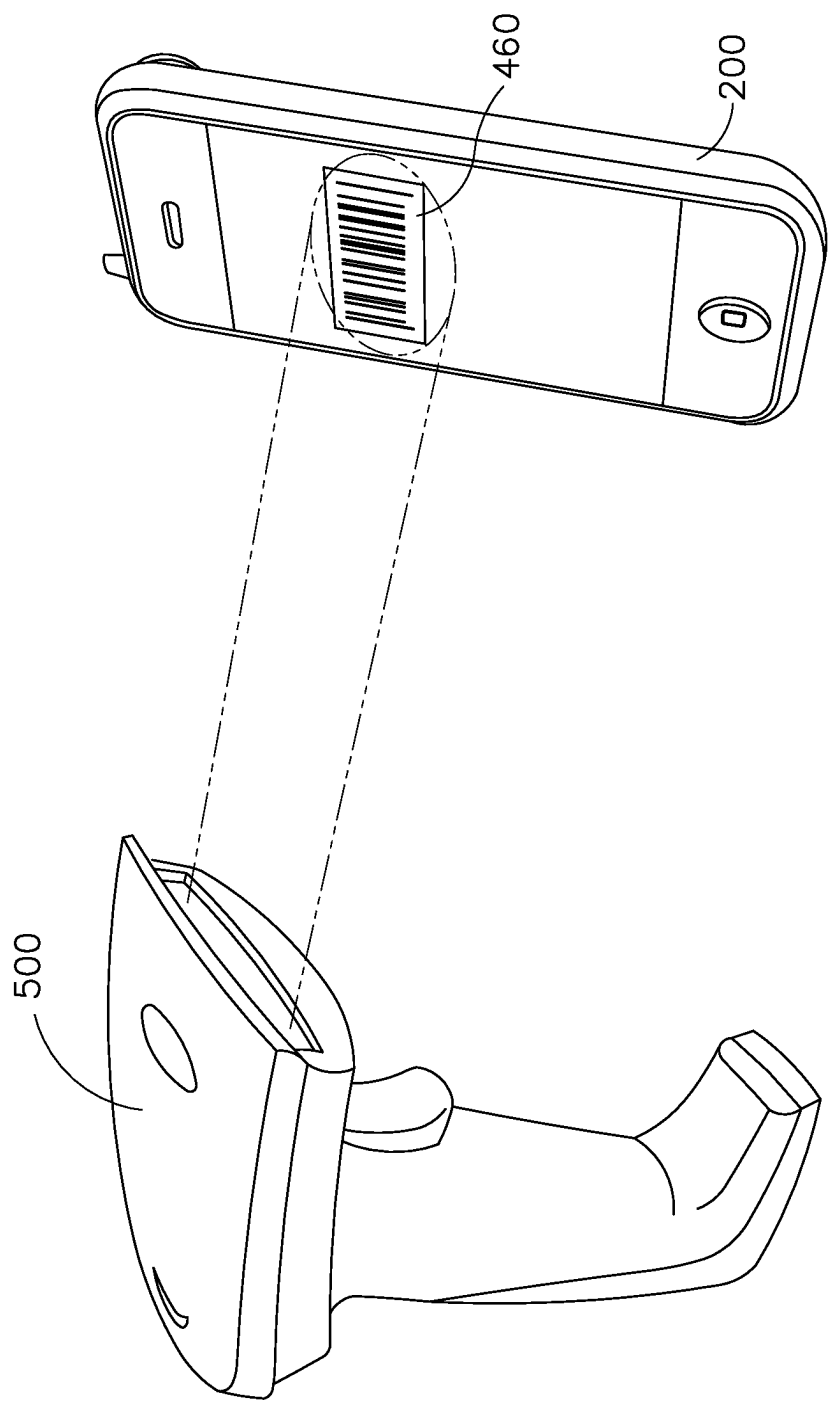


Fig. 1C

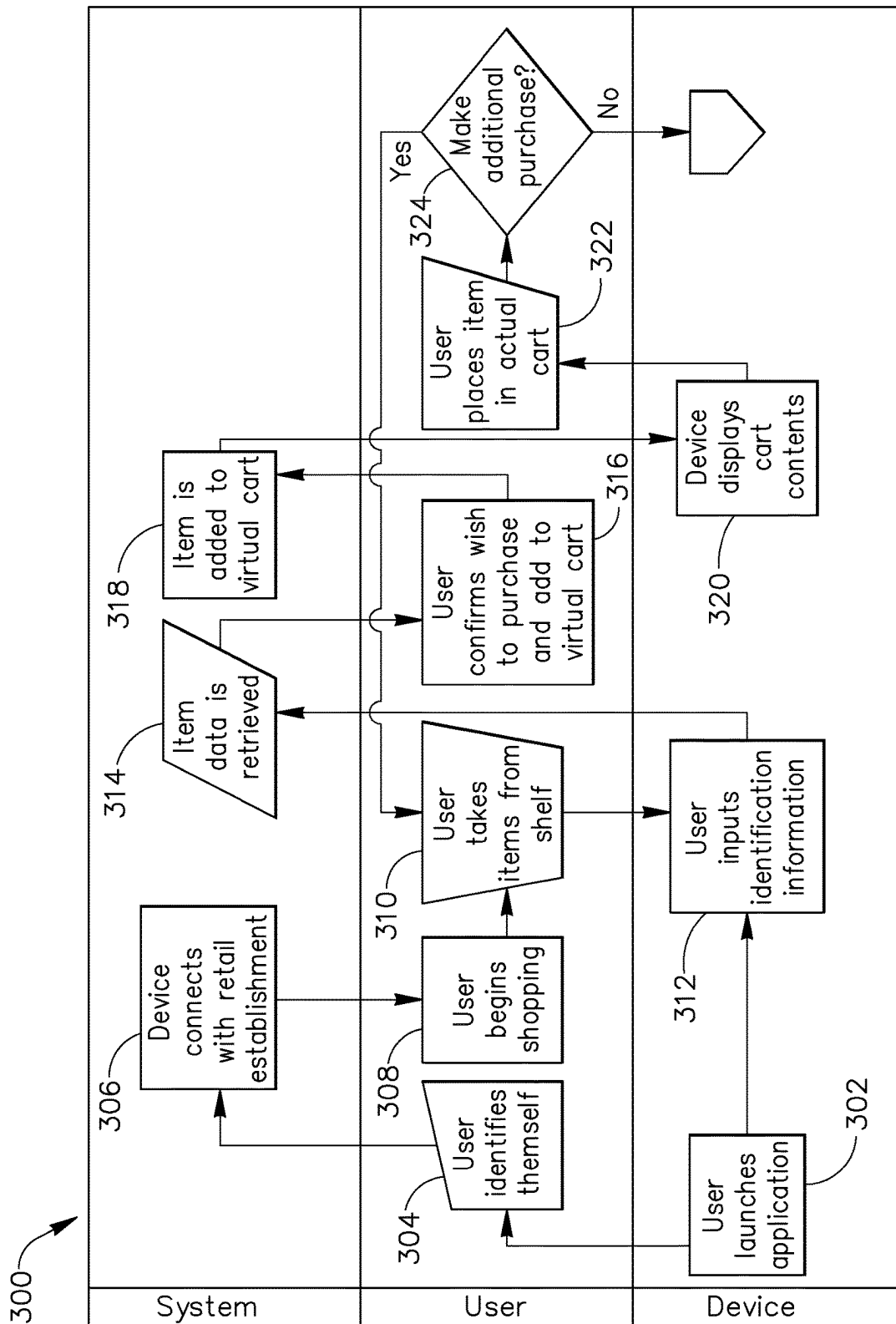


Fig.2A

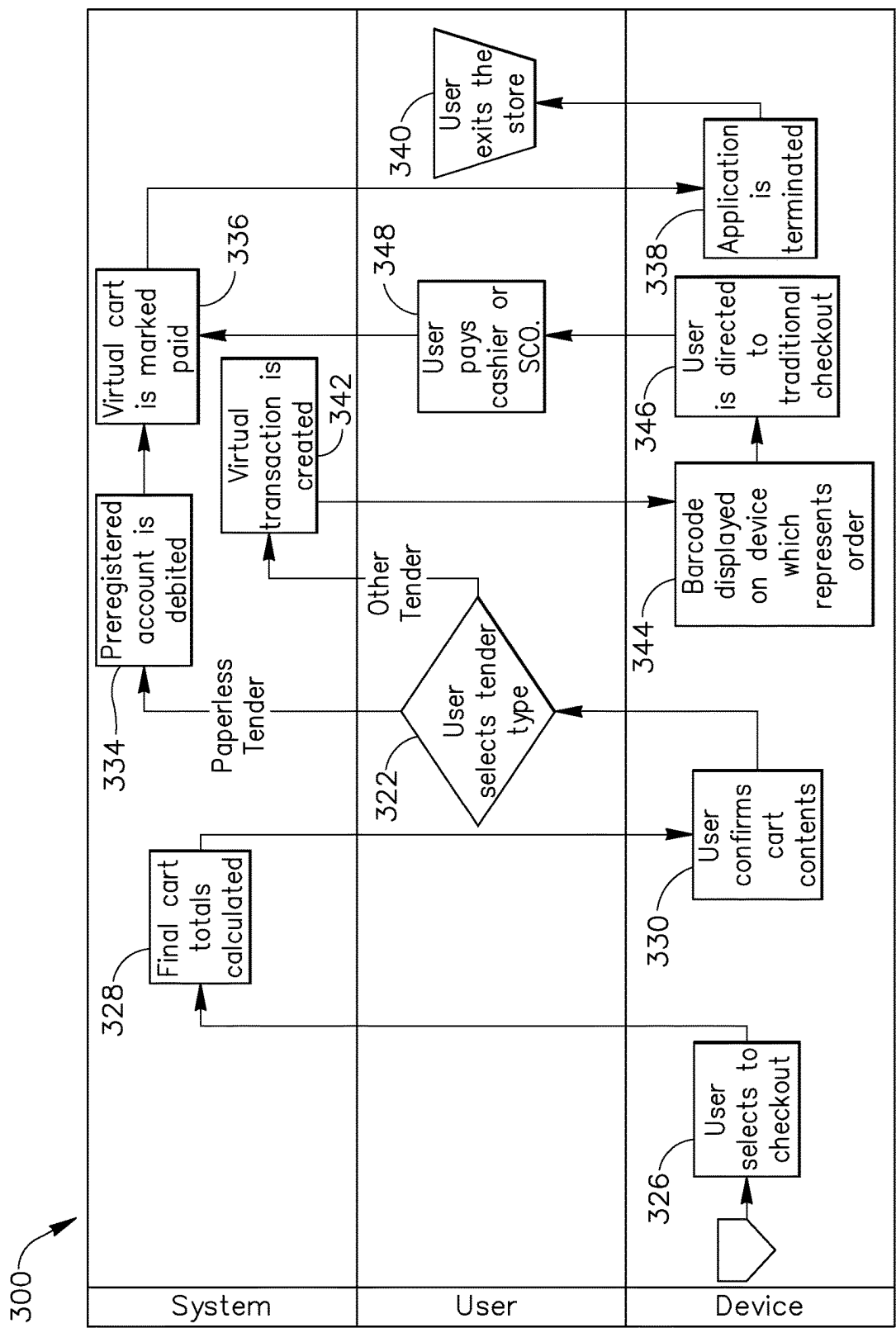


Fig. 2B

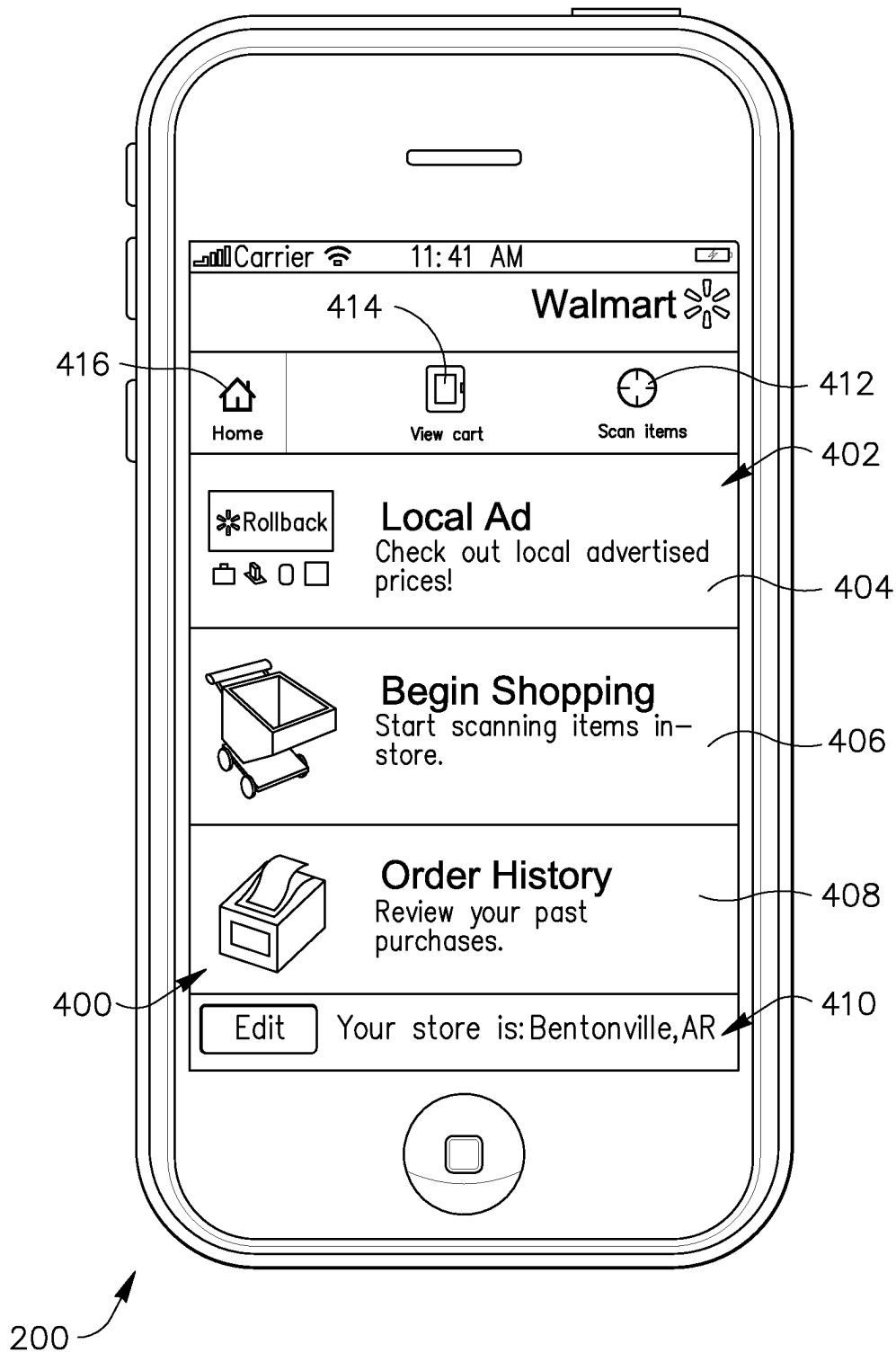


Fig.3

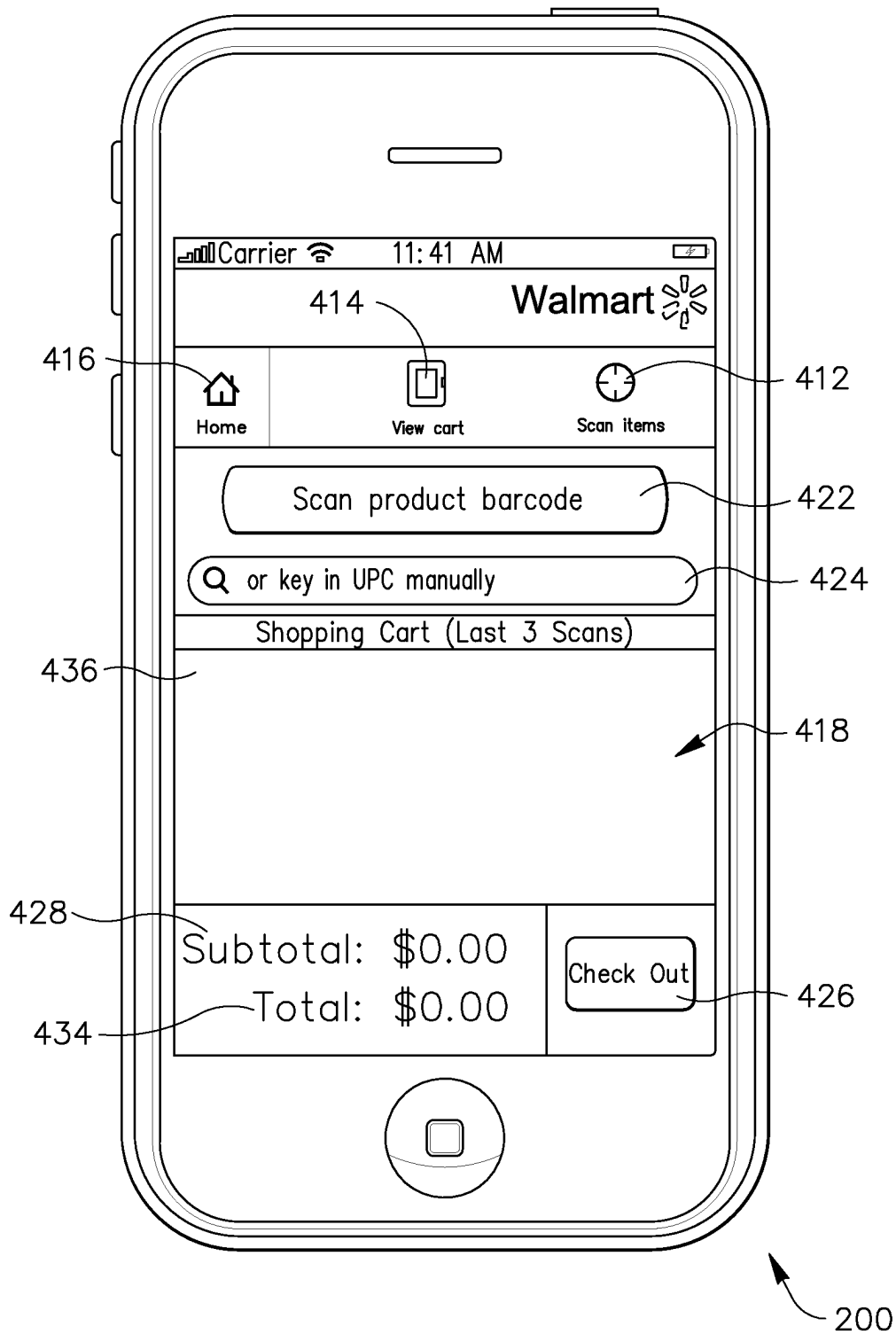


Fig.4

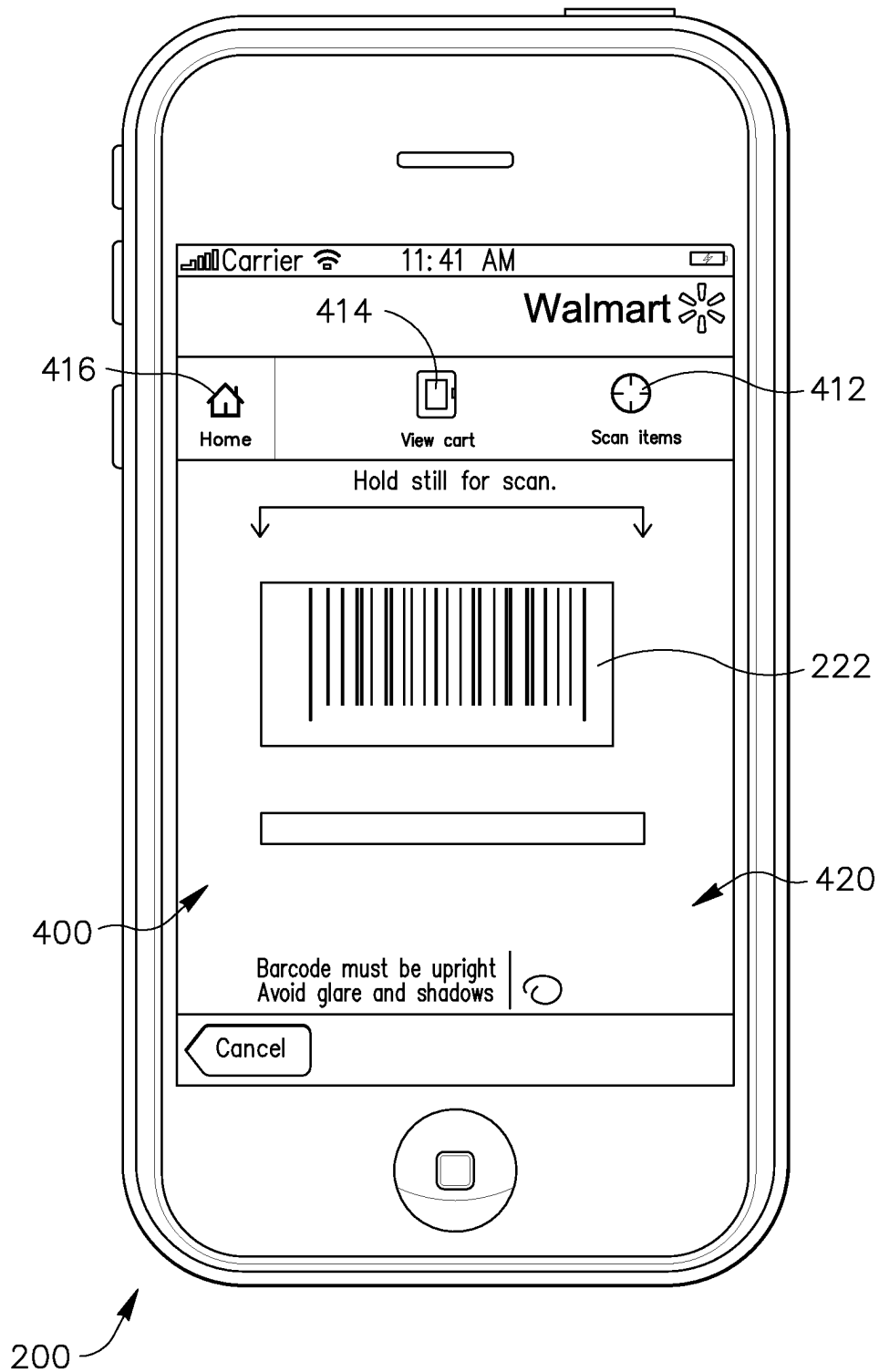


Fig.5

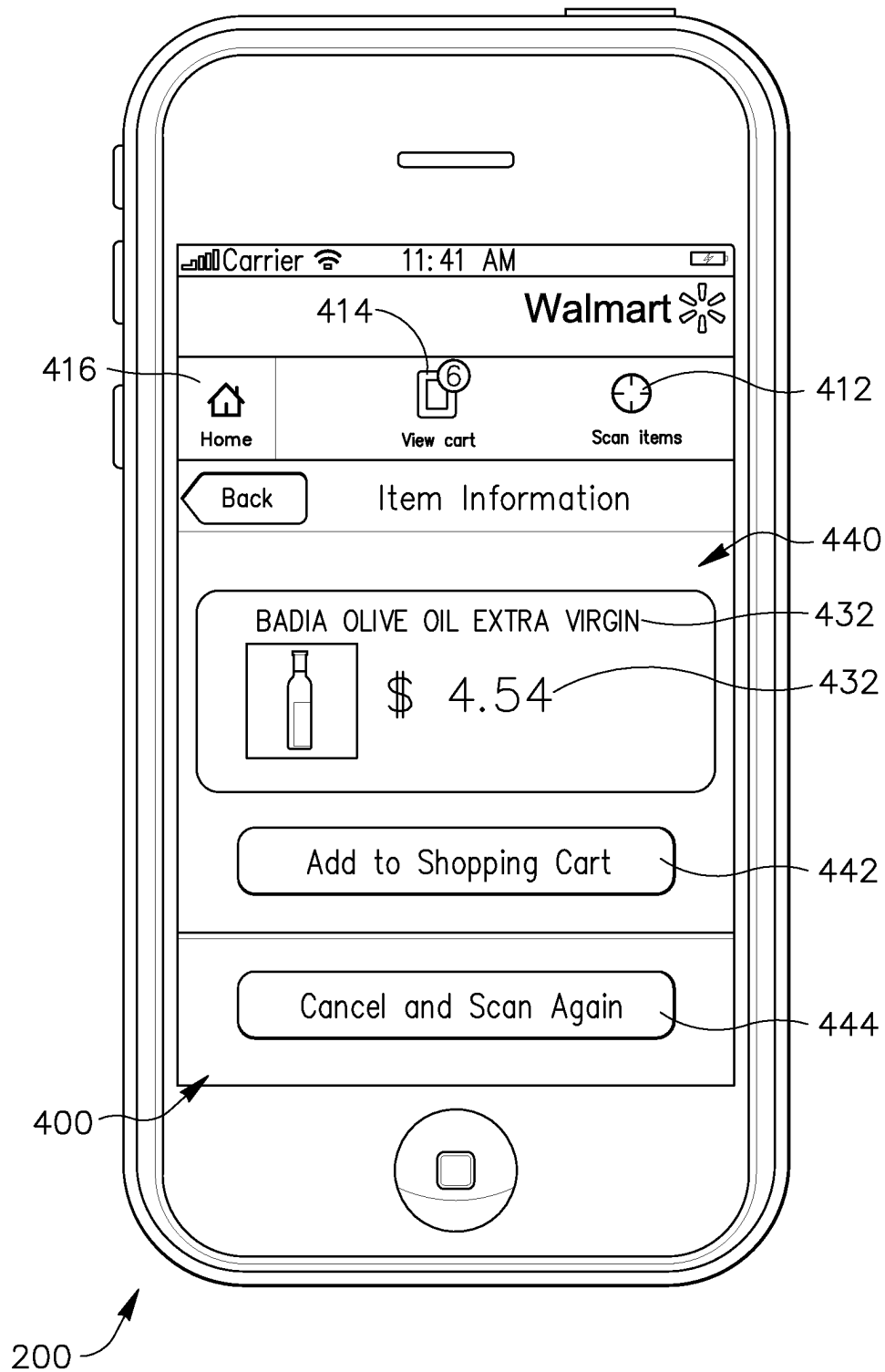


Fig.6

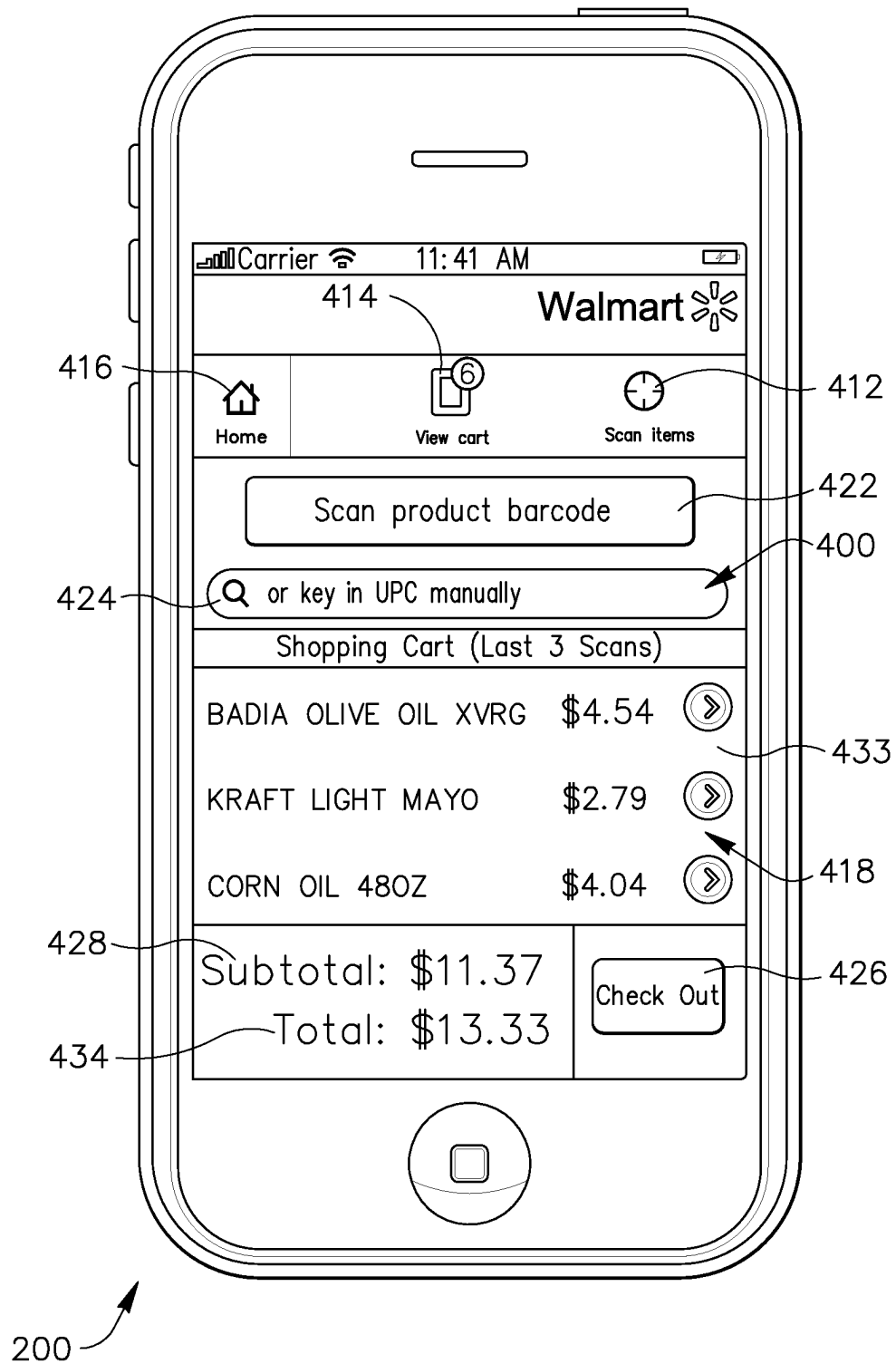


Fig.7

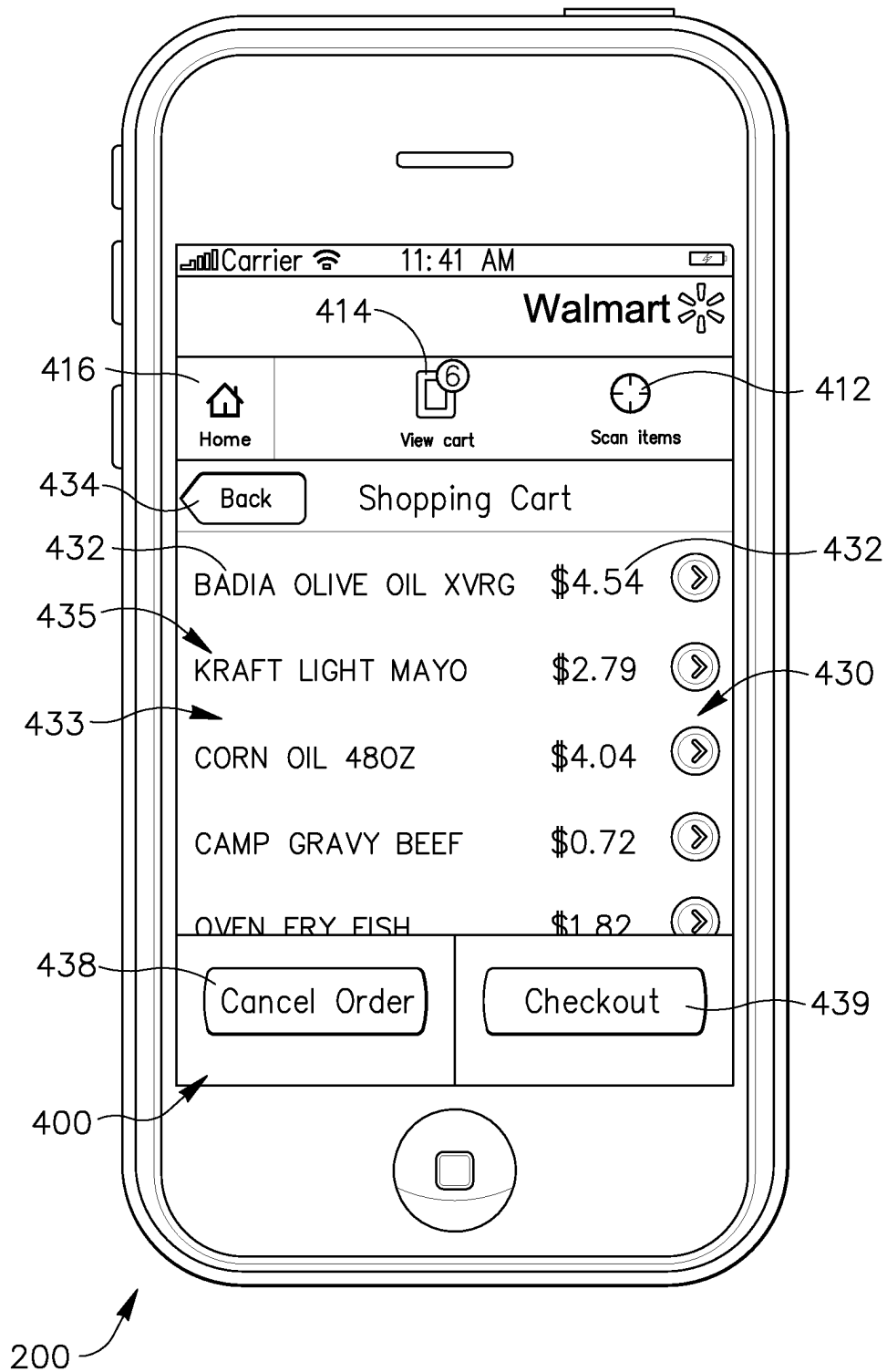


Fig.8

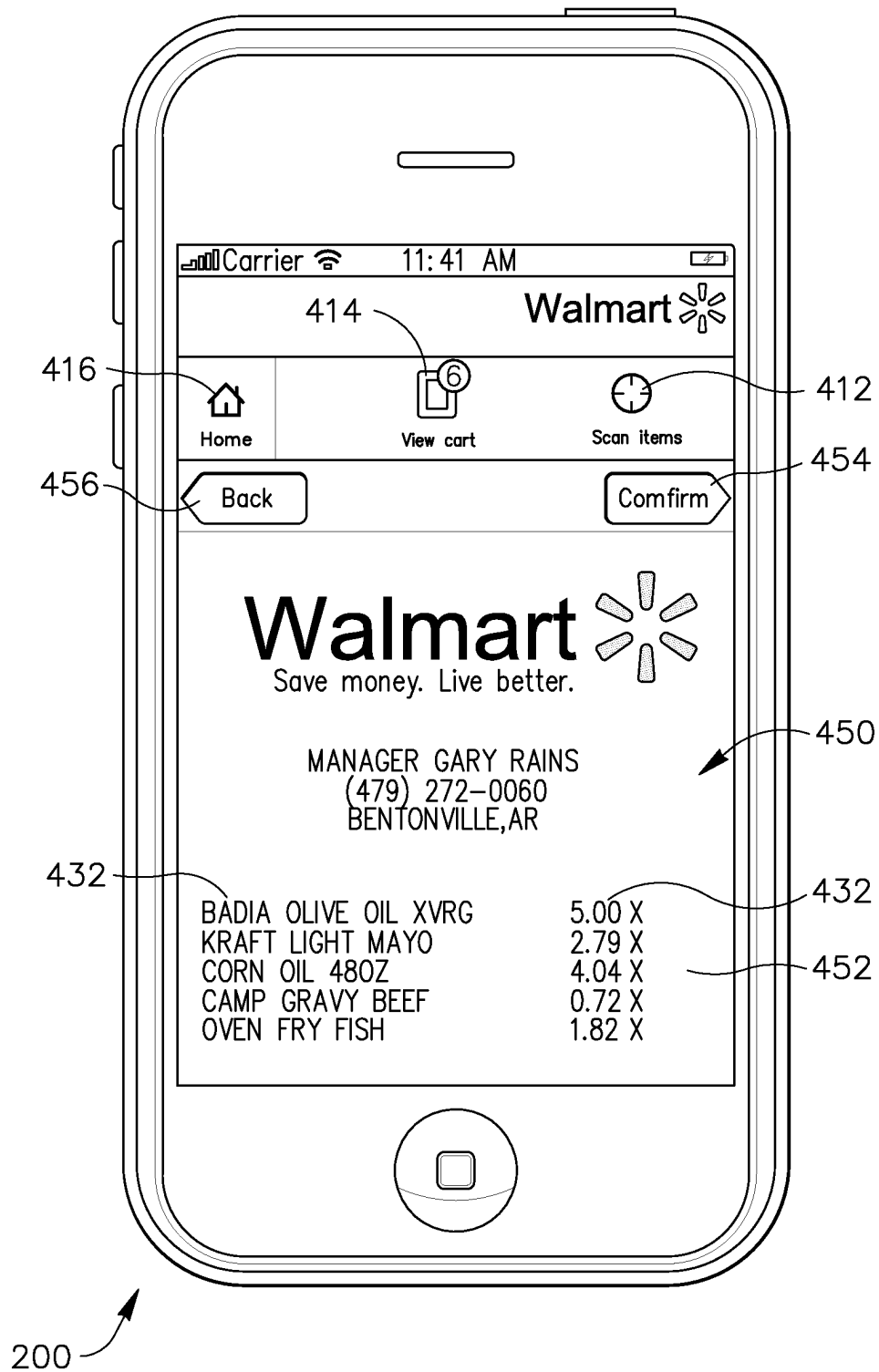


Fig.9

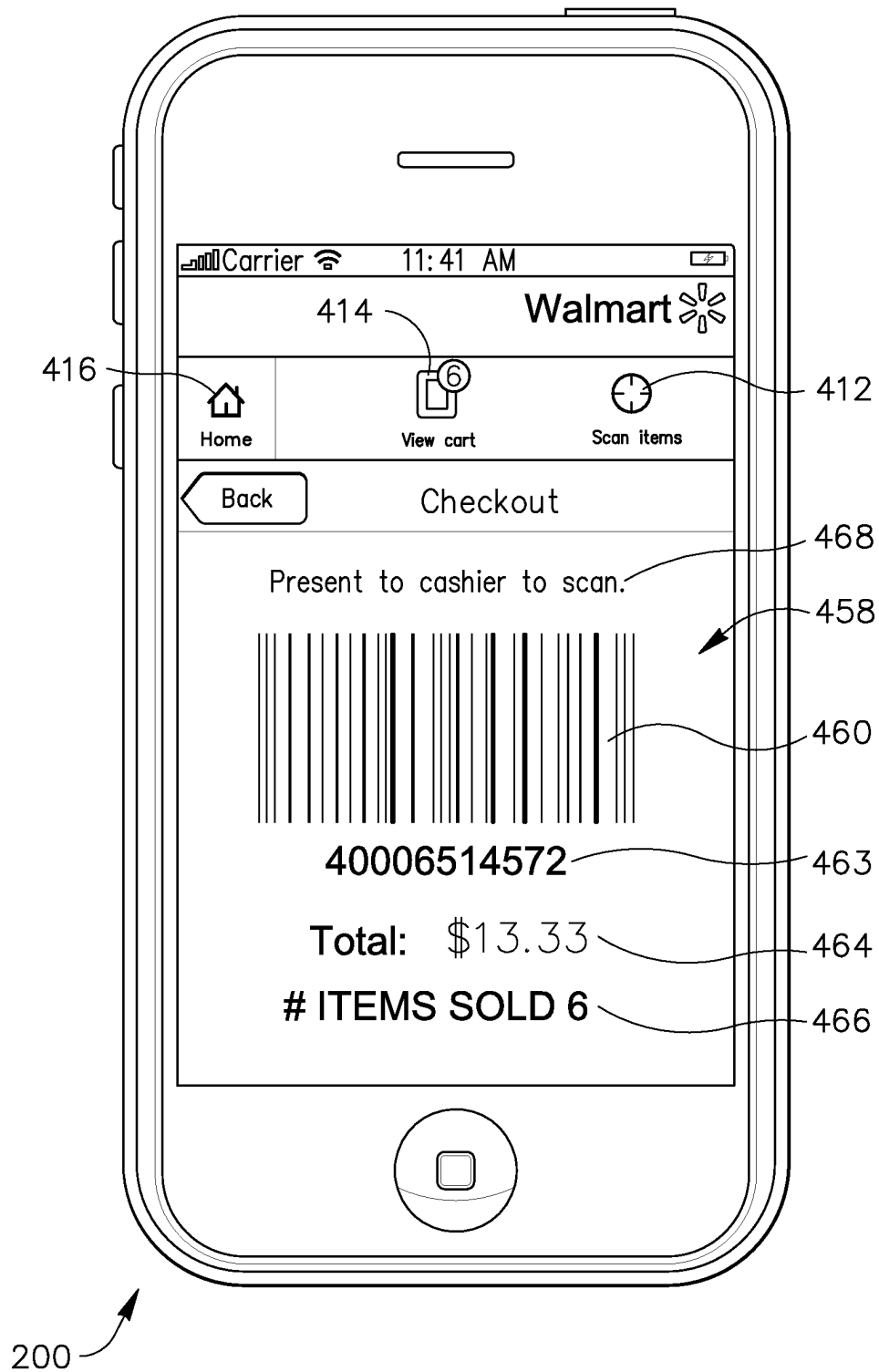


Fig.10

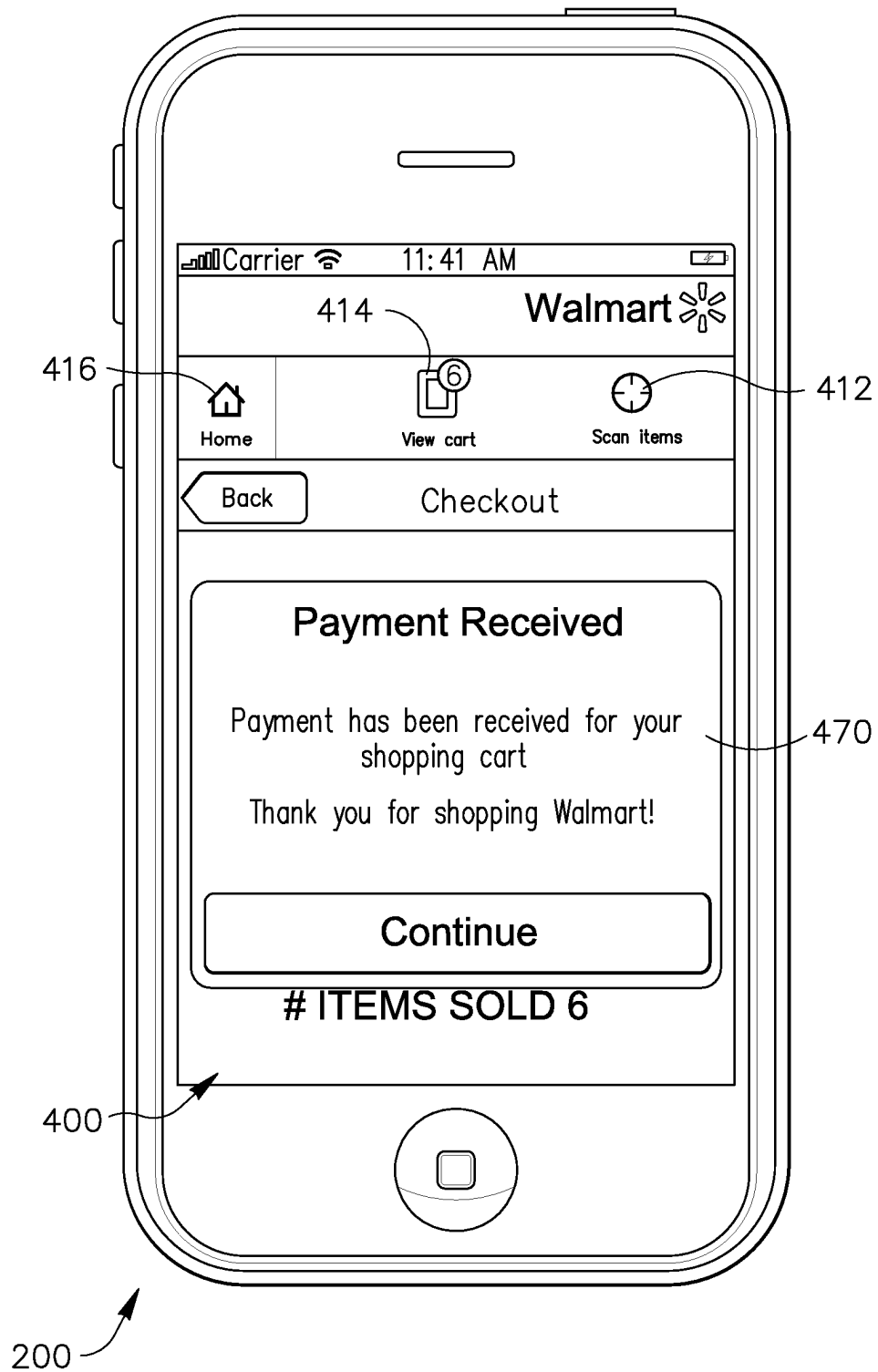


Fig. 11

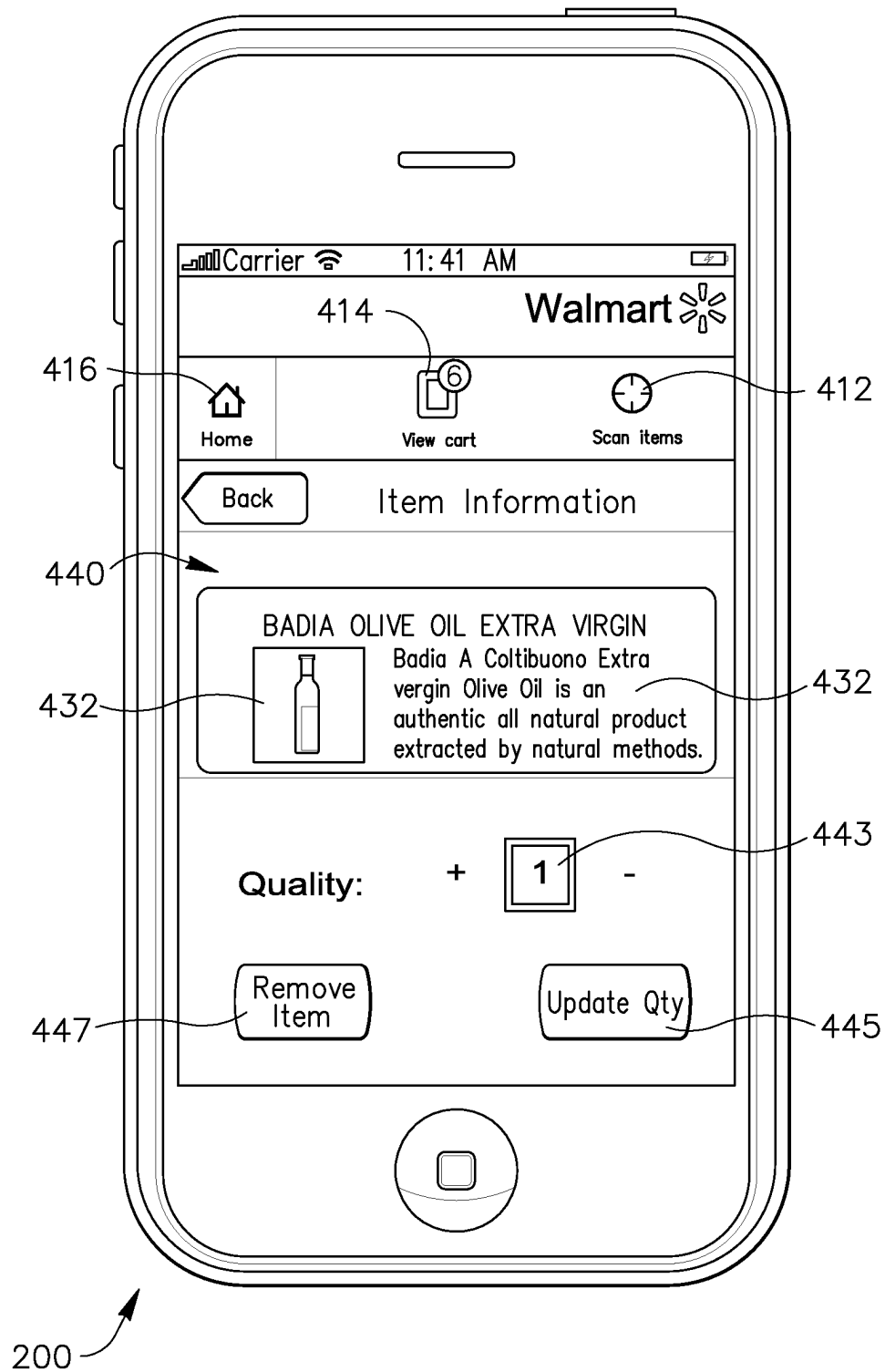


Fig.12

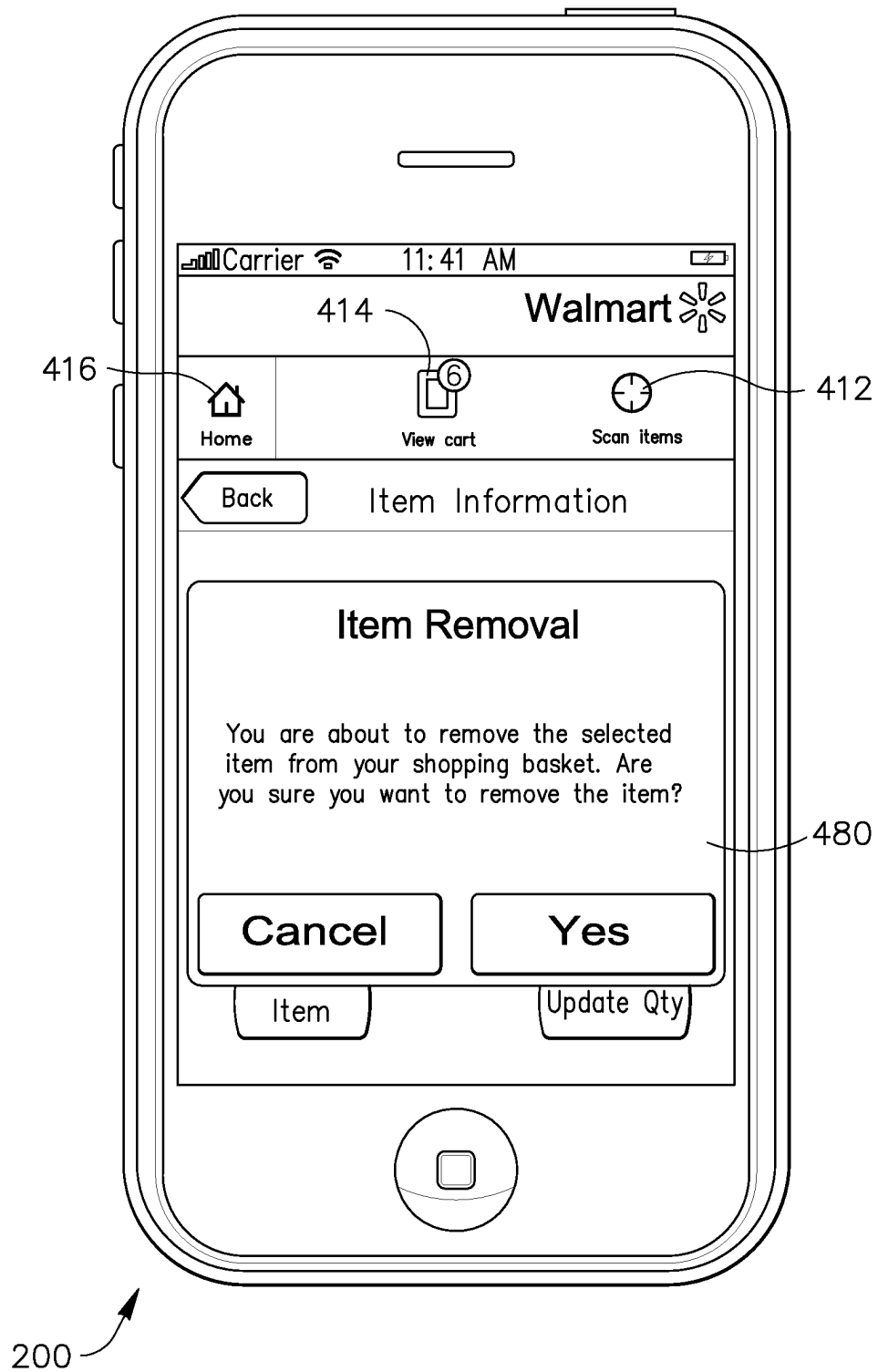


Fig.13

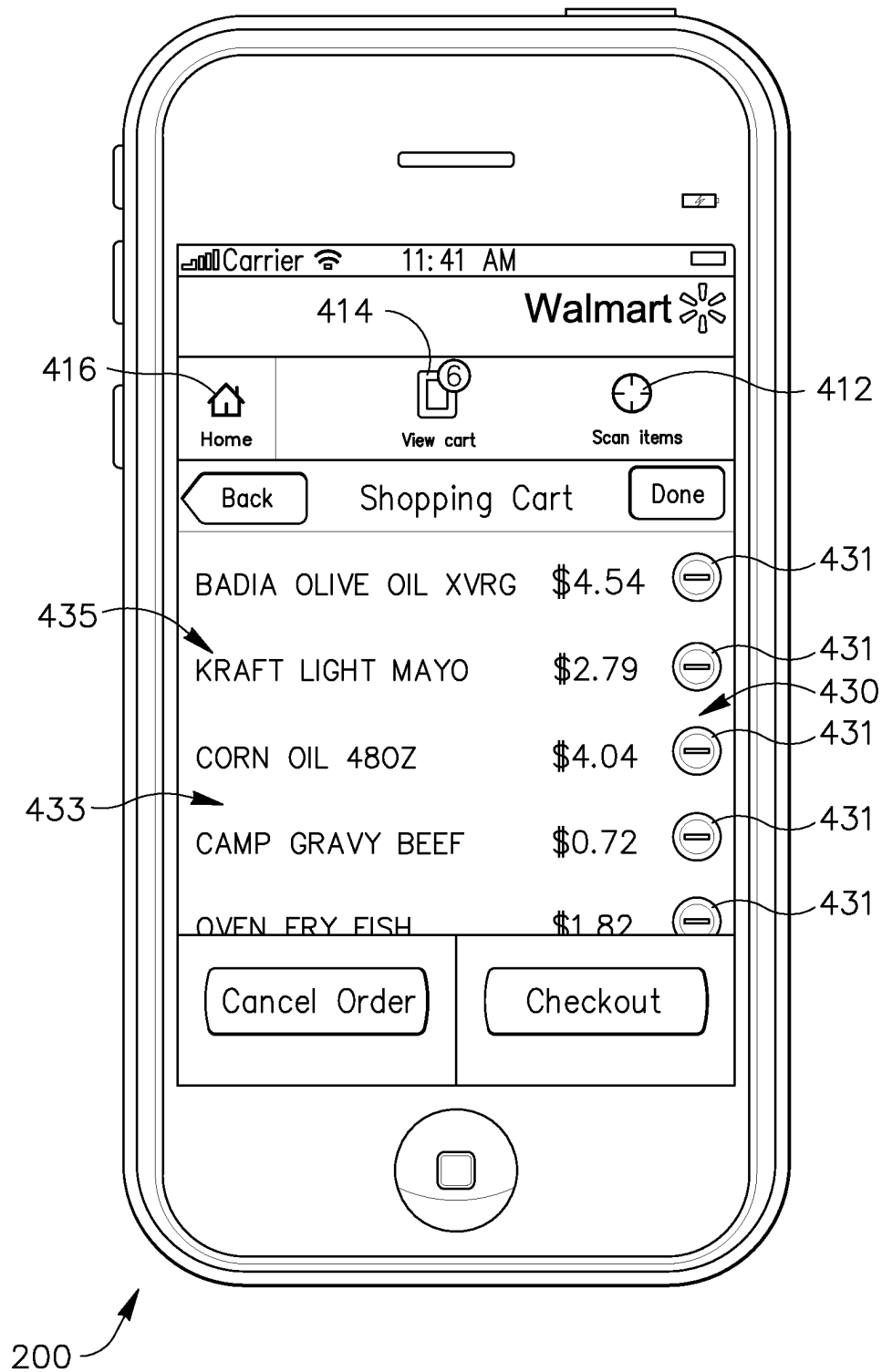


Fig. 14

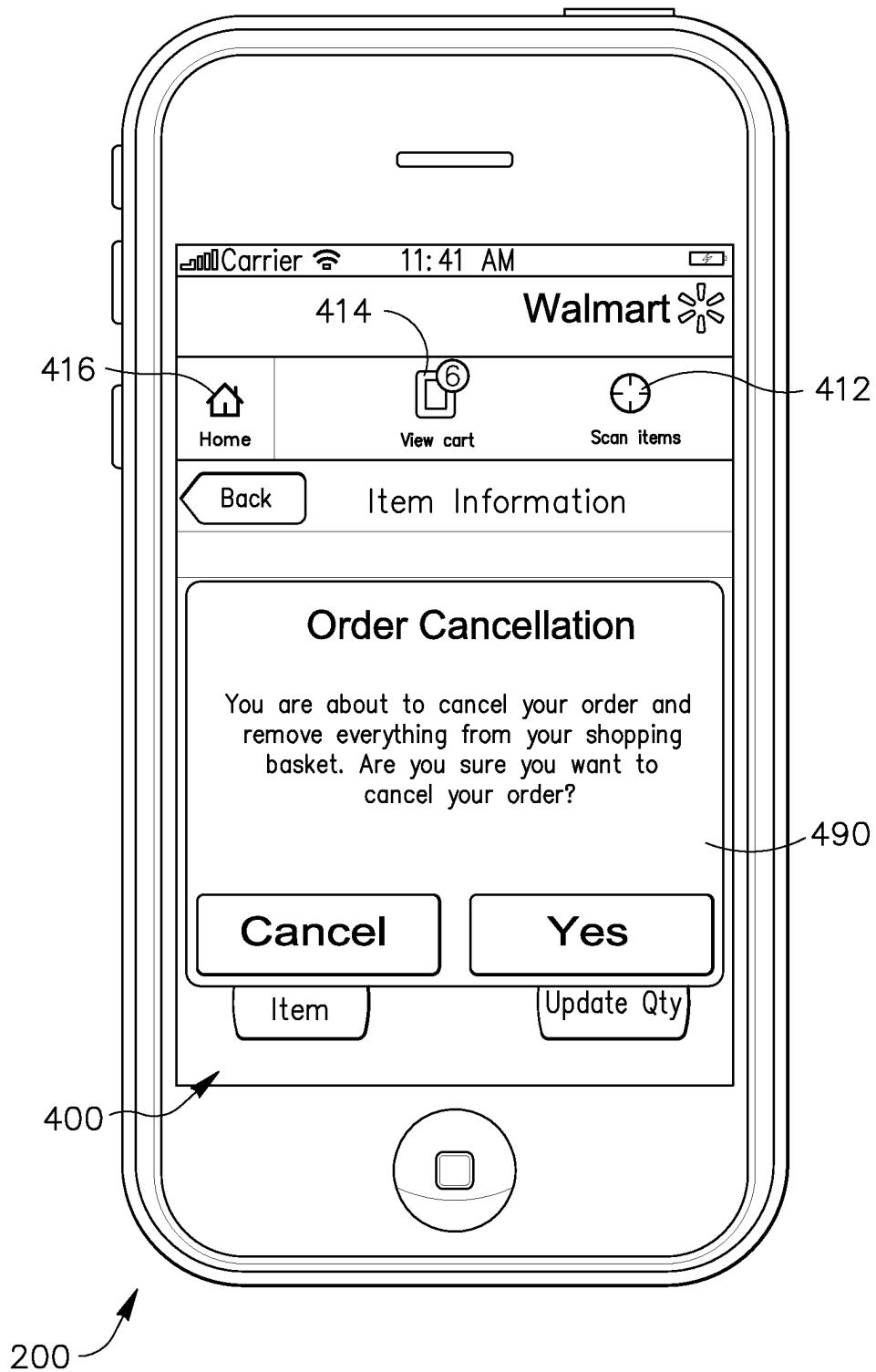


Fig.15

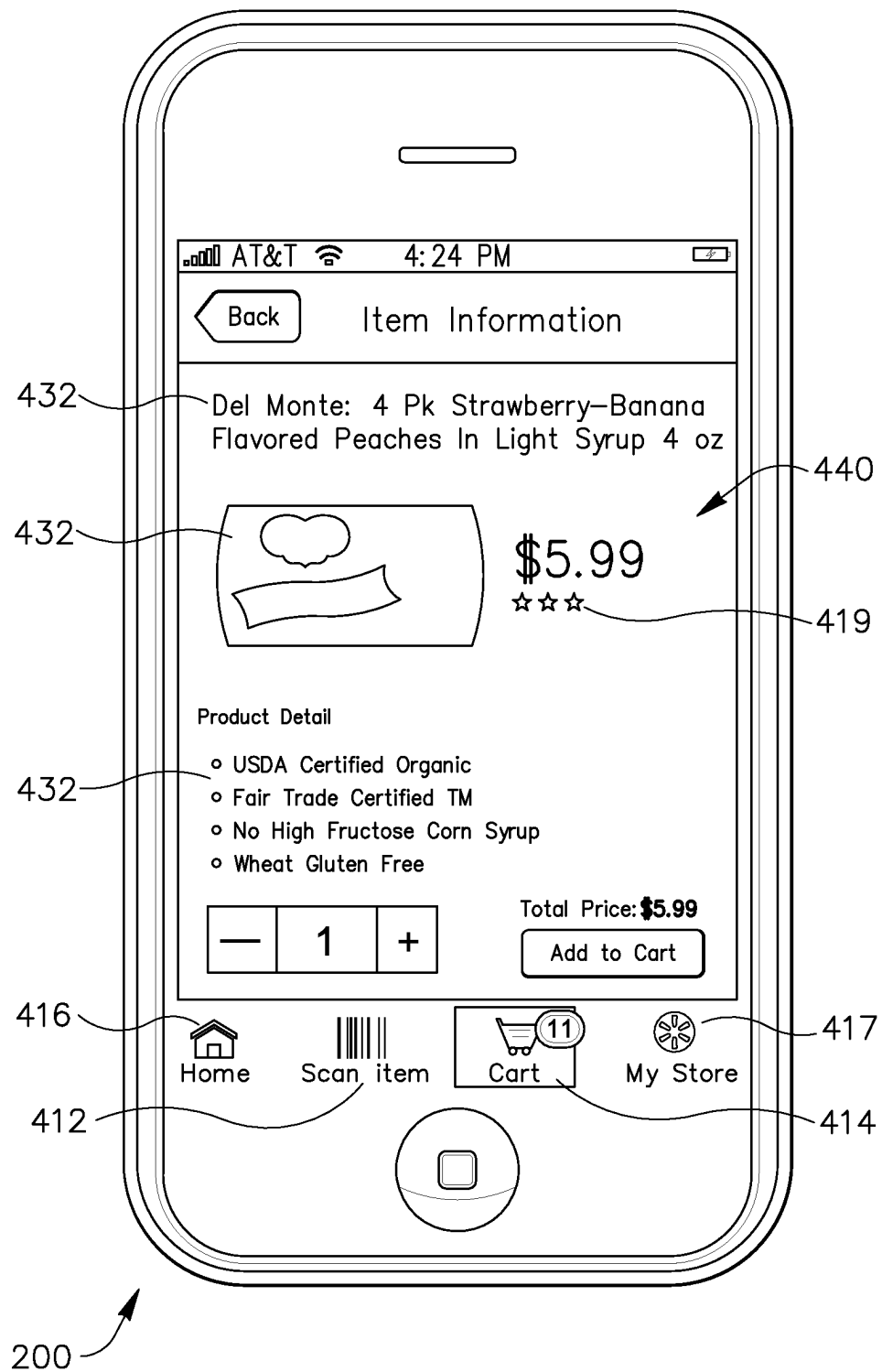


Fig.16

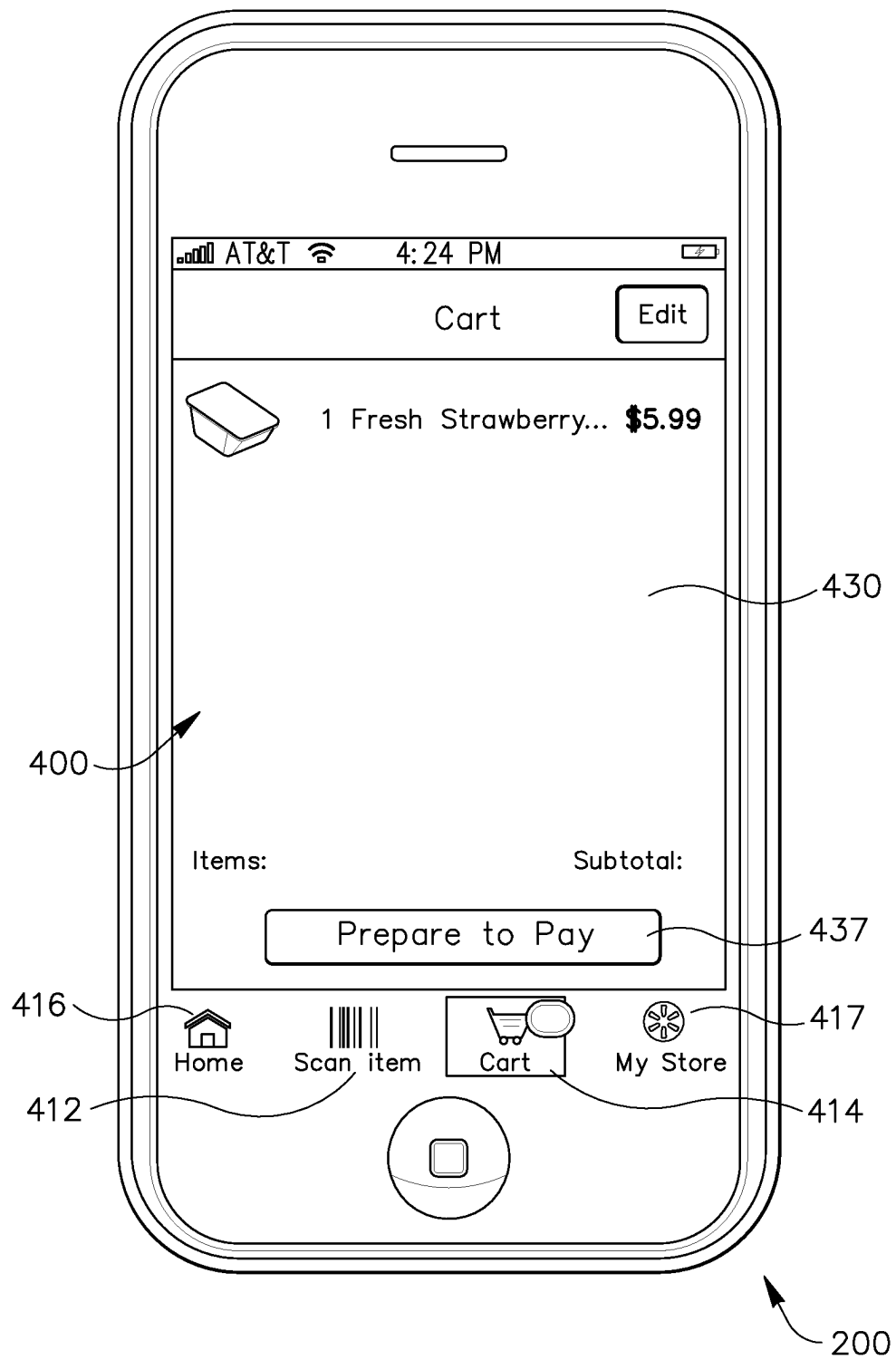


Fig.17

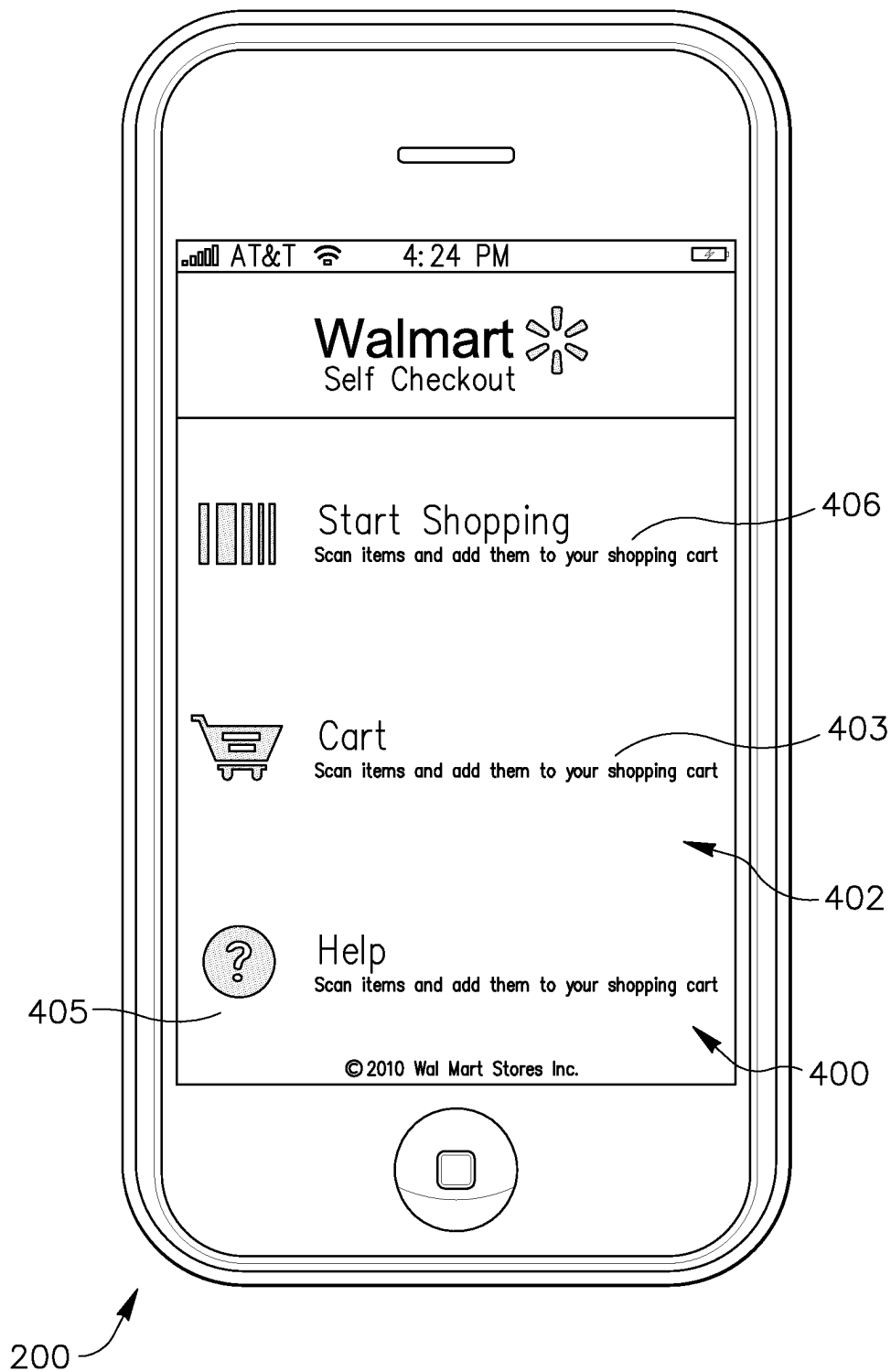


Fig.18

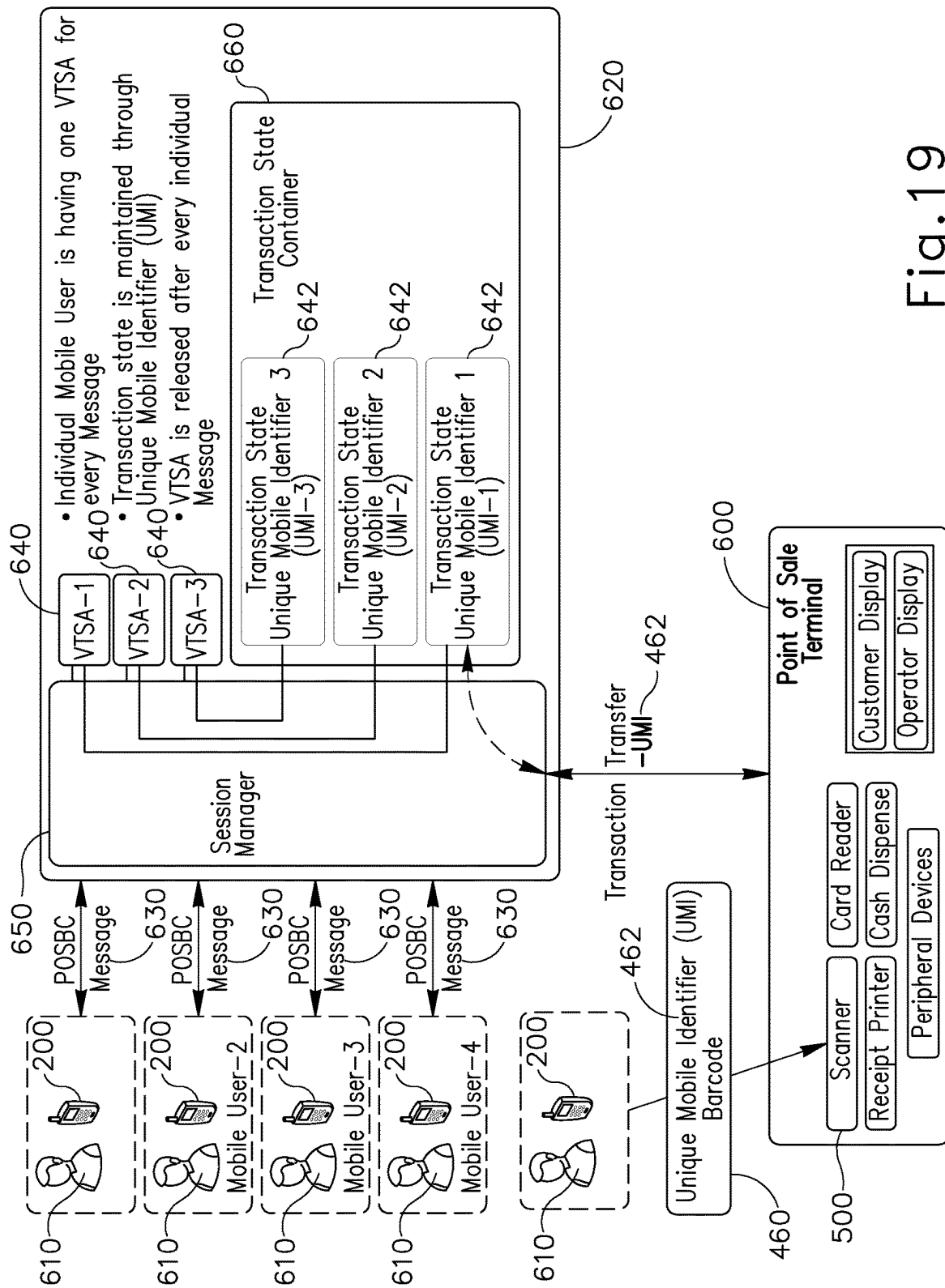
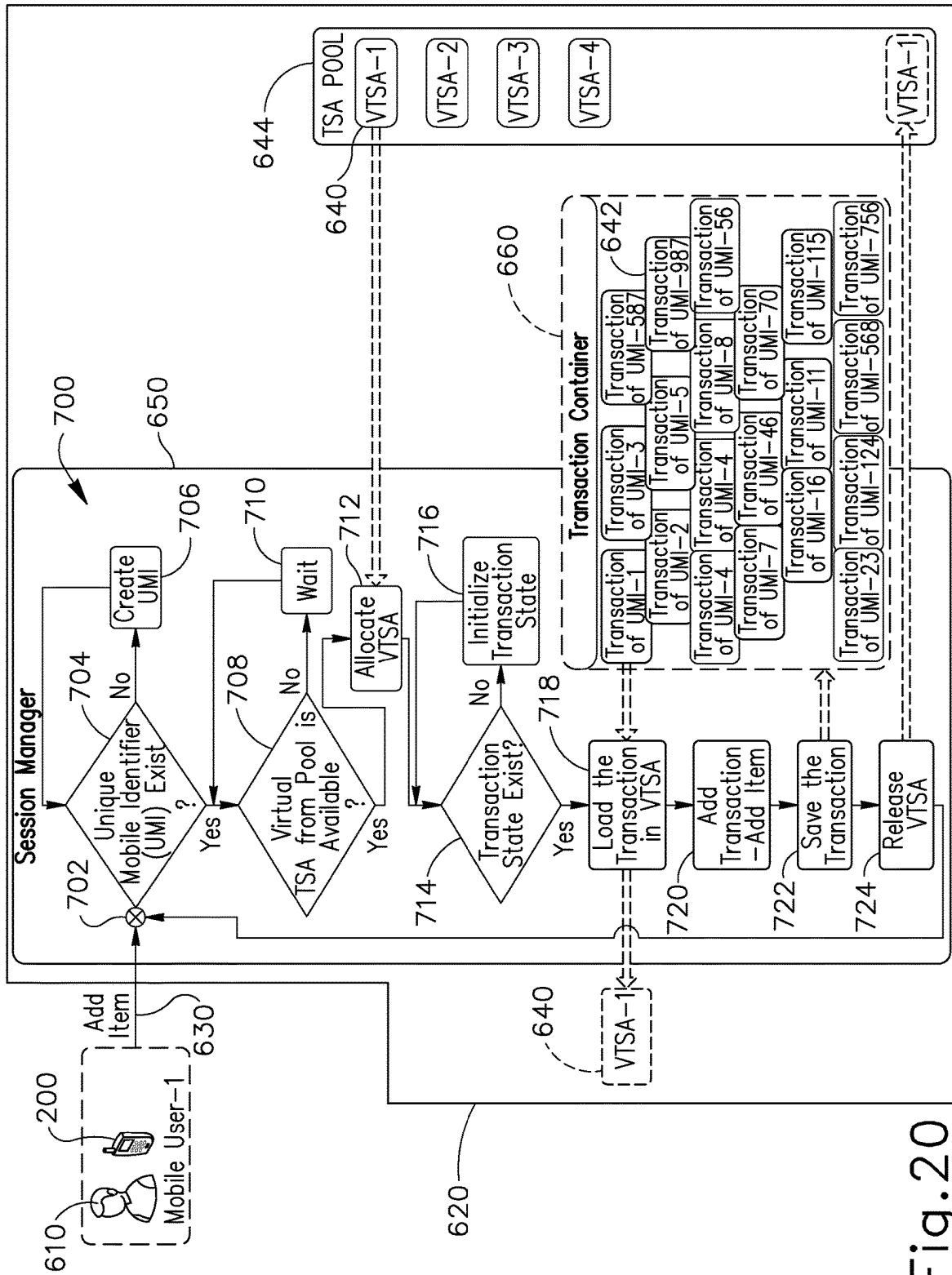


Fig. 19



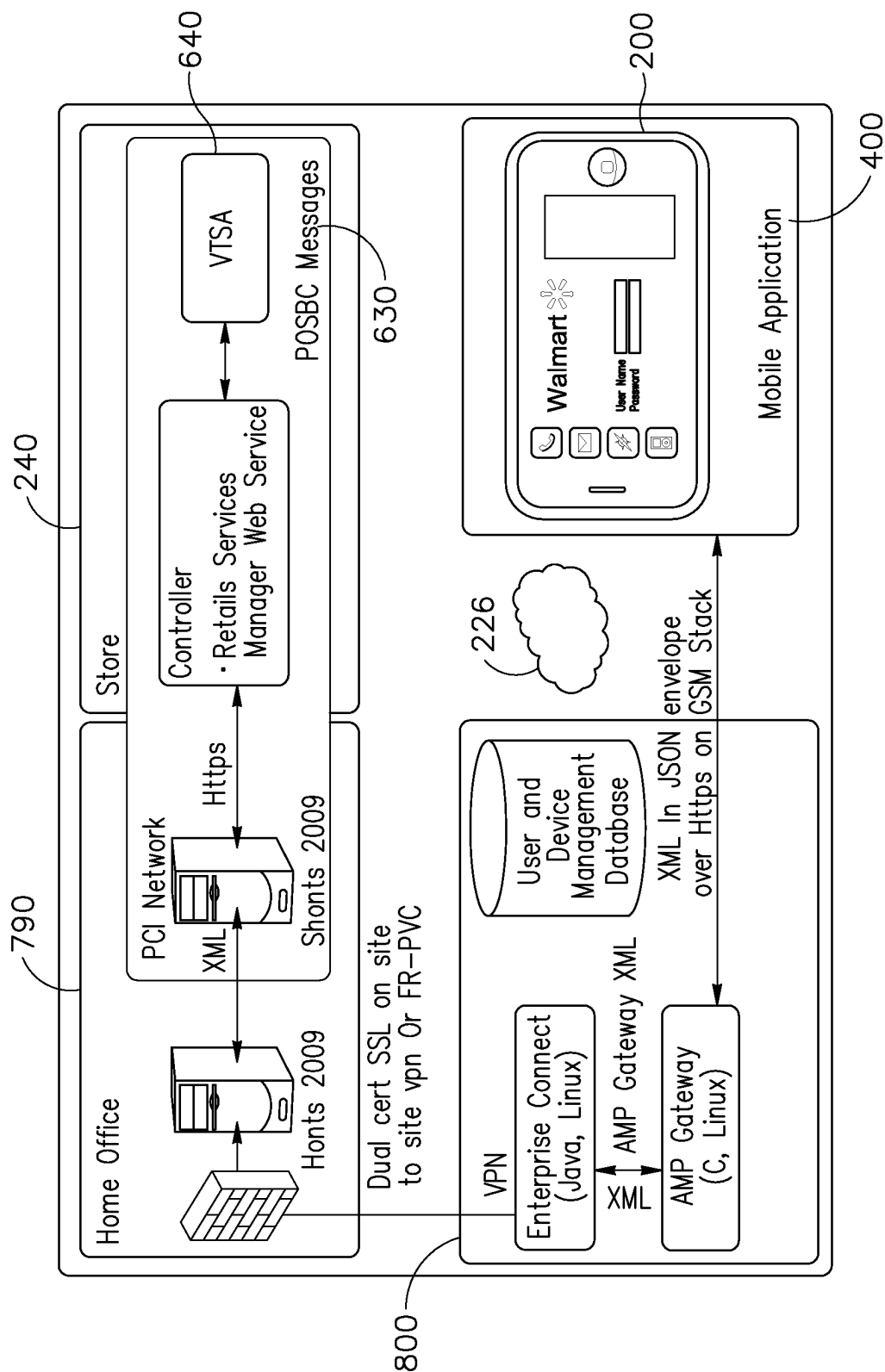


Fig. 21

METHOD FOR SELF-CHECKOUT WITH A MOBILE DEVICE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation application of U.S. patent application Ser. No. 18/217,945, filed on Jul. 3, 2023, which in turn is a continuation application of U.S. patent application Ser. No. 17/004,819, filed on Aug. 27, 2020, which in turn is a continuation application of U.S. patent application Ser. No. 16/730,567, filed on Dec. 30, 2019, now U.S. Pat. No. 10,803,435, which in turn is a continuation application of U.S. patent application Ser. No. 15/956,400, filed on Apr. 18, 2018, which in turn is a divisional application of U.S. patent application Ser. No. 12/947,545, filed on Nov. 16, 2010, now U.S. Pat. No. 10,121,133, which in turn claims the benefit of U.S. Provisional Patent Application No. 61/392,908, filed on Oct. 13, 2010, the contents of all of which are incorporated by reference herein in their entirety.

FIELD OF THE INVENTION

The present invention relates generally to self-checkout from a retail establishment. In particular, the invention relates a method for self-checkout from a retail establishment using a mobile device.

BACKGROUND

Many retail establishments currently allow a customer to check-out and pay for items either by using a traditional cashier who scans each item and then receives tender from the customer for payment of the items, or by using a self-checkout station (SCO) in which the customer scan each item individually at that self-checkout station and then tenders payment to the self-checkout station. In either way, the customer has to shop for items by placing each item in a shopping cart, then has to take the shopping cart to either a traditional cashier or a self-checkout station, often the customer has to wait in line, and then the customer has to remove each item from the shopping cart so that the traditional cashier or a self-checkout station can scan identification information from the item, such as a barcode, and then typically each item is then placed back in the shopping cart upon scanning.

As a result of this process, retail establishments have to invest in cashiers or self-checkout stations and additional time is required from the customer in requiring the customer to complete his purchase by having to go to a cashier or a self-checkout station. It would be desirable to provide a customer and a retail establishment with a method for self-checkout from the retail establishment which did not require having the customer go to a traditional cashier or a self-checkout station for scanning each item.

SUMMARY

In one aspect, a method for purchasing an item using a mobile device is provided. The method includes, but is not limited to, inputting identification information into the mobile device which identifies the item for purchase and generating an order for the item identified by the identification information. The method also includes, but is not limited to, displaying on a display of the mobile device an optical machine-readable representation of the order.

In one aspect, a computer readable memory medium comprising program instructions for purchasing an item using a mobile device having a digital camera is provided. The program instructions are executable by a processor to capture an image of an identifying portion of the item with the digital camera and analyze the identifying portion of the item and generating identification information which identifies the item for purchase. The program instructions are also executable by a processor to generate a virtual shopping cart containing item information representing the item. The item information is retrieved using the identification information. The program instructions are also executable by a processor to transmit payment information from the mobile device to a remote server for acceptance of an order for the item identified by the identification information.

In one aspect, a method for purchasing an item using a mobile device having a digital camera is provided. The method includes, but is not limited to, capturing an image of an identifying portion of the item with the digital camera and analyzing the identifying portion of the item and generating identification information which identifies the item for purchase. The method also includes, but is not limited to, generating a virtual shopping cart containing item information representing the item, generating an order for the item identified by the identification information and tendering payment for the item. The method also includes, but is not limited to, transmitting payment information to a remote server for acceptance, generating a receipt indicating acceptance of the payment information, and displaying the receipt on a display of the mobile device.

In one aspect, a computer readable memory medium comprising program instructions for maintaining and conducting a virtual transaction is provided. The program instructions are executable by a processor to access a session of a virtual terminal sales application upon receiving a point of sale message, load transaction state information into the virtual terminal sales application session, and process the transaction state information using the open session of the virtual terminal sales application.

The scope of the present invention is defined solely by the appended claims and is not affected by the statements within this summary.

BRIEF DESCRIPTION OF DRAWINGS

The invention can be better understood with reference to the following drawings and description. The components in the figures are not necessarily to scale, emphasis instead being placed upon illustrating the principles of the invention.

FIG. 1A depicts a block schematic diagram of an exemplary computing system, in accordance with one embodiment of the present invention.

FIG. 1B depicts an illustration of a mobile device receiving identification information from an item and communicating with a remote server, in accordance with one embodiment of the present invention.

FIG. 1C depicts an illustration of an optical scanning machine reading an optical machine-readable representation of an order display on a mobile device, in accordance with one embodiment of the present invention.

FIGS. 2A and 2B depict flowchart illustrations of methods, apparatus (systems) and computer program products, in accordance with one embodiment of the present invention.

FIGS. 3-18 depict various schematic representations of a mobile self-checkout device running a mobile self-checkout application, in accordance with one embodiment of the present invention.

FIGS. 19, 20 and 21 depict various schematic representations of methods, apparatus (systems) and computer program products, in accordance with one embodiment of the present invention.

DETAILED DESCRIPTION

The present invention makes use of the discovery that by using a mobile device with item identification information input capabilities and communications capabilities, a user can input identification information into the mobile device which identifies the item for purchase, generate an order for the item identified by the identification information and pay for the order without having to remove each item from a shopping cart and scan the item at a traditional cashier or a self-checkout station. The user has the option of paying for the order by simply providing a cashier with an optical machine-readable representation of an order number identifying an order displayed on a display of the mobile device or transmitting payment information directly from the mobile device to a remote server. In this manner, a user can save time by not having to remove items from a shopping cart for scanning at a traditional cashier or a self-checkout station, and the user can save additional time by bypassing the traditional cashier or self-checkout station completely and paying for an order directly with the mobile device.

In the description that follows, the subject matter of the application will be described with reference to acts and symbolic representations of operations that are performed by one or more computers, unless indicated otherwise. As such, it will be understood that such acts and operations, which are at times referred to as being computer-executed, include the manipulation by the processing unit of the computer of electrical signals representing data in a structured form. This manipulation transforms the data or maintains it at locations in the memory system of the computer which reconfigures or otherwise alters the operation of the computer in a manner well understood by those skilled in the art. The data structures where data is maintained are physical locations of the memory that have particular properties defined by the format of the data. However, although the subject matter of the application is being described in the foregoing context, it is not meant to be limiting as those skilled in the art will appreciate that some of the acts and operations described hereinafter can also be implemented in hardware, software, and/or firmware and/or some combination thereof.

With reference to FIG. 1A, depicted is an exemplary computing system for implementing embodiments. FIG. 1 includes a computer 100, which could be any one of a mobile device 200 or remote server 240. Computer 100 may be a portable device, wherein at least some or all of its components are formed together in a single device which can be carried around by a person. The computer 100 includes a processor 110, memory 120 and one or more drives 130. The drives 130 and their associated computer readable memory medium provide storage of computer readable instructions, data structures, program modules and other data for the computer 100. Drives 130 can include an operating system 140, application programs 150, program modules 160, and program data 180. Computer 100 further includes input devices 190 through which data may enter the computer 100, either automatically or by a user who enters commands and data. Input devices 190 can include an

electronic digitizer, a flatbed scanner, a barcode reader, a microphone, a camera, a video camera, a keyboard and a pointing device, commonly referred to as a mouse, a trackball or a touch pad, a pinpad, any USB device, any Bluetooth enabled device, an RFID or NFC device, and a debit card reader. Other input devices may include a joystick, game pad, satellite dish, scanner, and the like. In one or more embodiments, input devices 190 are portable devices that can direct display or instantiation of applications running on processor 110.

These and other input devices 190 can be connected to processor 110 through a user input interface that is coupled to a system bus 192, but may be connected by other interface and bus structures, such as a parallel port, game port or a universal serial bus (USB). Computers such as computer 100 may also include other peripheral output devices such as speakers, printers, and/or display devices, which may be connected through an output peripheral interface 194 and the like.

Computer 100 also includes a radio 198 or other type of communications device for wirelessly transmitting and receiving data for the computer 100 with the aid of an antenna. Radio 198 may wirelessly transmit and receive data using WiMAX™, 802.11a/b/g/n, Bluetooth™, 2G, 2.5G, 3G, and 4G, wireless standards.

Computer 100 may operate in a networked environment using logical connections to one or more remote computers, such as a remote server 240. The remote server 240 may be a personal computer, a server, a router, a network PC, a peer device or other common network node, and may include many if not all of the elements described above relative to computer 100. Networking environments are commonplace in offices, enterprise-wide computer networks, intranets and the Internet. For example, in the subject matter of the present application, computer 100 may comprise the source machine from which data is being migrated, and the remote computer may comprise the destination machine. Note, however, that source and destination machines need not be connected by a network or any other means, but instead, data may be migrated via any media capable of being written by the source platform and read by the destination platform or platforms. When used in a LAN or WLAN networking environment, computer 100 is connected to the LAN through a network interface 196 or an adapter. When used in a WAN networking environment, computer 100 typically includes a modem or other means for establishing communications over the WAN, such as radio 198, to environments such as the Internet. It will be appreciated that other means of establishing a communications link between computer 100 and other computers may be used.

With reference to FIG. 1B, illustrated is an exemplary representation of a mobile device 200 for purchasing an item 220 from a retailer at a retail establishment. Mobile device 200 includes any portable electronic device having a processor 201 for executing applications and a display 206 for displaying information connected with the processor 201, and includes such devices as a personal desktop assistant (PDA), a portable computer, a mobile telephone, a smartphone, a netbook, and a tablet computer. Display 206 can use any of a variety of types of display technologies, such as a liquid crystal display (LCD), a cathode-ray tube type display, an electronic ink display, a light emitting diode (LED) type display such as an OLED display, and a plasma display. Preferably, the mobile device 200 also includes a communications device 208 and a scanning module 202. The communications device 208 is connected with the processor 201 and capable of sending and receiving infor-

5

mation between one or more other computers connected with the mobile device 200. Preferably, communications device 208 is capable of wirelessly transmitting signals to another computer, such as remote server 240, using a radio transmitter and a radio receiver connected with an antenna. The scanning module 202 is capable of receiving identification information 222 from an item 220 and converting the identification information 222 into a format that the processor 201 can read, such as digital data. Preferably, Scanning module 202 includes any device which can capture, receive and process optical information, such as a barcode or any image, and includes devices such as a digital scanner, a digital camera, a video camera, a barcode reader, and any other type of digital or analog imaging device.

Preferably, communications device 208 communicates with another computer 100, such as remote server 240, via a network 226 using a network interface 209. Network interface 209 is connected with processor 201 and communications device 208, and preferably disposed within remote device 200.

Network 226 may include any type of network that is capable of sending and receiving communication signals, including signals for multimedia content, images, data and streaming video. Network 226 may include a data network, such as the Internet, an intranet, a local area network (LAN), a wide area network (WAN), a cable network, and other like systems that are capable of transmitting information, such as digital data, and the like. Network 226 may also include a telecommunications network, such as a local telephone network, long distance telephone network, cellular telephone network, satellite communications network, cable television network and other like communications systems that interact with computer systems to enable transmission of information between mobile device 200 and another computer such as remote server 240. Network 226 may include more than one network and may include a plurality of different types of networks. Thus, network 226 may include a plurality of data networks, a plurality of telecommunications networks, cable systems, satellite systems and/or a combination of data and telecommunications networks and other like communication systems.

Network 226 is connected with both mobile device 200 and remote server 240 and allows for information to be transmitted and shared between mobile device 200 and remote server 240. Remote server 240 includes any type of computer which can receive, store, process, and transmit information to another computer and includes devices such as a server based computer system capable of interacting with one or more other computer systems.

In one embodiment, mobile device 200 includes location information processing means which allows the mobile device 200 to determine its location. Location information processing means includes devices such as a Global Positioning System (GPS) based device, and methods such as using radio triangulation to determine the location of the mobile device 200. Preferably, mobile device 200 includes input means 210 for entering information from a user into the mobile device 200. Input means includes any device which can assist a user to enter information, such as a keyboard, a mouse, a touchpad, a touchscreen, a joystick, a button, and a dial.

Item 220 includes any item or service which is being sold by a retailer. Preferably, the item 220 is located with a retail establishment or store. Item 220 includes item identification information 222 which is any information on the item 220 which assists in identifying the item 220, such as a barcode 224, a serial number, a name of the item 220, and any text,

6

characters, illustrations, or images on the item 220 which can be used to identify the item 220.

With reference to FIGS. 2A and 2B, in operation, the mobile device 200 is brought into a retail establishment by a customer, or user, and a method 300 for self-checkout with the mobile device 200 is initiated by launching a mobile self-checkout application 400 within the mobile device 200 at block 302. Mobile self-checkout application 400 resides on the mobile device 200 and communicates with a terminal sales application 620 which resides on the server 240. Upon launching the mobile self-checkout application 400, the user may be first prompted to identify himself/herself by entering user identification information into the mobile device 200 via the input means 210. User identification information may include a username, a password, both a username and password. If user is launching the mobile self-checkout application 400 for the first time, the user may be prompted to enter additional user information in addition to user information. Additional user information includes any information which may be used to identify a user, such as user's name, a user's address, a user's telephone number, a user's credit card information or other user financial information, a user's social security number, a user's driver's license number, a user's birthdate, a user's passport number, and other identifying information which may or may not be stored by mobile device 200 or remote server 240.

Upon identifying the user, the method 300 then moves to block 306 where mobile device 200 is connected with a computer at the retail establishment, such as remote server 240, via network 226. Upon connecting mobile device 200 with remote server 240, the method 300 then moves to block 308 and the user begins shopping at the retail establishment by walking around the retail establishment and looking for items 220 to purchase. With reference to FIGS. 3 and 18, a home screen 402 is presented to the user by the mobile self-checkout application 400 upon launching the mobile self-checkout application 400 and entering user information. In one embodiment, the home screen 402 includes a local ad button 404 which can provide the user with advertisements specific to the retail establishment in which the user is in, a begin shopping button 406 which allows the user to communicate his intention to begin shopping to the mobile self-checkout application 400 and presents the user with a shopping screen 418, and an order history button 408 which allows the user to review past orders, purchases, or past virtual shopping carts. Additionally, the home screen 402 may display store identifying information 410 which provides information that identifies the specific retail establishment in which the user is in, for example, by stating the city and state in which the retail establishment is in or a store number for the retail establishment. Preferably, the mobile device 200 uses the location information processing means to determine the store identifying information 410.

With reference to FIG. 18, in one embodiment, the home screen 402 includes a cart button 403, which when selected, presents the user with a shopping cart screen 430, as shown in FIG. 8. In one embodiment, the home screen 402 includes a help button 405, which when selected, presents the user with a series of help screens which aid the user in using the mobile self-checkout application 400.

Additionally, mobile self-checkout application 400 includes a scan items button 412, a view cart button 414, and a home button 416. Preferably, buttons 412, 414, and 416 are all present on all or almost all of the screens of the mobile self-checkout application 400, as shown in FIGS. 3-15. When selected, the scan items button 412 activates the scanning module 202 and presents the user with a scan items

screen 420, as shown in FIG. 5, which allows the user to capture and input identification information 222 from an item 220. When selected, the view cart button 414 presents the user with a shopping cart screen 430, as shown in FIG. 8, which allows the user to view the contents of a virtual shopping cart 433 created by the user and which contains item information 432 representing the item 220. When selected, the home button 416 returns the user to the home screen 402.

With reference to FIG. 16, in one embodiment, mobile self-checkout application 400 includes a my store button 417 in addition to the scan items button 412, the view cart button 414, and the home button 416. When selected, the my store button 417 presents the user with a store information screen generated by mobile self-checkout application 400 which displays information about the retail establishment that the user is in. Specifically, the store information screen displays information including but not limited to an address of the retail establishment, a phone number of the retail establishment, hours of operation for the retail establishment, services and store number for the retail establishment, and other such information. Preferably, the my store button 417, when selected, also determines the geographic location of the retail location using location information processing means.

Preferably, once the user has begun shopping at the retail establishment, as shown in block 308, the user communicates his intention to begin shopping to the mobile self-checkout application 400, such as by activating the begin shopping button 406 on the home screen 402. With reference to FIGS. 4 and 7, when selected, the begin shopping button 406 presents the user with a shopping screen 418 having a scan product barcode button 422, a manual identification information box 424, a checkout button 426, subtotal information 428, total information 434, and a partial virtual shopping cart contents window 436. The scan product barcode button 422 activates the scanning module 202 and presents the user with a scan items screen 420, as shown in FIG. 5. The manual identification information box 424 allows a user to manually input identification information 222 into the mobile device 200 which identifies the item 220 for purchase, by inputting, for example, a Universal Product Code (UPC) for the item 220. In one embodiment, UPC information is either manually input or determined automatically from the scan of identification information 222 and then transmitted by mobile device to remote server 240 where a UPC lookup system resides which uses the UPC information to provide the user with item information 432. When selected, the checkout button 426 presents the user with a checkout screen 450, as shown in FIG. 9, which presents the user with the total contents of the virtual shopping cart 433 and the option to confirm the user's intention to purchase those contents. The subtotal information 428 presents the user with a subtotal amount for the contents of the virtual shopping cart 433, and the total information 434 presents the user with a total amount for the contents of the virtual shopping cart 433. The partial virtual shopping cart contents window 436 display a partial view of the contents of the virtual shopping cart 433, such as the last the items entered into the virtual shopping cart 433.

Upon spotting an item to purchase, method 300 then moves to block 310, wherein the user then takes and removes the item 220 from a shelf or stand within the retail establishment. Moving to block 312, the user then inputs identification information 222 into the mobile device 200 which identifies the item 220 for purchase. The identification information 222 can be input manually, such as by entering the information 222 into the identification information box

424, or the identification information 222 can be automatically input by scanning the information 222 using scanning module 202. The user may automatically input identification information 222 by activating scanning module 202. For example user may by activating scanning module 202 by either pressing scan product barcode button 422 or scan items button 412, either of which activates the scanning module 202 and presents the user with a scan items screen 420, as shown in FIG. 5. At scan items screen 420, the user initiates scanning module 202 which then receives identification information 222 from an item 220 and converts the identification information 222 into a format, such as a digital image, so that the processor 201 can read and process the identification information 222. In processing identification information 222, processor 201 may use optical character recognition (OCR) algorithms in order to determine identification information 222 such as a barcode number or a UPC for the item 220.

Upon inputting identification information 222 into the mobile device 200, the identification information 222 is transmitted to the remote server 240 via communications device 208 and received by the remote server 240 at block 314. The remote server 240 may then, at block 314, retrieve item information 432 representing the item 220 using the identification information 222 received from the mobile device 200. The item information 432 may include an item name, an item price, an item review, an item rating, item product details which is a description of the item 220, an image of the item 220, an item tax amount or percentage, or item location information as to where the item 220 is located in the retail establishment.

Upon retrieving item information 432, the remote server 240 may then transmit the retrieved item information 432 to the mobile device, whereupon mobile self-checkout application 400 may present the user with an item information screen 440 which displays item information 432 for item 220, as shown in FIGS. 6 and 12. In one embodiment, the user may be prompted to confirm that the item information 432 is related to the item 220 for purchase. Moving to block 316, the user may also be prompted to confirm the user's intent to purchase an item identified by item information 222. For example, the item information 222 may be for an item which is not item 220 selected by the user. In this case the user can inform the mobile self-checkout application 400 that the item information 222 is not accurate or does not relate to item 220.

With reference to FIGS. 6 and 12, in one embodiment, item information screen 440 provides the user with an add to shopping cart button 442 or a quantity select button 443 which when selected, generates a virtual shopping cart 433 containing item information 432 representing the item 220. The quantity select button 443 allows the user to add more than one item 220 in the virtual shopping cart 433 and then an update quantity button 445 allows the user to update the quantity of item 220 indicated by the virtual shopping cart 433. Additionally, the user may also be provided with a cancel and scan again button 444, which when selected, does not add item information 432 to a virtual shopping cart 433 and provides the user with the scan items screen 420 again. In one embodiment, a remove item button 447 is provided to the user, which when selected, removes the any item information 222 for item 220 from the virtual shopping cart 433. With reference to FIG. 13, if the remove item button 447 is selected, the user may be prompted with an item removal confirmation screen 480 which prompts the user to confirm his/her desire to remove item 220 along with item information 222 from virtual shopping cart 433.

With reference to FIG. 16, in one embodiment, item information screen 440 provides the user with the option 419 to rate the item 220 and read reviews from other users for the item 220. In one embodiment, mobile self-checkout application 400 provides the user with the option to provide stock information to the remote server 240 indicating whether or not item 220 is in stock at the retail establishment.

If the user confirms his/her intent to purchase item 220, a virtual shopping cart 433 containing item information 432 representing the item 220 is then generated at block 318. In one embodiment, the virtual shopping cart 433 is generated by the mobile device 200 and then transmitted to the remote server 240. In another embodiment, the virtual shopping cart 433 is generated by the remote server 240 and specifically, generated by the terminal sales application 620. Preferably, the virtual shopping cart 433 is generated by the remote server 240, specifically generated by the terminal sales application 620, stored on the remote server, and transmitted to the mobile device 200 for display to the user via display 206. Preferably, any contents in the virtual shopping cart 433, which represent items 220, are kept in the virtual shopping cart 433 for a predefined period of time, after which the contents are deleted. If the method 300, and the shopping process, is interrupted for any reason (for example a phone call is received by the mobile device 200, a text message is received by the mobile device 200, the user wishes to perform another function with the mobile device 200, etc.) the user can resume the method 300 and continue using the virtual shopping cart 433 representing the already selected items 220 for a predefined period of time. In one embodiment, mobile self-checkout application 400 provides the user with the option to create and save a shopping list having all the items 220 represented in the virtual shopping cart 433. The shopping list can be created any place, such as a user's home. When using the shopping list in a retail establishment as part of method 300, items 220 may be paired up and removed from the shopping list and then added to the virtual shopping cart 433.

In one embodiment, upon retrieving item information 432, the remote server 240 may then transmit the retrieved item information 432 to the mobile device, whereupon mobile self-checkout application 400 may automatically generate a virtual shopping cart 433 containing item information 432 representing the item 220. Additionally, in one embodiment, upon retrieving item information 432, the remote server 240 may automatically generate a virtual shopping cart 433 containing item information 432 representing the item 220 and transmit that shopping cart along with its contents to the mobile device 200 for display to the user.

Upon generating a virtual shopping cart 433 containing item information 432 representing the item 220 and transmitting that information between the mobile device 200 and the remote server 240, the mobile device 200 then displays a virtual shopping cart 433 along with all of its contents 435 to the user at block 320. In one embodiment, the mobile self-checkout application 400 presents the user with the shopping cart screen 430, as shown in FIG. 8, displaying the contents of the virtual shopping cart 433 to the user using item information 432.

With reference to FIGS. 8 and 17, the shopping cart screen 430 may present the user with the option to purchase the item 220 represented by item information 432 in the virtual shopping cart 433 by pressing a checkout button 439 or a prepare to pay button 437. The shopping cart screen 430 may present the user with the option to continue shopping by pressing a back button 434, wherein the user is presented

with the shopping screen 418 again. The shopping cart screen 430 may also present the user with the option to cancel an order 452 altogether by pressing a cancel order button 438, wherein the contents of the virtual shopping cart 433 are emptied and the user is presented with either the shopping screen 418 or the home screen 402 again. With reference to FIG. 15, in one embodiment, when the cancel order button 438 is selected, the user may be prompted with an order cancellation screen 490 which prompts the user to confirm his/her desire to cancel order 452 and empty the virtual shopping cart 433.

As used herein, an order 452 is any request to purchase an item 220. Order 452 is comprised of information received from the mobile self-checkout program 400, such as identification information 222, quantity information indicating how many units of a particular item 220 in the virtual shopping cart 433 a user would like to purchase, user identification information, additional user information, an order number 463, and store identifying information 410 which provides information that identifies the specific retail establishment in which the user is in.

With reference to FIG. 14, in one embodiment, the shopping cart screen 430 includes an item removal button 431 associated with each item 220 represented in the virtual shopping cart 433. When the item removal button 431 is selected, the user may be prompted with item removal confirmation screen 480 which prompts the user to confirm his/her desire to remove item 220 along with item information 222 from virtual shopping cart 433, as shown in FIG. 13.

Upon displaying the contents of the virtual shopping cart 433 to the user at block 320, the user then places the item 220 in an actual shopping cart at block 322. The user is then prompted by the mobile self-checkout application 400 at block 324 and asked if the user would like to continue shopping and purchases additional items 220 or not. In one embodiment, the user is presented with checkout button 439, back button 434, and/or cancel order button 438. If the user wishes to continue shopping, method 300 moves back to block 310. If the user does not wish to continue shopping, method 300 moves to block 326, wherein the user selects the option of checking out, using for example checkout button 439. In one embodiment, instead of pressing a button, such as checkout button 439, back button 434, and/or cancel order button 438, the user may make a hand gesture, a swipe, or other such gesture to activate a feature of mobile self-checkout application 400.

Upon selecting the option to checkout, a total amount for all the contents of the virtual shopping cart 433 is calculated and an order 452 is generated for the item 220 identified by the identification information 222, listed in the virtual shopping cart 433 at block 328. Preferably, an order number 463 associated with the order 452 is also generated. In one embodiment, the order 452 is generated by the mobile device 200 and transmitted to the remote server 240, in another embodiment the order 452 is generated by the remote server 240 and then transmitted to the mobile device 200 for confirmation by the user. Upon generating the order 452 at block 328, the user is then presented with the order 452 at block 330 and asked to confirm the accuracy of the order 452. Preferably, an order confirmation screen 450 is generated by the mobile self-checkout application 400, as shown in FIG. 9. The order confirmation screen 450 includes a confirm button 454 which allows the user to confirm the accuracy of the order 452 and the user's intent to purchase the items 220 identified in the order 452 via identification information 432. The order confirmation screen 450 also

11

includes a back button **456** which when pressed allows the user to go back to the shopping cart screen **430**.

Upon confirming a user's intent to purchase items **220** identified by item information **432**, method **300** then moves to block **332** wherein the user selects the type of tender and the payment method used to pay for the order **452** which was generated. The user has the choice of providing a variety of types of tender, such as cash, credit card, direct debit from a bank, payment using an online payment service such as PayPal™ or Google Checkout, a gift card, store credit, personal check, money order, or other payment means. The user also has the choice of payment method, either transmitting payment via the mobile device **200** using mobile self-checkout application **400** or providing payment at either a traditional cashier or self-checkout station.

If the user decides to provide payment for order **452** using a traditional cashier or self-checkout station, which may occur when the user wishes to pay with cash, then a virtual transaction is created at block **342** and an optical machine-readable representation **460** of the order **452** which identifies the order **452** is generated and displayed on the display **206** of the mobile device **200** at block **344**. The optical machine-readable representation **460** encodes a unique identifier **462** such as an order number **463** or a unique mobile identifier (UMI) which is associated with the mobile device **200** in order to identify order **452**, since each optical machine-readable representation **460** generated is unique to each order **452**. Each unique mobile identifier (UMI) is specific to a particular mobile device **200**, and contains information identifying that particular mobile device **200**, such as unique device identifier (UDID) for the mobile device **200**, a serial number of the mobile device **200**, or some other identifying information for a particular mobile device **200**.

The optical machine-readable representation **460** of the order **452** represents the order **452** and encodes identifying information or a unique identifier **462**, which includes things such as: an order number **463**; customer identifying information which can be used to identify a customer, such as a customer name, a customer number, and a social security number; transaction identifying information which can be used to identify a specific transaction, such as a number or other alphanumeric code; device identifying information that can identify a specific device such as a serial number or a unique device identifier (UDID) for a device such as a mobile phone or mobile computer; and any alphanumeric code or any symbol which may be used associated with and used to identify and retrieve a specific order **452**. Preferably, an order number **463** along with the optical machine-readable representation **460** of the order **452** is displayed on the display **206** of the mobile device **200**.

The optical machine-readable representation **460** of the order **452** is encoded in such a manner as to allow an optical scanning machine **500**, such as barcode reader, to read the optical machine-readable representation **460**. A barcode reader, or barcode scanner, is an electronic device for reading printed barcodes. Like a flatbed scanner, a barcode reader consists of a light source, a lens, and a light sensor translating optical impulses into electrical ones. As a result, the optical machine-readable representation **460** of the order **452** is preferably in a form of barcode, such as a linear or 1D barcode, or a matrix 2D barcode, allowing for a standard barcode reader to read and convert the optical machine-readable representation **460** into digital data which any computer can read. In one embodiment, the optical machine-readable representation **460** of the order **452** is preferably in

12

a form of 1D barcode, such as a 1D barcode having one of the following formats: UPC-A, UPC-E, EAN-13, EAN-8, Code-128, Code-39 and ITF.

Preferably, upon generating the optical machine-readable representation **460** of the order **452**, a machine-readable representation screen **458** is generated by the mobile self-checkout application **400**, as shown in FIG. **10**. The machine-readable representation screen **458** includes an order number **463** along with the optical machine-readable representation **460** of the order **452**, a total payment amount **464** for all the items **220** ordered, and a total amount **466** of items **220** sold. The user is directed by the mobile self-checkout application **400** to present the optical machine-readable representation **460** to a cashier to scan in order to complete a checkout process via message **468** on machine-readable representation screen **458** at block **346**. In one embodiment, a unique identifier **462** is provided to the user and displayed on display **206**, either alone or printed below the optical machine-readable representation **460**. In this embodiment, the user may manually input or key in the unique identifier **462** at a point of sale terminal **600** or provide the unique identifier **462** to a cashier who then can manually enter the unique identifier **462** at the point-of-sale terminal **600**.

With reference to FIGS. **1C** and **2B**, at block **348**, the user goes to a traditional cashier or a self-checkout station and presents the optical machine-readable representation **460** to an optical scanning machine **500** which reads the optical machine-readable representation **460**, and specifically the unique identifier **462** encoded in the optical machine-readable representation **460**. Preferably, the unique identifier **462** is then transmitted to remote server **240** in order to retrieve the order **452** from the remote server **240**. Order **452**, along with the total payment amount **464**, is then transmitted from the remote server **240** to a cash register or other computer operable by the traditional cashier or the self-checkout station. Moving to block **348**, the user is then prompted by either the traditional cashier or the self-checkout station to tender payment for the total payment amount **464**, whereupon the user then tenders payment to either the traditional cashier or the self-checkout station for the total payment amount **464**.

Moving the block **336**, upon the user tendering payment, payment information is generated by the cash register or other computer operated by the traditional cashier, or by the self-checkout station and then transmitted to the remote server **240**, whereupon the virtual shopping cart **433** is then marked paid and a receipt is generated by the remote server **240**, the traditional cashier, and/or the self-checkout station. The receipt is then provided to the user.

Preferably, an acknowledgment that payment has been received is sent to the mobile device **200** from the remote server **240** and displayed to the user via a payment received screen **458** which is generated by the mobile self-checkout application **400**, as shown in FIG. **10**. In one embodiment, a digital copy of the receipt is transmitted to the user from the remote server **240**. Preferably, the digital copy of the receipt is transmitted from the remote server **240** to the mobile device **200** and displayed via the mobile self-checkout application **400**. In one embodiment, the digital copy of the receipt is transmitted from the remote server **240** in an email to an email address of the user. Preferably, upon generating and sending an acknowledgment that payment has been received, the server **240** then changes the current virtual shopping cart into a past virtual shopping cart, retrievable by the user via the order history button **408**. Sales transaction and receipt information is stored on the remote

13

server **240** and is thus accessible for a variety of purposes: including but not limited to: asset protection solutions, return validation, customer reference, and the like.

If at block **332** the user decides to provide payment for order **452** by transmitting payment via the mobile device **200** using mobile self-checkout application **400**, then application **400** either debits a preregistered payment account of the user's or the user is prompted for payment account information. If the user has already created a preregistered payment account with mobile self-checkout application **400**, by already entering payment account information which could include, credit card information, online payment service information such as a user account name and password, store credit information, or gift card information, then mobile self-checkout application **400** debits the preregistered account at block **334**. If the user has not already created a preregistered payment account with mobile self-checkout application **400**, the user is prompted to create a payment account by providing payment account information. Upon creating the payment account, the mobile self-checkout application **400** then debits the payment account for the total payment amount **464**. Upon debiting either the preregistered payment account or the just created payment account, the method **300** then moves to block **336**, whereupon the whereupon the virtual shopping cart **433** is then marked paid and a receipt is generated by the remote server **240**, the traditional cashier, and/or the self-checkout station. The receipt is then provided to the user. Sales transaction and receipt information is stored on the remote server **240** and is thus accessible for a variety of purposes: including but not limited to: asset protection solutions, return validation, customer reference, and the like.

Moving to block **338**, upon marking that the virtual shopping cart **433** is paid, mobile self-checkout application **400** is terminated and the user exits the retail establishment at block **340**.

With reference to FIGS. **19**, **20** and **21**, a terminal sales application **620** for maintaining and conducting a virtual transaction is shown. Terminal sales application **620** includes a session manager **650** in communication with a Virtual Terminal Sales Application (VTSA) **640** and a transaction state container **660**. A virtual transaction contains all of the information found within order **452** or virtual shopping cart **433**, but is conducted entirely with an application, such as a VTSA **640** run within terminal sales application **620**. Unlike real transactions which are processed on a point of sale terminal **600** having peripheral devices, a virtual transaction is processed with an application which mimics the functionality of a point of sale terminal **600** and is run on a computer which lacks the peripheral devices point of sale terminal **600**. Terminal sales application **620** resides on a computer, such as remote server **240**, and interacts with mobile self-checkout application **400**.

With reference to FIG. **19**, terminal sales application **620** communicates with mobile self-checkout application **400** by receiving requests via Point of Sale (POS) messages **630** from the mobile self-checkout application **400** and by transmitting POS message **630** back to the mobile self-checkout application **400**. POS messages **630** are messages which contain point-of-sale purchase information, such as Point of Sale Business Component messages used by point of sale terminals such as IBM™ Self Checkout Systems. In particular, session manager **650** of terminal sales application **620** communicates with mobile self-checkout application **400** by sending and receiving POS messages **630**. POS messages **630** received by terminal sales application **620** contain a request to process information required for order

14

452, such as identification information **222**, quantity information indicating how many units of a particular item **220** in the virtual shopping cart **433** a user **610** would like to purchase, user identification information, additional user information, an order number **463**, store identifying information **410** which provides information that identifies the specific retail establishment in which the user is in, and a unique identifier **462** which identifies a specific virtual transaction, such as an order number **463** or a unique mobile identifier (UMI).

Each virtual transaction created or processed by terminal sales application **620** has a unique identifier **462** associated with it which identifies a specific order **452**. The unique identifier **462** may be an order number **463** or a piece of identifying information which identifies a specific user **610** or a specific mobile device, such as a unique mobile identifier (UMI) or a user name or ID number. In one embodiment, the unique identifier **462** is a unique mobile identifier (UMI). By associating a unique identifier **462** with a specific virtual transaction, virtual transactions can be identified for each user **610** or mobile device **200** interacting with terminal sales application **620**. Additionally, each POS message **630** received or transmitted by the terminal sales application **620** also includes a unique identifier associated with a virtual transaction. In this manner, each POS message **630** can be associated with and applied to the virtual transaction it was intended for.

Session manager **650** transmits and receives POS messages **630** to and from the mobile self-checkout application **400**, and communicates the POS messages **630** to the VTSA **640**. Additionally, session manager **650** interacts with a transaction state container **660** and communicates transaction state information **642** between the transaction state container **660** and the VTSA **640**. Preferably, session manager **650** receives and prioritizes POS messages **630** received from mobile self-checkout application **400** in a queue and determines when a POS messages **630** is communicated to a given VTSA **640** and which VTSA **640** to communicate a given POS messages **630** to. Each time a POS message **630** is received from the mobile self-checkout application **400**, the POS message **630** is routed by session manager **650** to an open session of an available VTSA **640** within terminal session manager **620**. The unique identifier **462** embedded within POS message **630** is used by session manager **650** to locate transaction state information **642** associated with that unique identifier **462** used to conduct the virtual transaction for user **610**. The transaction state information **642** is then sent by the session manager **650** to the VTSA **640** to which the POS message **630** having the same unique identifier **462** was routed, and then the session manager **650** instructs the VTSA **640** to start or further modify the virtual transaction for user **610**. Preferably, session manager **650** can access an open session of a VTSA **640** from any one of a plurality of VTSA's **640** running within terminal sales application **620**.

VTSA **640** is an application which represents a virtual point of sale terminal and is run on a computer, such as remote server **240**. VTSA **640** mirrors the functions of a traditional terminal sales application run on a point of sale terminal **600** without requiring the associated peripheral devices of a traditional point of sale terminal **600**, such as a scanner or a bar code scanner, a receipt printer, a credit card reader, a cash dispenser, a customer display, an operator display, and other peripheral devices. Preferably, terminal sales application **620** includes a plurality of VTSA's **640**, each of which is running a session. By operating a plurality of VTSA's **640** all within a single terminal sales application

15

620, substantial cost and money may be saved by not having to purchase multiple dedicated hardware devices, such as traditional point of sale terminals 600, each running only a single instance of a terminal sales application.

Each VTSA 640 maintains a virtual transaction by generating and maintaining transaction state information 642 associated with each virtual transaction using a unique identifier 462. Since none of the peripheral devices associated with a traditional point of sale terminal 600 are required, multiple VTSAs 640 can run simultaneously on a general-purpose computer, such as remote server 240, in a virtual environment which simulates a traditional point of sale terminal 600 at less cost. Since multiple point of sale terminals 600 can be replaced in-part by a single computer, such as remote server 240, running multiple instances of VTSA 640, such as VTSA-1 640, VTSA-2 640, VTSA-3 640, as shown in FIG. 19, significant cost savings can be incurred by using VTSAs 640 in addition to added flexibility with how the virtual transaction can be created and maintained.

Every time a virtual transaction is created or modified in any way, for example when the user 610 creates a virtual shopping cart 433 in mobile-self checkout application 400 or adds items 220 to the virtual shopping cart 433, a POS message 630 is transmitted from mobile-self checkout application 400 to session manager 650. POS messages 430 received by session manager 650 are then put in a queue with an associated priority. Typically, POS messages 430 are prioritized in the order they were received. As a session of a VTSA 640 becomes available or open, the session manager 650 can access the open session and load transaction state information 642 along with a particular POS message 430 into the open session of VTSA 640. The VTSA 640, upon receiving the transaction state information 642 and the POS message 630, acts upon the contents of the POS message 630 and updates the transaction state information 642 associated with that particular POS message 430. For example, if a particular POS message 430 includes information that an additional item 220 has been added to a user 610's virtual shopping cart 433, then the transaction state information 642 associated with that particular POS message 430 is updated to reflect this new information.

Every time a POS message 430 is received by an open session of VTSA 640, transaction state information 642 associated with the POS message 430 is retrieved from the transaction state container 660 and transmitted to the same open session of VTSA 640. Transaction state information 642 defines the current state of a virtual transaction, and includes information such as a listing of every item 220 in virtual shopping cart 433 or specific virtual transaction, item information 432 for each item 220 in virtual shopping cart 433 or specific virtual transaction, quantity information indicating how many units of a particular item 220 in the virtual shopping cart 433 or specific virtual transaction a user 610 would like to purchase, user identification information, additional user information, an order number 463, store identifying information 410 which provides information that identifies the specific retail establishment in which the user is in, and a unique identifier 462 which identifies a specific user, such as a unique mobile identifier (UMI), amount totals for all the items 220 in virtual shopping cart 433 or specific virtual transaction, tax information for items 220 in virtual shopping cart 433 or specific virtual transaction, a receipt, a receipt identifier, and any other information or data that VTSA 640 may require to process a virtual transaction.

16

Transaction state information 642 associated with a unique identifier 462 changes each time a POS message 430 having that same unique identifier 462 is received by a user 610 or transmitted to a user 610 by session manager 650 using VTSA 640. Once a virtual transaction has been processed by the VTSA 640, any change in the transaction state information 642 is saved by VTSA 640 in the transaction state container 660 and the transaction state information 642 is exported from the VTSA 640. New transaction state information 642 for another virtual transaction is then imported into and opened by VTSA 640. By constantly importing and exporting transaction state information 642 upon processing each received POS message 630, a terminal sales application 620 can use multiple VTSAs 640 to maintain virtual transactions for a large number of users 610. For example, ten VTSAs 640, can manage and process as many as hundreds of virtual transactions, each with their own unique transaction state information 642, since not all of the virtual transactions need be active at any given time.

By importing, changing, and then exporting the transaction state information 642 each time a POS message 630 is received from a mobile device 200, instead of just keeping transaction state information 642 constantly open, the VTSA 640 allows the server 240 to support more users 610 and ultimately more orders 452. The process of being able to import and export the transaction state for each VTSA 640 by sending and receiving a unique identifier 462 is also known as transaction transfer, as opposed to a suspend/retrieve transaction which reprocesses every item in a virtual shopping cart. Transaction transfer allows for a real time import and export of the transaction state for each VTSA 640.

If the session manager 650 did not use transaction transfer, users 610 would be limited to the number of sessions of VTSA 640 running on the server 240 since each mobile user 610 would be statically connected to an instance of a VTSA 640 until the order 452 was complete. If, for example, ten sessions were running on the remote server 240, then remote server 240 would only be able to handle the virtual transactions of only ten mobile users 610 at one time.

Using transaction transfer allows the session manager 650 to import the transaction state information 642, if it exists for a specific virtual transaction, to an open session of a VTSA 640, process the current request within POS message 630 (such as adding another item 220 to a user 610's virtual shopping cart 433), save and export the transaction state information in a storage device or drive 130 by associated the virtual transaction or order 452 with a unique identifier 462, and then free that session of VTSA 640 up for the next mobile user 610. This allows the remote server 240 to support many users 610 having many virtual transactions using only a limited number of, such as ten, sessions of VTSA 640 running on the server 240. Response time is the only variant, as opposed to being blocked until another user 610 is finished shopping.

Upon updating the transaction state information 642 associated with a particular POS message 430, a new POS message 630 is generated by the VTSA 640 and sent to the session manager 650. Depending on transaction state information 642, the POS message 630 is either routed back to the mobile device 200 and the mobile-self checkout application 400 or to a point of sale terminal 600. For example, if a user 610 has completed shopping and decided to transmit payment via the mobile device 200, or if the user is still shopping and adding items to the virtual shopping cart 433, then POS message 630 is routed back to the mobile device 200. However, if the user 610 has completed shopping and

decided to transmit payment via a traditional cashier or self-checkout station, then POS message 630 is routed to point of sale terminal 600 whereupon the user 610 provides identifying information, such as a unique identifier 462 or optical machine-readable representation 460, to the point of sale terminal 600 and payment for order 452 is processed.

With reference to FIG. 21, transaction state information 642 can be maintained within remote server 240, transferred to another remote server 240 at another retail establishment, transferred to a home office server 790, or transferred to an enterprise server 800 and maintained at an enterprise level for access by multiple remote servers at multiple retail establishments across an enterprise. In one embodiment, POS messages 630 can be received by a terminal sales application 620 running in an enterprise server 800 at the enterprise level, from a multitude of sources, such as a mobile shelf-checkout application 400 running on a mobile device 200, a point of sale terminal 600, a user accessing a website on the Internet, or from a web-based application running on a browser. In this manner, a virtual transaction can be maintained and modified from a plurality of POS messages 630 arriving from a multitude of sources. As a result, a first POS message 630 related to a particular virtual transaction may be generated by a mobile shelf-checkout application 400 running on mobile device 200 and a second POS message 630 related to the same transaction may be generated by a user accessing a website. For example, a first item 220 can be added to a virtual shopping cart 433 (represented by a particular virtual transaction) via mobile self-checkout application 400, and a second item 220 can be added to the same virtual shopping cart 433 via a user accessing a website.

In one embodiment, a virtual transaction representing a gift registry having a first item 220 and a second item 220 is generated. A first user 610 may purchase and pay for the first item 220, using a first method of purchase, such as for example the mobile self-tr application 400, and a second user 610 can purchase and pay for the second item 220 using a second method of purchase, such as for example a website. Methods of purchase include mobile self-checkout application 400, a point of sale terminal 600, web-based applications, or information sent via a website.

In one embodiment, a first user 610 may purchase and pay for a first portion of a first item 220, using a first method of purchase, such as for example the mobile self-checkout application 400, and a second user 610 can purchase and pay for a second portion of first item 220 using the first method purchase or a second method of purchase, such as for example a website. A portion of an item 220 refers to a portion of the purchase price of the item 220. For example, if a purchase price of the first item 220 is \$500, the first user 610 may pay for a \$200 portion of the \$500 purchase price and a second user may pay for a \$300 portion of the purchase price, completing the transaction.

In one embodiment, if a user 610 decides to provide payment for order 452 using a point of sale terminal 600, a POS message 630 is transmitted to the terminal sales application 620 indicating the desire for the user to conduct a virtual transaction for order 452, where information for order 452 resides in the remote server 240 and an optical machine-readable representation 460 is generated by the terminal sales application 620 which is unique to each order 452. The optical machine-readable representation 460 is later transmitted to the mobile device 200 and displayed on the display 206.

In one embodiment, upon receiving the POS message 630, terminal sales application 620 generates a Virtual

Terminal Sales Application (VTSA) 640 for each virtual transaction conducted by each user 610. Each VTSA 640 has its own unique identifier 462 generated for it, wherein each VTSA 640 and each unique identifier 462 associated with it represent a unique order 452. Transaction state information 642 is maintained for each VTSA 640 using the unique identifier 462. Upon generating a unique identifier 462 for an order 452, the unique identifier 462 is then transmitted to mobile device 200 via application 400 in the form of optical machine-readable representation 460, and also transmitted to the point of sale terminal 600, where the unique identifier 462 is used to identify order 452 and a user 610 associated with unique order 452.

The self-checkout station, in addition to the cashier's point-of-sale terminal, also uses transaction transfer to import the transaction state using optical machine-readable representation 460 encoding a unique identifier 462 from the remote server 240. The optical machine-readable representation 460 scanned from the mobile device 200 with scanning machine 500 encodes the unique identifier 462 which is then used by the point of sale terminal 600 to find the associated order 452 that resides on the remote server 240.

With reference to FIG. 20, a POS message 630 is generated as a result of input from a user 610 and received by terminal sales application 620, and specifically session manager 650, and a method 700 for maintaining and conducting a virtual transaction is initiated at block 702. Method 700 is performed by terminal sales application 620 which resides on the server 240. Upon initiating method 700 at block 702, the method 700 awaits to receive a POS message 630. Upon receiving a POS message 630, method 700 then moves to block 704, whereupon it is determined if a unique identifier 462 or a unique mobile identifier (UMI) exists or not for a particular virtual transaction being referred to by user 610 within received POS message 630. If a unique identifier 462 or a unique mobile identifier (UMI) does not exist, then one is created by session manager 650 or retrieved from mobile device 200 by session manager 650 at block 706. The unique identifier 462 or unique mobile identifier (UMI) is then associated with the virtual transaction and method 700 moves from block 706 back to block 704. If a unique identifier 462 or a unique mobile identifier (UMI) does exist, then method 700 moves to block 708, whereupon it is determined if a Virtual Terminal Sales Application (VTSA) 640 is available for use. In one embodiment, a plurality of VTSA 640 are operating within the terminal sales application 620, the plurality of VTSA 640 referred to herein as a TSA Pool 644. In this embodiment, at block 708, it is determined if a VTSA 640 from TSA Pool 644 is available. If no VTSA 640 is available, then method 700 moves to block 710 and waits.

If a VTSA 640 is available, then method 700 moves to block 712 and allocates a VTSA 640 for conducting a virtual transaction in relation to the received POS message 630 and associated with the unique identifier 462 or UMI. When the VTSA 640 is allocated, a new session of the VTSA 640 is opened. Upon allocating a VTSA 640, method 700 then moves to block 714, whereupon it is determined if transaction state information 642 exists for the virtual transaction or not. If transaction state information 642 does not exist for the virtual transaction, then transaction state information 642 is initialized and generated at block 716 and then the method 700 moves back to block 714.

If transaction state information 642 does exist for the virtual transaction, then the method 700 moves to block 718, whereupon the transaction state information 642 is retrieved and loaded into the VTSA 640. If transaction state informa-

tion **642** already existed and resides in transaction container **660**, then the transaction state information **642** is retrieved from the transaction container **660** and loaded into the VTSA **640** at block **718**.

Upon loading the transaction state information **642** into the VTSA **640**, the method **700** then moves to block **720**, whereupon a transaction item, received within POS message **630**, is added to or processed within the virtual transaction by the VTSA **640**. A transaction item is any portion of a virtual transaction, and includes such things as adding or removing an item **220** from virtual shopping cart **433**, totaling items **220** within virtual shopping cart **433**, a payment for all items **220**, some items **220**, one item **220**, or a portion of one item **220** within virtual shopping cart **433** or a specific virtual transaction, indicating a payment method for an item **220** within virtual shopping cart **433** or a specific virtual transaction, or any other activity or bit of information associated with the virtual transaction.

Upon processing or adding the transaction item using VTSA **640**, method **700** moves to block **722**, whereupon the transaction state information **642** is saved by VISA **640** within the transaction container **660**. Transaction container **660** represents a location within a storage device for storing transaction state information **642**. Upon saving the transaction state information **642**, method **700** moves to block **724** and releases the VTSA **640** back into the TSA Pool **644**, and closing the open VTSA **640** session, making VTSA **640** available for use by session manager **650** to process another request within a received POS message **630**. Method **700** then moves back to block **702** and awaits the receipt of another POS message **630**.

Those having skill in the art will recognize that the state of the art has progressed to the point where there is little distinction left between hardware and software implementations of aspects of systems; the use of hardware or software is generally (but not always, in that in certain contexts the choice between hardware and software can become significant) a design choice representing cost vs. efficiency tradeoffs. Those having skill in the art will appreciate that there are various vehicles by which processes and/or systems and/or other technologies described herein can be effected (e.g., hardware, software, and/or firmware), and that the preferred vehicle will vary with the context in which the processes and/or systems and/or other technologies are deployed. For example, if an implementer determines that speed and accuracy are paramount, the implementer may opt for a mainly hardware and/or firmware vehicle; alternatively, if flexibility is paramount, the implementer may opt for a mainly software implementation; or, yet again alternatively, the implementer may opt for some combination of hardware, software, and/or firmware. Hence, there are several possible vehicles by which the processes and/or devices and/or other technologies described herein may be effected, none of which is inherently superior to the other in that any vehicle to be utilized is a choice dependent upon the context in which the vehicle will be deployed and the specific concerns (e.g., speed, flexibility, or predictability) of the implementer, any of which may vary. Those skilled in the art will recognize that optical aspects of implementations will typically employ optically-oriented hardware, software, and or firmware.

The foregoing detailed description has set forth various embodiments of the devices and/or processes via the use of block diagrams, flowcharts, and/or examples. Insofar as such block diagrams, flowcharts, and/or examples contain one or more functions and/or operations, it will be understood by those within the art that each function and/or

operation within such block diagrams, flowcharts, or examples can be implemented, individually and/or collectively, by a wide range of hardware, software, firmware, or virtually any combination thereof. In one embodiment, several portions of the subject matter described herein may be implemented via Application Specific Integrated Circuits (ASICs), Field Programmable Gate Arrays (FPGAs), digital signal processors (DSPs), or other integrated formats. However, those skilled in the art will recognize that some aspects of the embodiments disclosed herein, in whole or in part, can be equivalently implemented in integrated circuits, as one or more computer programs running on one or more computers (e.g., as one or more programs running on one or more computer systems), as one or more programs running on one or more processors (e.g., as one or more programs running on one or more microprocessors), as firmware, or as virtually any combination thereof, and that designing the circuitry and/or writing the code for the software and or firmware would be well within the skill of one of skill in the art in light of this disclosure. In addition, those skilled in the art will appreciate that the mechanisms of the subject matter described herein are capable of being distributed as a program product in a variety of forms, and that an illustrative embodiment of the subject matter described herein applies regardless of the particular type of signal bearing medium used to actually carry out the distribution. Examples of a signal bearing medium include, but are not limited to, the following: a computer readable memory medium such as a magnetic medium like a floppy disk, a hard disk drive, and magnetic tape; an optical medium like a Compact Disc (CD), a Digital Video Disk (DVD), and a Blu-ray Disc; computer memory like random access memory (RAM), flash memory, and read only memory (ROM); and a transmission type medium such as a digital and/or an analog communication medium like a fiber optic cable, a waveguide, a wired communications link, and a wireless communication link.

The herein described subject matter sometimes illustrates different components contained within, or connected with, different other components. It is to be understood that such depicted architectures are merely exemplary, and that in fact many other architectures can be implemented which achieve the same functionality. In a conceptual sense, any arrangement of components to achieve the same functionality is effectively “associated” such that the desired functionality is achieved. Hence, any two components herein combined to achieve a particular functionality can be seen as “associated with” each other such that the desired functionality is achieved, irrespective of architectures or intermediate components. Likewise, any two components so associated can also be viewed as being “operably connected”, or “operably coupled”, to each other to achieve the desired functionality, and any two components capable of being so associated can also be viewed as being “operably couplable”, to each other to achieve the desired functionality. Specific examples of operably couplable include but are not limited to physically mateable and/or physically interacting components and/or wirelessly interactable and/or wirelessly interacting components and/or logically interacting and/or logically interactable components.

Those skilled in the art will recognize that it is common within the art to implement devices and/or processes and/or systems in the fashion(s) set forth herein, and thereafter use engineering and/or business practices to integrate such implemented devices and/or processes and/or systems into more comprehensive devices and/or processes and/or systems. That is, at least a portion of the devices and/or

21

processes and/or systems described herein can be integrated into comprehensive devices and/or processes and/or systems via a reasonable amount of experimentation. Those having skill in the art will recognize that examples of such comprehensive devices and/or processes and/or systems might include—as appropriate to context and application—all or part of devices and/or processes and/or systems of (a) an air conveyance (e.g., an airplane, rocket, hovercraft, helicopter, etc.), (b) a ground conveyance (e.g., a car, truck, locomotive, tank, armored personnel carrier, etc.), (c) a building (e.g., a home, warehouse, office, etc.), (d) an appliance (e.g., a refrigerator, a washing machine, a dryer, etc.), (e) a communications system (e.g., a networked system, a telephone system, a Voice over IP system, etc.), (f) a business entity (e.g., an Internet Service Provider (ISP) entity such as Comcast Cable, Quest, Southwestern Bell, etc.); or (g) a wired/wireless services entity such as Sprint, Cingular, Nextel, etc.), etc.

While particular aspects of the present subject matter described herein have been shown and described, it will be apparent to those skilled in the art that, based upon the teachings herein, changes and modifications may be made without departing from the subject matter described herein and its broader aspects and, therefore, the appended claims are to encompass within their scope all such changes and modifications as are within the true spirit and scope of the subject matter described herein. Furthermore, it is to be understood that the invention is defined by the appended claims. Accordingly, the invention is not to be restricted except in light of the appended claims and their equivalents.

The invention claimed is:

1. A method of providing a point-of-sale checkout process using a mobile device, the method comprising:
 - executing a mobile self-checkout application on the mobile device, the mobile self-checkout application configured to enable a user of the mobile device to select items in a store and pay for the selected items via a user interface rendered by the mobile self-checkout application on a display of the mobile device;
 - rendering the user interface on the display of the mobile device by the mobile self-checkout application;
 - in response to an activation by the user via the user interface, activating a scanning mode of the mobile self-checkout application and causing the mobile device to render a scan item interface, obtaining identification information associated with each of a plurality of physical objects, wherein the obtaining the identification information comprises activating an imaging device of the mobile device to capture the identification information associated with each of the plurality of physical objects, wherein the scan item interface comprises a barcode position indication area and a back action option to cancel the scanning mode;
 - obtaining, by the mobile device based on the identification information, item information associated with each of the plurality of physical objects;
 - generating a virtual shopping cart to save the item information associated with the plurality of physical objects;
 - adding the item information of the plurality of physical objects to the virtual shopping cart;
 - rendering, by the mobile device on the display, an added to cart confirmation in response to the item information of each of the plurality of physical objects having been virtually added to the virtual shopping cart;
 - rendering, by the mobile device on the display, a view cart option in the scan item interface;

22

- rendering, in response to detecting a selection of the view cart option, a shopping cart interface on the display, wherein the shopping cart interface comprises a listing of the item information corresponding to the plurality of physical objects having been added to the virtual shopping cart and a checkout option, wherein the item information comprises an item name and an item price; wherein a first view of the shopping cart interface renders a listing of at least some of the item information corresponding to one or more of the plurality of physical objects while further enabling scrolling by the user to view the at least some of the item information corresponding to at least one of the plurality of physical objects not previously visible;
- rendering, by the mobile device on the display, a quantity adjustment option configured to, in response to activation by the user, change a quantity of a first physical object in the virtual shopping cart, and a remove item option configured to, in response to activation by the user, remove the first physical object from the virtual shopping cart;
- maintaining state information comprising user identification data, store identification data, transaction identification data, and a state of a transaction; and
- receiving an instruction via the user interface of the mobile self-checkout application to complete the transaction of a sale of the plurality of physical objects identified in the virtual shopping cart.
2. The method of claim 1, further comprising:
 - updating, by a server, the state information to include the item information of each of the plurality of physical objects in the virtual shopping cart.
3. The method of claim 1, further comprising:
 - rendering, by the mobile device on the display, an electronic receipt.
4. The method of claim 1, further comprising:
 - displaying a manual identification information box to receive identification information associated with a second physical object of one or more of the plurality of physical objects manually entered by the user through the manual identification information box.
5. The method of claim 1, further comprising:
 - optically scanning, by an optical scanning device in communication with a point-of-sale system, a machine readable element rendered on the display of the mobile device;
 - retrieving, by the point-of-sale system using the transaction identification data, a total payment amount due to complete the sale of the plurality of physical objects; and
 - receiving, through the point-of-sale system, tendered payment of the total payment amount of the transaction.
6. The method of claim 1, further comprising:
 - rendering, by the mobile device on the display, a machine readable element that encodes customer identifying information of the user, the machine readable element to be scanned at an optical scanning device within the store, wherein the machine readable element is used by the optical scanning device to extract the customer identifying information to be used at least in part to look up the transaction including the virtual shopping cart to confirm payment and complete the transaction, the transaction including the plurality of physical objects identified in the virtual shopping cart.
7. The method of claim 1, wherein the obtaining the item information associated with the plurality of physical objects comprising retrieving, by the mobile device, the item infor-

23

mation stored in memory using the identification information, and the mobile device generating the virtual shopping cart.

8. The method of claim 1, wherein maintaining the state information comprises storing, by a server, the state information in a state container.

9. The method of claim 1, further comprising:
accessing, through the mobile device, item rating information corresponding to at least the first physical object of the plurality of physical objects; and
displaying the item rating information on the display of the mobile device.

10. The method of claim 1, further comprising:
activating relative to the transaction, in response to receiving the instruction to complete the transaction, a first virtual terminal sales application (VTSA) of a plurality of VTSA's on a server, causing the first VTSA to import the state information in completing the transaction of the sale of the plurality of physical objects and export, by the first VTSA, changed state information to a state container based at least on the completion of the transaction, wherein the first VTSA is deactivated relative to the transaction following the exporting of the changed state information.

11. The method of claim 1, further comprising:
activating, in response to each selection of each of the plurality of physical objects, one of a plurality of virtual terminal sales applications (VTSA), causing the activated one of the plurality of VTSA's to import the state information corresponding to the transaction, and export to a state container by the activated one of the plurality of VTSA's changed state information corresponding to the selected physical object of the plurality of physical objects, wherein the activated one of the plurality of VTSA's is deactivated relative to the transaction following the exporting of the changed state information.

12. A retail self-checkout system comprising:
a mobile self-checkout application configured to be executed on a mobile device and to allow a user of the mobile self-checkout application to select items in a store and pay for the selected items via a user interface rendered by the mobile self-checkout application on a display of the mobile device, wherein the mobile device in executing the mobile self-checkout application is configured to:

render the user interface on the display of the mobile device by the mobile self-checkout application;

active, in response to an activation by the user via the user interface, a scanning mode of the mobile self-checkout application and cause the mobile device to render a scan item interface, obtain identification information associated with each of a plurality of physical objects through activation of an imaging device of the mobile device to capture the identification information associated with each of the plurality of physical objects, wherein the scan item interface comprises a barcode position indication area and a back action option to cancel the scanning mode;

obtain, by the mobile self-checkout application based on the identification information, item information associated with each of the plurality of physical objects; add the item information of the plurality of physical objects to a virtual shopping cart;

24

render, on the display, an added to cart confirmation in response to the item information of each of the plurality of physical objects having been virtually added to the virtual shopping cart;

render, on the display, a view cart option in the scan item interface;

render, in response to detecting a selection of the view cart option, a shopping cart interface on the display, wherein the shopping cart interface comprises a listing of the item information corresponding to the plurality of physical objects having been virtually added to the virtual shopping cart and a checkout option, wherein the item information comprises an item name and an item price; wherein a first view of the shopping cart interface renders a listing of at least some of the item information corresponding to one or more of the plurality of physical objects while further enabling scrolling by the user to view the at least some of the item information corresponding to at least one of the plurality of physical objects not previously visible;

render, on the display, a quantity adjustment option configured to, in response to activation by the user, change a quantity of a first physical object in the virtual shopping cart, and a remove item option configured to, in response to activation by the user, remove the first physical object from the virtual shopping cart; and receive an instruction via the user interface of the mobile self-checkout application to complete a transaction of a sale of the plurality of physical objects identified in the virtual shopping cart.

13. The retail self-checkout system of claim 12, wherein the mobile self-checkout application executed on the mobile device is further configured to render, on the display, an electronic receipt.

14. The retail self-checkout system of claim 12, wherein the mobile self-checkout application executed on the mobile device is further configured to:

display a manual identification information box to receive identification information associated with a second physical object of one or more of the plurality of physical objects manually entered by the user through the manual identification information box.

15. The retail self-checkout system of claim 12, further comprising:

a point-of-sale system; and
an optical scanning device communicatively coupled with the point-of-sale system;

wherein the mobile self-checkout application is configured to cause the display of the mobile device to render a machine readable element that encodes identifying information;

wherein the optical scanning device is configured to optically scan the machine readable element rendered on the display; and

wherein the point-of-sale system is configured to:
retrieve, based on the identifying information, a total payment amount due; and
confirm tendered payment is received of the total payment amount due of the transaction.

16. The retail self-checkout system of claim 12, wherein the mobile self-checkout application executed on the mobile device is further configured to render, on the display of the mobile device, a machine readable element that encodes customer identifying information of the user, the machine readable element configured to be scanned at an optical scanning device within the store to extract the customer identifying information to be used at least in part to look up

25

the transaction including the virtual shopping cart to confirm payment and complete the transaction, the transaction including the plurality of physical objects identified in the virtual shopping cart.

17. The retail self-checkout system of claim 12, further comprising:

a server configured to maintain and update over time state information related to the transaction, wherein the state information comprises the item information in the virtual shopping cart of each of the plurality of physical objects having been added to the virtual shopping cart.

18. The retail self-checkout system of claim 17, wherein the server causes the state information to be maintained in a state container.

19. The retail self-checkout system of claim 17, further comprising:

the server, wherein the server is configured to activate relative to the transaction, based on the instructions to complete the transaction, a first virtual terminal sales application (VTSA) of a plurality of VTSA's on the server, causing the first VTSA to import the state information in completing the transaction of the sale of

26

the plurality of physical objects and export, by the first VTSA, changed state information to a state container based at least on the completion of the transaction, wherein the first VTSA is deactivated relative to the transaction following the exporting of the changed state information.

20. The retail self-checkout system of claim 12, further comprising:

a server, wherein the server is configured to activate, in response to each selection of each of the plurality of physical objects, one of a plurality of virtual terminal sales applications (VTSA), causing the activated one of the plurality of VTSA's to import state information corresponding to the transaction, and export to a state container, by the activated one of the plurality of VTSA's, changed state information corresponding to a selected physical object of the plurality of physical objects, wherein the activated one of the plurality of VTSA's is deactivated relative to the transaction following the exporting of the changed state information.

* * * * *